

### TECHNICAL PRE-QUALIFICATION REQUIREMENT

**Name of Project:** 765/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Name of Customer:** NTPC Limited

**Name of Item:** 132kV Isolator/ Disconnectors

#### **TECHNICAL PRE-QUALIFICATION REQUIREMENT**

(a) The Bidder should have manufactured and supplied minimum five (05) nos. of three phase Disconnectors/ Isolators suitable for Air Insulated Substation/ Switchyard of 132kV or above class which should have been in successful operation for minimum two (2) years prior to the date of Techno-Commercial bid opening (i.e. 10.07.2024).

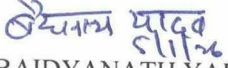
#### **SUPPORTING DOCUMENTS TO BE ATTACHED**

| Sr | Required Criteria    | Supporting Documents to be submitted by bidder along with technical bid                                                                                             |
|----|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Manufacturing        | Approved Drawings/ GTP/ Approved Quality Plan/ Factory Inspection Test Report etc. of offered item.                                                                 |
| 2  | Supply               | Purchase Order (PO)/ Dispatch clearance/ Lorry Receipt (LR)/ Material Receipt certificate at site / installation or commissioning certificate etc. of offered item. |
| 3  | Successful operation | Successful operation means certificate issued by the Employer/Utility certifying the operation without any adverse remark.                                          |

Notes (General points):

1. Consideration of offer shall be subject to customer's approval of bidder's, if applicable.
2. Bidder to submit all supporting documents in English. If documents submitted by bidder are in language other than English, a self- attested English translated document should also be submitted.
3. Notwithstanding anything stated above, BHEL reserves the right to assess the capabilities and capacity of the bidder to perform the contract, should the circumstances warrant such assessment in the overall interest of BHEL.
4. After satisfactory fulfilment of all the above criteria/ requirement, offer shall be considered for further evaluation as per NIT and all the other terms of the tender.
5. Bidder to submit complete documents complying above Pre-qualification Criterion (PQR) along with duly filled 3K document (format attached).

PREPARED BY

  
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MANAGER-TBEM

REVIEWED BY

  
SAI K SONI  
DGM-TBEM

APPROVED BY

  
VIVEK KAPIL  
AGM-TBEM

**Sub : Sub-Qualifying Requirements for the 132 KV Disconnectors**

**5.14.2.B.III. For 132 KV Disconnectors :** In order to satisfy the requirement as indicated in Technical Specifications, Clause No. 5.14.2.B.III, of Sub-Section-IA, Part-A, Section-VI, We declare that M/s ..... is a manufacturer who have manufactured and supplied atleast five (05) nos. of Three phase Disconnectdor of 132 KV or above Voltage Class suitable for Air Insulated Substation/Switchyard which are in successful operation for minimum two (2) years. The details are given below :

**For 132 kV or above class Disconectors**

| Sl. No. | Item Description                                                                                           | Details |
|---------|------------------------------------------------------------------------------------------------------------|---------|
| 1.      | Name and Address of Manufacturer and Name of contact person with email ID, Telephone & Fax Nos.            |         |
| 2       | Location of manufacturer plant & its address, Telephone & Fax Nos.                                         |         |
| 3.      | Client Name and its Address, including Tel. No. and Fax no.                                                |         |
| 4       | Name of the plant / project & its location for which the manufacturer has supplied the 132KV Disconnectors |         |
| 5.      | Name and designation of the responsible person in Clients organisation.                                    |         |
| 6       | Name & locatin of the substation / switchyard                                                              |         |
| 7.      | i)Contract No. & Date                                                                                      |         |
|         | ii)Brief scope of work                                                                                     |         |
|         | iii) Value of Order                                                                                        |         |
| 08      | Scope of work for the aforesaid contract includes the following                                            |         |
|         | (i) Manufactured                                                                                           | Yes/No  |
|         | (ii) Supplied                                                                                              | Yes/No  |
|         | iii)Type test                                                                                              | Yes/No  |

|     |                                                                                                                                                                                                                     |            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 09  | i)No. of Three phase disconnectors supplied                                                                                                                                                                         |            |
|     | ii)Voltage Level (in KV)                                                                                                                                                                                            |            |
| 10  | Date of Commissioning                                                                                                                                                                                               |            |
| 11  | Date of commencement of successful Operation & No of years of service                                                                                                                                               |            |
| 12  | Whether Three phase disconnectors supplied which are in successful operation for at least two (2) years as on the date of consideration for approval but not later than six months after award date of the package. | Yes / No   |
| 13. | Client(s) certificate(s) enclosed in support of stated experience above at Annexure.....to this Attachment-3K.(Use separate sheet for each experience/Contract)                                                     | YES* / NO* |

-----

Date : (Signature).....

Place : (Printed Name).....

(Designation).....

(Common Seal).....

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               | <b>BHARAT HEAVY ELECTRICALS LIMITED</b><br><b>TRANSMISSION BUSINESS ENGINEERING MANAGEMENT</b><br><b>NOIDA</b> |             |           |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
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| <b>DOCUMENT NO.</b> TB-435-316-TS-004B<br><b>TYPE OF DOC.</b> TECHNICAL SPECIFICATION<br><b>Title:</b><br><b>132kV Isolators with/ without Earth Switches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>REV 00</b> | Prepared                                                                                                       | Checked     | Approved  |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>NAME</b>   | BY                                                                                                             | JK          | VK        |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>SIGN</b>   |                              |             |           |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>DATE</b>   | 05.01.26                                                                                                       | 05.01.26    | 05.01.26  |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>GROUP</b>  | TBEM                                                                                                           |             |           |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>WO No.</b> | CS-8003-001-2-FC/SC/TC-COA-7491/7492/7493                                                                      |             |           |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
| <b>CUSTOMER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NTPC Limited  |                                                                                                                |             |           |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
| <b>PROJECT</b> 765/400/132kV AIS Switchyard Extension at NTPC SIPAT STPP Stage-III (1 x 800 MW)<br><b>Contents</b><br><table border="1"> <thead> <tr> <th>Section No.</th> <th>Description</th> <th>No of Pages</th> </tr> </thead> <tbody> <tr> <td>Section-1</td> <td>Scope, Technical Requirements and Quantities</td> <td></td> </tr> <tr> <td>Section-2</td> <td>Equipment Specification under Scope of Supplies</td> <td></td> </tr> <tr> <td>Section-3</td> <td>Project details and general technical requirements (For all equipment under the Project)</td> <td></td> </tr> <tr> <td>Section-4</td> <td>Annexures</td> <td></td> </tr> <tr> <td></td> <td>Annexure-A: Compliance Certificate to Technical Specification</td> <td></td> </tr> <tr> <td></td> <td>Annexure-B: Deviation/ Change Request to Technical Specification</td> <td></td> </tr> <tr> <td></td> <td>Annexure-C: Technical Checklist</td> <td></td> </tr> <tr> <td></td> <td>Annexure-D: Guaranteed Technical Particulars</td> <td></td> </tr> </tbody> </table> <p><b>Remarks:</b> Bidder to note that data and details of guaranteed technical particulars shall not be reviewed during technical evaluation/ scrutiny, hence compliance of guaranteed technical particulars in line with technical specification shall be bidder's responsibility.</p> | Section No.   | Description                                                                                                    | No of Pages | Section-1 | Scope, Technical Requirements and Quantities |  | Section-2 | Equipment Specification under Scope of Supplies |  | Section-3 | Project details and general technical requirements (For all equipment under the Project) |  | Section-4 | Annexures |  |  | Annexure-A: Compliance Certificate to Technical Specification |  |  | Annexure-B: Deviation/ Change Request to Technical Specification |  |  | Annexure-C: Technical Checklist |  |  | Annexure-D: Guaranteed Technical Particulars |  |  |  |  |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rev. No.      | Date                                                                                                           | Altered     | Checked   | Approved                                     |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
| Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |                                                                                                                |             | To        |                                              |  |           |                                                 |  |           |                                                                                          |  |           |           |  |  |                                                               |  |  |                                                                  |  |  |                                 |  |  |                                              |  |  |  |  |  |
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### CHECKLIST FOR TECHNICAL EVALUATION

Along with the technical offer/ bids, the bidder should submit this checklist confirming the inclusion of the enclosures as listed below,

| Sl. No. | Documents to be enclosed                                                                                   | Bidder to confirm<br>(Please tick "Confirmed") |
|---------|------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1.      | Supporting documents for compliance of Technical Qualifying Requirement (as per 3K format, as applicable). | Confirmed                                      |
| 2.      | Unpriced BOQ duly mentioning "Quoted" for all the items, signed and sealed.                                | Confirmed                                      |
| 3.      | Annexure- A duly signed and sealed & Annexure- B duly filled, signed and sealed.                           | Confirmed                                      |

Note: Any bidder not meeting the above requirement shall be liable for non-evaluation.

The above checklist is reviewed and verified for,

NIT Reference No.:

Name of Bidder:

Name of Project: **400kV AIS Switchyard at Singareni Thermal Power Project Stage- II (1x800MW)**

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

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## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

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### SECTION 1: SCOPE, PROJECT SPECIFIC TECHNICAL REQUIREMENTS & BILL OF QUANTITIES

#### 1. Scope

This technical specification covers the requirements of design, manufacture, inspection including third party inspection and testing at manufacturer's work before supply, proper packing and delivery to project of equipment (**132kV Isolators with/ without Earth Switches**) complete with all fittings, accessories, mandatory spares, as applicable of the equipment complete in all respects for efficient & trouble-free working mentioned under this specification.

This section covers the specific technical requirements of equipment. This constitutes minimum technical parameters for the above item as specified by the BHEL/ NTPC. The offered equipment shall also comply with the Section-3 (Project Details and General technical requirements for all equipment under the Project) of this specification.

The specification comprises of following sections:

- Section-1 : Scope, Project Specific Technical Requirements & Bill of Quantities
- Section-2 : Equipment Specification under scope of Supplies/ Service
- Section-3 : Project Details & General Technical Requirements (For all equipment under the Project)
- Section-4 : Annexures
  - Annexure-A: Compliance Certificate to Technical Specification
  - Annexure-B: Deviation/ Change Request to Technical Specification
  - Annexure C: Technical Checklist
  - Annexure-D: Guaranteed Technical Particulars

The following order of priority/ precedence shall be followed. In case of conflict between requirements specified in various documents, the more stringent one shall be followed. BHEL/ NTPC concurrence shall, however, be obtained before taking a final decision in such matters.

#### 1. Statutory Regulations

In particular, the latest version of the following statutory regulations, as applicable, shall be followed for system,

- o Indian Electricity Act
- o CEA regulations
- o The Factory Act
- o Requirements of other statutory bodies as applicable, e.g. CEA etc.

#### 2. Section-1

#### 3. Section-2

#### 4. Section-3

#### 5. Codes & Standards

Bidder shall furnish list of conflicts/ ambiguities/ deviations, if any, along with their technical offer and also furnish the basis that is considered for submitting technical offer. BHEL/ NTPC will

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

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resolve listed conflicts prior to award. In case of ambiguity, bidder shall inform BHEL/ NTPC of their interpretation. In case bidder fails to convey the same prior to award, BHEL/ NTPC decision on interpretation shall be considered final if need arises during the execution. No additional cost or extra time on account of conflicts/ ambiguities/ deviations shall be admissible.

In general, no deviation from the requirements specified in various clauses of this specification shall be allowed and hence, a certificate to this effect shall have to be furnished along with the offer (Annexure-A), however bidder shall furnish list of conflicts/ ambiguities/ deviations (Annexure-B), if any. Any conflicts/ ambiguities/ deviations mentioned elsewhere in technical offer shall not be reviewed.

The equipment is required for the following project:

|                         |   |                                                                                   |
|-------------------------|---|-----------------------------------------------------------------------------------|
| Name of the Customer    | : | <b>NTPC Limited (NTPC Ltd.)</b>                                                   |
| Name of Main Contractor | : | <b>Bharat Heavy Electricals Limited (BHEL)</b>                                    |
| Name of the Project     | : | <b>765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)</b> |

**The terms used in this specification namely, “Employer” refers to “NTPC Ltd.”, “PURCHASER” refers to BHEL, “Contractor” refers to successful Bidder and “EQUIPMENT” refers to “132kV Isolators with/ without Earth Switches”.**

The scope of supplies shall be as per commercial terms and conditions enclosed separately with the notice inviting tender/ enquiry.

## 2. Codes & Standards

- The Isolator shall comply with the latest editions and amendments of the following standards as applicable, unless otherwise specified elsewhere in this specification,
 

|                      |                                                                      |
|----------------------|----------------------------------------------------------------------|
| IEC 62271-102        | Alternating current isolators (dis-connectors) and earthing switches |
| IS:9921              | Alternating current isolators (dis-connectors) and earthing switches |
| IS:2544              | Insulators                                                           |
| IS 13947             | Degree of protection provided by enclosure                           |
| IS/IEC60034-5 (2000) | Degree of protection provided by enclosure                           |
| IS:325               | Three phase induction motors                                         |
| IS:4722              | Rotating electrical machines                                         |
| IS:262               | Recommended practice for hot dip galvanizing of iron & steel         |
| IS:4759              | Hot dip galvanization coating on structural steel                    |
| IS:2633              | Method of testing weight, thickness and uniformity                   |
| IS:1573              | Electroplated coating of zinc on iron and steel                      |
| IS:6735              | Spring washers                                                       |
| IS:2016              | Plain washers                                                        |
- For the purpose of this specification, all technical terms used hereinafter shall have the meaning as per IEC/ ISS specification.
- The equipment meeting with the requirements of other authoritative standards, which ensure equal or better quality than the standards mentioned above shall also be acceptable. Where the equipment offered by the bidder confirms to other standards, salient points of difference

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

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between the standards adopted and the specified standards shall be clearly brought out in the offer.

4. In case of imported equipment, standards of the country of origin shall be applicable, if these standards are equivalent or stringent than the applicable Indian standards.
5. The equipment shall also conform to the provisions of Indian Electricity Rules, 1956 and other statutory regulations currently in force in the country.
6. In case Indian standards are not available for any equipment, standards issued by IEC/ BS/ VDE/ IEEE/ NEMA or equivalent agency shall be applicable.

### 3. Specific Technical Requirements

1. Isolator shall perform satisfactorily under various other electrical, electromechanical and meteorological conditions of the site of installation.
2. Equipment shall be able to withstand all external and internal mechanical, thermal and electromechanical forces due to various factors like wind load, temperature variation, ice & snow, (wherever applicable) short circuit etc. for the equipment.
3. The equipment shall also comply to facilitate erection of equipment, all items to be assembled at site shall be “match marked”.

| Sl. No. | Description                                                             | Unit                                                                               | Technical Parameters (132kV system)                                                                                                  |
|---------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Rated voltage                                                           | kVrms                                                                              | 132                                                                                                                                  |
| 2       | Rated Maximum Voltage                                                   | kVrms                                                                              | 145                                                                                                                                  |
| 3       | Rated frequency                                                         | Hz                                                                                 | 31.5                                                                                                                                 |
| 4       | No. of poles                                                            | Nos.                                                                               | 3                                                                                                                                    |
| 5       | Design ambient temperature                                              | °C                                                                                 | 50                                                                                                                                   |
| 6       | Type                                                                    |                                                                                    | Outdoor, off-load type<br>Isolator: Horizontal Centre Break (HCB)<br>Earth Switch: Single/ double earth switch/ without Earth Switch |
| 6       | Rated current at 50°C ambient temperature                               | A                                                                                  | 1250                                                                                                                                 |
| 7       | Rated short time withstand current of isolator and earth switch         | kA                                                                                 | 31.5 for 1sec                                                                                                                        |
| 8       | Rated dynamic short time withstand current of isolator and earth switch | kAp                                                                                | 80                                                                                                                                   |
| 9       | Temperature rise over design ambient temperature                        | As per Table-3 of IEC 62271-1/ IEC 60694 for an ambient design temperature of 50°C |                                                                                                                                      |
| 10      | Rated mechanical terminal load                                          | N                                                                                  | As per table III of IEC-62271-102                                                                                                    |
| 11      | Mechanical Endurance Class                                              | Isolator-M2 as per IEC<br>E/S-M0/ M1 as per IEC                                    |                                                                                                                                      |
| 12      | Operating mechanism of isolator/earth switch                            | AC/DC/universal motor                                                              |                                                                                                                                      |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

|     |                                                                                          |                                                                                                                                                                                                              |                                  |
|-----|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 13  | Minimum no. of auxiliary contacts on each isolator                                       | Isolator: In addition to requirements of specification/ Equipment Schematics (Min.8NO+8NC contacts per pole/isolator), 12NO +12NC+4MBB contacts wired to Terminal Blocks (Reversible) for future use.        |                                  |
| 14  | Minimum no. of auxiliary contacts on each earthing switch                                | Earth switch: In addition to requirements of specification/ Equipment Schematics (Min.6NO+6NC contacts per pole/isolator), 12 NO+ 12 NC+4 MBB contacts wired to Terminal Blocks (Reversible) for future use. |                                  |
| 15  | Max. Operating time                                                                      | secs                                                                                                                                                                                                         | Not more than 12sec              |
| 16  | Number of terminals in control cabinet                                                   | All contacts & control circuits are to be wired up to control cabinet plus 24 spare terminals evenly distributed.                                                                                            |                                  |
| 17  | Rated Insulation levels                                                                  |                                                                                                                                                                                                              |                                  |
| a)  | Full wave impulse withstand voltage (1.2/50 microsec.)                                   |                                                                                                                                                                                                              |                                  |
| i)  | between line terminals and ground                                                        | kVpeak                                                                                                                                                                                                       | ±650                             |
| ii) | between terminals with isolator open                                                     | kVpeak                                                                                                                                                                                                       | ±750 (across isolating distance) |
| b)  | Switching impulse withstand voltage (250/2500 micro-second) dry and wet                  |                                                                                                                                                                                                              |                                  |
| i)  | between line terminals and ground                                                        | kVpeak                                                                                                                                                                                                       | Not Applicable                   |
| ii) | between terminals with Isolator open                                                     | kVpeak                                                                                                                                                                                                       | Not Applicable                   |
| c)  | One-minute power frequency dry withstand voltage                                         |                                                                                                                                                                                                              |                                  |
| i)  | between line terminals and ground                                                        | kV rms                                                                                                                                                                                                       | 275                              |
| ii) | between terminals with isolator open                                                     | kV rms                                                                                                                                                                                                       | 315 (across isolating distance)  |
| 18  | Minimum Corona extinction voltage with Isolator in all positions                         | kV rms                                                                                                                                                                                                       | 92                               |
| 19  | Max. radio interference voltage for frequency between 0.5 MHz and 2 MHz in all positions | µV                                                                                                                                                                                                           | 1000µV at 92kVrms                |
| 20  | Minimum Creepage distance                                                                | mm                                                                                                                                                                                                           | 3625mm (i.e. 25mm/kV)            |
| 21  | Seismic acceleration                                                                     |                                                                                                                                                                                                              | 0.3g/ As per IS 1893             |
| 22  | Thermal Rating of Auxiliary Contacts                                                     | A                                                                                                                                                                                                            | 10A at 220 V DC                  |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

|    |                                                                     |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|---------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | Breaking Capacity of auxiliary contacts                             |    | 2A DC with circuit time constant not less than 20 ms                                                                                                                                                                                                                                                                                                                                                                                |
| 24 | Distance between support structures foundations (within same phase) | m  | 6.0                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 25 | System neutral earthing                                             |    | Effectively Earthed                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 26 | Local remote switch indication of Isolator for SAS/ SCADA           |    | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 27 | Phase to phase spacing (mm) (min)                                   | mm | 2100                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 28 | Rated mechanical Terminal load                                      |    | As per IEC – 62271-102 or value calculated                                                                                                                                                                                                                                                                                                                                                                                          |
| 29 | Number of terminals in Control cabinet                              |    | (1) All contacts and control circuits are to be wired up to control cabinet including potential free auxiliary contacts of Isolator/Earth switch.<br>(2) Additional 24 nos. evenly distributed spare TBs in Master to be provided for interpole cabling of Aux contacts (employer's use).<br>(3) Sufficient TBs shall be provided in control cabinet for looping of AC supply from master to follower of Isolator and earth switch. |
| 30 | DC operated Electrical Interlocking coil                            |    | To be included for Isolator and Earth Switch                                                                                                                                                                                                                                                                                                                                                                                        |
| 31 | TB size & Type                                                      |    | (1) TBs shall be stud type<br>(2) Power Cable- Each TB Should be suitable for terminating two cores of 10 Sq. mm size on each side<br>(3) Control cable- Each TB shall be suitable for terminating two wires of 2.5 Sq. mm size on each side                                                                                                                                                                                        |

**4. Pitch Circle Diameter (PCD) details for 132kV, 6kN Insulator: (For Isolator)-BHEL scope of supply**

|              |         |
|--------------|---------|
| Top PCD      | 127     |
| No. of holes | 4 x Ø16 |
| Bottom PCD   | 254     |
| No. of holes | 8 x Ø18 |

*All dimensions are in mm*

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

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5. The earthing switches shall be capable of discharging trapped charges of the associated lines. Isolator and earth switches shall be able to bear on the terminals the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.
6. The isolator shall be capable for making / breaking normal currents when no significant change in voltage occurs across the terminals of each pole of the isolator on account of making /breaking operation.
7. Disconnecting Switches with Single Earth switch shall be so constructed that Earth Switch can be installed on either side of the disconnect switch at site.
8. All the hardware (Nut, bolt & washer etc.) required for the fixing the isolator & its accessories to structure shall be included in the bidder's scope.
9. Any other applicable parameters, not mentioned here, please refer section-2 of this specification.

#### 4. General Technical Requirements

The general technical requirements for Equipment shall be as follows,

1. Bidder shall provide adequate power distribution circuit & terminal block in Isolator/ earth switch marshalling box. Bidder shall provide Power circuit diagram for all isolator/ earth switch during detailed engineering stage. Duplicated wiring on TB shall not be allowed.
2. Flexible Cu braided earthing for Isolator/ earth switch marshalling box (2 number/ box), size equivalent to 75X12 mm GS flat. Bidder to provide minimum cross section of 270Sq mm flexible copper braid on each side of marshaling box. Hole to hole distance for flexible copper braid shall be minimum 200mm. Hardwires for installation of flexible Cu braid for fixing on marshalling box shall be under bidder's scope.
3. Earth switch shall have grounded connection with minimum 2 number's 155sqmm Cu flexible braid.
4. The commencement of erection, testing and commissioning activities of Isolators shall be communicated to manufacturers from time to time as per front readiness/ availability at site.
5. Any special tools and tackle, which are specifically required for the equipment and are proprietary in nature shall be included in offer. List of such special tools and tackle should be clearly listed along with the technical offer. Any special tool which is not listed in the list but required during the erection/ testing/ commissioning of equipment shall also be deemed in bidder's scope.
6. Supervision of erection, testing and commissioning of all supplied Isolator with earth switch shall be carried out under the supervision of the Isolator manufacturer's representative. The commissioning report shall be prepared and signed by the manufacturer's representative. However, BHEL shall extend all the support including manpower, general tools & plants required for erection work.
7. The bidder must fill up all the details required for offered item/s. Instead of indicating "refer drawing, or as per IS/IEC", the exact value/s must be filled in.

#### 5. Bill of Quantities

1. Bill of quantities shall be as per following, however, any item not appearing herein but required for completeness of the work and mentioned elsewhere in technical specification is deemed to

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be included in bidder's scope.

| Sl. No.   | Item Description                                                                                                                                                                                                                                                                                                          | Unit | Quantity |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|
| <b>A.</b> | <b>Supply: Main Items</b>                                                                                                                                                                                                                                                                                                 |      |          |
| 1         | Supply: 132kV, 1250, 3-Ø, HCB type isolator (mechanically ganged motor operated) line isolators/ dis-connectors with 2 earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors         | Set  | 02       |
| 2         | Supply: 132kV, 1250, 3-Ø, HCB type isolator (mechanically ganged motor operated) line isolators/ dis-connectors with 1 earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors         | Set  | 02       |
| 3         | Supply: 132kV, 1250, 3-Ø, HCB type Tandem isolator (mechanically ganged motor operated) line isolators/ dis-connectors without earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors | Set  | 02       |
| <b>B.</b> | <b>Service</b>                                                                                                                                                                                                                                                                                                            |      |          |
| 1         | Services: 132kv, 1250, 3-Ø, HCB type Isolator-Supervision of Erection, Testing and Commissioning including Alignment Check                                                                                                                                                                                                | Nos. | 06       |

- Vendor to quote individual spare items against the above BOQ separately, even if the items are covered in any other spare items of this BOQ. Vendor to mention, spares which are not applicable as per the design the of the offered Isolator. However, in latter stage, if found that the spares are applicable for the offered circuit breakers, the same should be provided without any price implication.
- Flexible Braided Copper (2 Nos. of 300 mm Length, 75sq.mm Cross Section for each MOM Box) along with Lugs for earthing of each Isolator & Earth Switch Shall Be Included in Equipment Prices.
- Prices for all applicable accessories, fixing hardware of Isolators, Post Insulators, Operating Rod Insulators and MOM box on structure of isolators and earth switches, as applicable shall be included in the equipment (against above BOQ) prices.
- Erection, testing and commissioning of all 132kV Isolators and earth switches shall be carried out under the supervision of the Isolator manufacturer's representative. However, required

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unskilled men power/labour, tools (other than special tools and tackles which shall be in bidder's scope) shall be provided by BHEL.

6. Respective dates for the commencement of erection, testing and commissioning activities of Isolators shall be communicated to manufacturers from time to time as per the readiness of respective sites. Bidder to note that erection, testing and commissioning of all of the supplied breakers shall not be done in one go. Multiple visits may be required.
7. The quantities in BOQ may vary up to  $\pm 10\%$  in line with quantity variation clause. However, individual quantities may be deleted or vary up to any extent.

### 6. Drawings / Documents required for Technical Clearance for Manufacturing

The engineering drawings/ documents, shall be used for providing technical clearance for manufacturing of the equipment, which shall be used for delay analysis, if applicable for respective group.

|   |                                                                       |
|---|-----------------------------------------------------------------------|
| 1 | 132kV Isolator- Outline General Arrangement, BOM & Control Schematics |
| 2 | 132kV Isolator- Guaranteed Technical Particulars/ Datasheet           |
| 3 | 132kV Isolator- Type Test Reports                                     |
| 4 | 132kV Isolator- Quality Assurance Plan                                |

Technical clearance for manufacturing shall be issued after approval of drawings in category-I (approval without any comments)/ category-II (approval with comments) from customer/ BHEL. In case drawing/ document are not duly stamped in category-1/ category-2 by customer, BHEL stamp in Category-1 & 2 shall be treated final to proceed further.

The successful bidder shall have to extend all possible supports like timely submission/ re-submission of drawings, visit to end customer to facilitate documents approval without any commercial implications to BHEL. Acceptance of bidder's documents shall be subject to end customer/ NTPC approval.

### 7. Type Testing

All equipment to be supplied shall be of type tested design. During detail engineering, the contractor/ bidder shall submit for Owner/ NTPC approval the reports of all the type tests as listed in this specification. The validity period of reports shall be as per CEA Guidelines for the validity period of Type test (s) conducted on Major Electrical Equipment in power Transmission- May2020 & with latest amendments from **10.07.2024**. These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test (s) should have been either conducted at an independent laboratory or should have been witnessed by a Client/ NTPC.

However, if contractor/ bidder is not able to submit report of the type test(s) conducted as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical equipment in power Transmission-May'2020 & with latest amendments from **10.07.2024**, or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor/ bidder shall conduct all such tests under this contract at no additional cost to the owner either at third

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party lab or in presence of client/ owner/ customer representative and submit the reports for approval.

All acceptance and routine tests as per the specification and relevant standards shall be carried out. Charges for these shall be deemed to be included in the equipment price.

### 8. Quality Plan

The successful bidder shall submit Quality Assurance Plan with in-process inspection methods, tests, records, etc. for BHEL/ NTPC approval. Customer/ NTPC hold points will also be included in the plan, which shall be mutually agreed by the BHEL/ NTPC. In case bidder has reference, Quality Assurance Plan agreed with BHEL/ NTPC same shall be submitted for specific project to BHEL/ NTPC approval. There shall be no commercial implication to BHEL/ NTPC on account of Quality Plan approval.

Superior quality control system shall be adopted to assure high product quality. Raw materials of the best commercial grade quality and high reliability shall be used in the manufacture of the equipment. All materials shall be procured, manufactured, inspected and tested by vendor/ sub-vendor as per approved quality plan. The supplier shall perform all tests necessary to ensure that the material and workmanship conform to the relevant standards and comply with the requirements of the specification. Charges for all tests for the equipment shall be deemed to be included in bidder's scope.

### 9. Inspection & Testing

1. The equipment shall be subject to inspection by BHEL/ NTPC or authorized representative at bidder/ manufacturers' works. Hence, Bidder shall furnish all necessary information concerning the supply to BHEL/ NTPC.
2. Routine and acceptance tests as listed in relevant standard and section-2, technical specifications shall be complied.
3. Bidder shall also furnish factory acceptance test (FAT) from manufacturers for BHEL/ NTPC approval in line with specific requirements mentioned in section-2, technical specification.

### 10. Packing and Dispatch

1. The equipment shall be properly packed for selected mode of transportation i.e. sea, rail and road in such a manner that it is protected against the climatic conditions and for any damage during transportation, transit and storage. The panels shall be wrapped in polyethylene sheets before being placed in wooden crates/ cases to prevent damage to the finish. Crates/ cases shall have skid bottoms for handling. Special notations such as 'Fragile', 'This side up', 'Weight', 'Owner's particulars\ 'PO nos.' etc., shall be clearly marked on the package together with other details as per purchase order.
2. All equipment shall be suitably protected, coated, covered or boxed and crated to prevent damage or deterioration during transit, handling and outdoor storage (for a minimum period of 1 years) at site till the time of erection. While packing all the materials, the limitations from the point of view of availability of transportation facilities in India should be considered. The Bidder shall be responsible for any loss or damage during transportation, handling and storage.
3. The Bidder shall include and provide for security, protection and packing the equipment so as

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to avoid loss or damage during transport by any mode.

4. All packing shall allow for easy removal and checking at site. Wherever necessary, proper arrangement for attaching slings for lifting shall be provided. All packages shall be clearly marked for with signs showing 'UP' and 'DOWN' side of boxes, and handling and unpacking instructions as considered necessary. Special precautions shall be taken to prevent rusting of steel and iron parts during transit and storage. Gas seals or other methods proposed to be adopted for protection against moisture during transit shall be to the satisfaction of the purchaser.
5. Each package delivered under the contract shall be marked by the Bidder at his expense and such marking must be distinct (all previous irrelevant marking being carefully obliterated). Such marking shall show the description and quantity of contents, the name of consignee and address, the gross and net weights of the package, the name of Bidder with a distinctive number of mark sufficient for purpose of identification. All markings shall be carried out with such materials as to ensure quickness of drying, fastness and legibility.
6. Each Package shall contain a note quoting specifically the name of the Bidder, the number and date of contract or order and the name of office placing the contract, nomenclature of the stores and include a schedule of parts for each complete equipment giving the parts number with reference to the General Arrangement/ Assembly drawing and the quantity of each part, drawing number and tag numbers.
7. All equipment/ material shall be suitably packed for transport, carriage at site and outdoor storage during transit. The Bidder shall be responsible for any damage to the equipment during transit. The contents of each package shall bear marking that can be readily identified from the package list and packing shall provide complete protection from moisture, termites and mechanical shocks etc.
8. Any material found short inside the packing cases shall be supplied by the Bidder without any extra cost.
9. Notwithstanding anything stated in this clause the Bidder shall be entirely responsible for any loss, damage or depreciation to the stores.

### 11. Exceptions

Followings are not in bidder's scope of supply (BHEL supplied items)

- a. Equipment support structure
- b. Terminal Connector

### 12. Abbreviations Used

The following terminology/ acronym hereunder and elsewhere in the technical specification used and their grammatical variations shall unless repugnant to the subject or context thereof, have the following full form hereunder respectively assigned to them, namely,

|      |                     |
|------|---------------------|
| AC:  | Alternating Current |
| DC:  | Direct Current      |
| kV:  | Kilovolt            |
| Hz:  | Hertz               |
| IP:  | Ingress Protection  |
| BOQ: | Bill of Quantities  |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

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|       |                                                 |
|-------|-------------------------------------------------|
| QAP:  | Quality Assurance Plan                          |
| NIT:  | Notice Inviting Tender                          |
| PI:   | Post Insulator                                  |
| OEM:  | Original Equipment Manufacturer                 |
| BHEL: | Bharat Heavy Electricals Limited                |
| BIS:  | Bureau of Indian Standards                      |
| BS:   | British Standard                                |
| ANSI: | American National Standards Institute           |
| ASTM: | American Society for Testing and Materials      |
| IS:   | Indian Standards                                |
| IEC:  | International Electro Technical Commission      |
| IEEE: | Institute of Electrical & Electronics Engineers |
| CEA:  | Central Electricity Authority                   |
| NEMA: | National Electrical Manufacturers Association   |

Bharat Heavy Electricals Limited

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SECTION 2:

|                                   |   |
|-----------------------------------|---|
| Customer/ NTPC specification..... | 2 |
|-----------------------------------|---|

|                                                                           |                                                                                                                                 |                                   |                     |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------|
|                                                                           | <div>TECHNICAL REQUIREMENTS</div> <div></div> |                                   |                     |
|                                                                           |                                                                                                                                 |                                   |                     |
| 3.00.00                                                                   | DISCONNECTOR : GENERAL                                                                                                          |                                   |                     |
| SIPAT SUPER THERMAL POWER PROJECT<br>STAGE- III (1X800 MW)<br>EPC PACKAGE | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B                                                                                 | SUB-SECTION :<br>B-17: SWITCHYARD | Page<br>E- 19 of 98 |

| <div style="text-align: center;"> <b>TECHNICAL REQUIREMENTS</b>  </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                 |                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------|
| 3.01.01                                                                                                                                                   | The isolators and accessories shall conform in general to relevant IEC 62271-102 (or equivalent Indian Standard) except to the extent explicitly modified in specification. Earth switches shall be provided on isolators as marked on SLD. Isolators shall be horizontal centre break type.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                 |                                                                            |
| 3.01.02                                                                                                                                                   | The isolators and earth switches shall be A. C / D.C. motor operated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                 |                                                                            |
| 3.02.00                                                                                                                                                   | <b>DUTY REQUIREMENTS</b><br><br>Isolators and earth switches shall be capable of withstanding the dynamic and thermal effects of the maximum possible short circuit current of the system in their closed position. They shall be constructed such that they do not open under influence of short circuit current and wind pressure together. The earth switches wherever provided shall be constructional interlocked so that the earth switches can be operated only when the isolator is open and vice-versa.                                                                                                                                                                                                                                           |                                                 |                                                                            |
| 3.02.01                                                                                                                                                   | The earth switches wherever provided shall be constructional interlocked so that the earth switches can be operated only when the isolator is open and vice-versa. Mechanical Endurance : M2 type of duty as per IEC for 765kV and earthing switches shall be of class M0 / M1 duty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                 |                                                                            |
| 3.02.02                                                                                                                                                   | In addition to the constructional interlock, isolator and earth switches shall have provision to prevent their electrical and manual operation unless the associated and other interlocking conditions are met. All these interlocks shall be of failsafe type. Suitable individual interlocking coil arrangements shall be provided. The interlocking coil shall be suitable for continuous operation from DC supply and within a variation range as stipulated in relevant section. The interlock coil shall be provided with adequate contacts for facilitating permissive logic for 'DC' control scheme of the isolator as well as for AC circuit of the motor to prevent opening or closing of isolators when the interlocking coil is not energised. |                                                 |                                                                            |
| 3.02.03                                                                                                                                                   | The earthing switches shall be capable of discharging trapped charges of the associated lines. Isolator and earth switches shall be able to bear on the terminals the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                 |                                                                            |
| 3.02.04                                                                                                                                                   | The isolator shall be capable for making / breaking normal currents when no significant change in voltage occurs across the terminals of each pole of the isolator on account of making / breaking operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                 |                                                                            |
| 3.03.00                                                                                                                                                   | <b>CONSTRUCTIONAL FEATURES :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                 |                                                                            |
| 3.03.01                                                                                                                                                   | The isolators shall be provided with high pressure current carrying contacts on the hinge/ jaw ends and all contact surfaces shall be silver plated. The thickness of silverplating should not be less than 25 microns. The contacts shall be accurately machined and self aligned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                 |                                                                            |
| 3.03.02                                                                                                                                                   | The isolator shall be provided with a galvanised steel base provided with holes and designed for mounting on a lattice/pipe support structure. The base shall be rigid and self supporting. The position of movable contact system (main blades) of each of the isolator and earthing switch shall be indicated by a mechanical indicator at the lower end of the vertical rod of shaft for the isolator and earthing switch. The indicator shall be of metal and shall be visible from operating level.                                                                                                                                                                                                                                                   |                                                 |                                                                            |
| SIPAT SUPER THERMAL POWER PROJECT<br>STAGE- III (1X800 MW)<br>EPC PACKAGE                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B | SUB-SECTION :<br><b>B-17: SWITCHYARD</b><br><br>Page<br><b>E- 20 of 98</b> |

|                                                                                                | <b>TECHNICAL REQUIREMENTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                               |                                                 |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|
| <b>3.03.03</b>                                                                                 | All metal parts shall be of non-rusting and non-corroding metal. Current carrying parts shall be from high conductivity electrolytic copper/aluminium. Bolts, screws and pins shall be provided with lock washers. Keys or equivalent locking facilities, if provided on current carrying parts, shall be made of copper silicon alloy or equivalent. The live parts shall be designed to eliminate sharp joints, edges and other corona producing surfaces.                                                                                                                                                                                                                                                                                                                           |                                                               |                                                 |
| 3.03.04                                                                                        | The isolators shall be so constructed that the switch blade will not fall to the closed position if the operating shaft gets disconnected. Isolators and earthing switches including their operating parts shall be such that they cannot be dislodged from their open or closed positions by gravity, wind pressure, vibrations shocks or accidental touching of the connecting rods of the operating mechanism. The switch shall be designed such that no lubrication of any part is required except at very infrequent intervals.                                                                                                                                                                                                                                                   |                                                               |                                                 |
| 3.03.05                                                                                        | The insulator of the isolator shall conform to the requirements stipulated under clause no: 08.00.00 as specified in the specification and shall have a min. cantilever strength of 800 kg for 765kV and 600kg for 132kV. Pressure due to the contact shall not be transferred to the insulators after the main blades full close. The insulators shall be so arranged that leakage current will pass to earth and not between terminals of the same pole or between phases. The terminal connectors shall conform to requirements stipulated under clause no:08.00.00 as specified else where in the specification.                                                                                                                                                                   |                                                               |                                                 |
| 3.04.00                                                                                        | <b>EARTHING SWITCHES:</b><br><br>Where earthing switches are specified these shall include the complete operating mechanism and auxiliary contacts. The earthing switches shall form an integral part of the isolator and shall be mounted on the base frame of the isolator. Earthing switches shall be suitable for local operation only. The earthing switches shall be constructional interlocked with the isolator so that the earthing switches can be operated only when isolator is open and vice versa.                                                                                                                                                                                                                                                                       |                                                               |                                                 |
| 3.05.00                                                                                        | <b>OPERATING MECHANISM AND CONTROL</b><br><br>The Contractor shall offer, motor operated switches having padlock arrangement on both 'ON' and 'OFF' positions. Limit switches for control shall be fitted on the isolator/ earth switch shaft, within the cabinet to sense the open and close positions of the isolators and earth switches. It shall not be possible, after final adjustment has been made for any part of the mechanism to be displaced at any point in the travel sufficient enough to allow improper functioning of the isolator when the isolator is opened or closed at any speed. Control cabinet/operating mech. box shall conform to requirements stipulated in clause no: 08.00.00 else where in the specification and IS:5039/IS 8623/IEC 439 as applicable |                                                               |                                                 |
| <b>3.05.01</b>                                                                                 | <b>OPERATION</b><br><br>Isolator shall be gang operated for main blades and earth switches. The operation of the three poles shall be well synchronised and interlocked.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                               |                                                 |
| 3.05.02                                                                                        | The design shall be such as to provide maximum reliability under all service conditions. All operating linkages carrying mechanical loads shall be designed for negligible deflection. The length of inter insulator and interpole operating rods shall be capable of adjustments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                               |                                                 |
| 3.05.03                                                                                        | The design of linkages and gears be such so as to allow one man to operate the handle with ease for isolator and earth switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                               |                                                 |
| <b>3.06.00</b>                                                                                 | <b>TESTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                               |                                                 |
| <b>SIPAT SUPER THERMAL POWER PROJECT</b><br><b>STAGE- III (1X800 MW)</b><br><b>EPC PACKAGE</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>TECHNICAL SPECIFICATION</b><br><b>SECTION – VI, PART-B</b> | <b>SUB-SECTION :</b><br><b>B-17: SWITCHYARD</b> |
|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                               | <b>Page</b><br><b>E- 21 of 98</b>               |

|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                             |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--|-------------|-----------|----|------------------|-------------------------------------------------------------------------------------------------------------|----|--------------|---------------|----|----------------------|------------------|----|-----------------|--------|----|--------------------------------|--------------------------------------------------------------|---|-----------------------------------------------------------------|--------------------------------------------|----|--------------------------------------------------|-----------------------|----|---------------------------|---------|----|---------------------------|-------------------|----|--------------------------|-----------------------------------------------------------------------------------------------------|---|------------------|-----------------------------------------------------------|
|                                                                           | TECHNICAL REQUIREMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                             |  |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.07.01                                                                   | In continuation to the requirements stipulated under Part-I the isolator alongwith operating mechanism shall conform to the type tests and shall be subjected to routine tests and acceptance tests in accordance with IEC 62271-102. Minimum 50 nos. mechanical operations will be carried out on 1 (one) isolator assembled completely with all accessories as acceptance test. During final testing of isolator sequential closing/ opening of earth switch shall also be checked only after isolator is fully open/close. Acceptance test shall be carried out with operating box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                             |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.06.02                                                                   | The insulator shall conform to all the type tests as per IEC60168. In addition to all type, routine and acceptance tests, as per IEC-60168, the following additional routine/ acceptance tests shall also be carried out:<br>a)Bending load test in four directions at 50% min. bending load guaranteed in all insulators.<br>b) Bending load test in four directions at 100% min. bending load guaranteed on sample insulators in a lot. c) Torsional test on sample insulator of a lot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                             |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.7.00                                                                    | General PARAMETERS:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                             |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|                                                                           | <table><tr><td></td><td>Description</td><td>Parameter</td></tr><tr><td>a)</td><td>Type of isolator</td><td>Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV</td></tr><tr><td>b)</td><td>No. of poles</td><td>Three(3poles)</td></tr><tr><td>c)</td><td>Rated operating time</td><td>Not &gt; than 12sec</td></tr><tr><td>d)</td><td>Control voltage</td><td>220VDC</td></tr><tr><td>e)</td><td>Auxiliary contacts on isolator</td><td>Min.8NO &amp; 8NC contacts per pole/isolator for 800kV &amp; 145kV .</td></tr><tr><td>f</td><td>Auxiliary contacts on earth/ safety/grounding/high speed switch</td><td>Min.6NO &amp; 6NC contacts per pole/isolator .</td></tr><tr><td>g)</td><td>Operating mechanism of isolator and earth switch</td><td>AC/DC/universal motor</td></tr><tr><td>h)</td><td>Minimum creepage distance</td><td>25mm/Kv</td></tr><tr><td>i)</td><td>Rated ambient temperature</td><td>50 degree Celsius</td></tr><tr><td>j)</td><td>Support structure height</td><td>Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level</td></tr><tr><td>k</td><td>Temperature rise</td><td>As per Table III of IEC 60694 for an ambient of 50 deg. C</td></tr></table> |                                                                                                             |                                                                                    |  | Description | Parameter | a) | Type of isolator | Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV | b) | No. of poles | Three(3poles) | c) | Rated operating time | Not > than 12sec | d) | Control voltage | 220VDC | e) | Auxiliary contacts on isolator | Min.8NO & 8NC contacts per pole/isolator for 800kV & 145kV . | f | Auxiliary contacts on earth/ safety/grounding/high speed switch | Min.6NO & 6NC contacts per pole/isolator . | g) | Operating mechanism of isolator and earth switch | AC/DC/universal motor | h) | Minimum creepage distance | 25mm/Kv | i) | Rated ambient temperature | 50 degree Celsius | j) | Support structure height | Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level | k | Temperature rise | As per Table III of IEC 60694 for an ambient of 50 deg. C |
|                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameter                                                                                                   |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| a)                                                                        | Type of isolator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| b)                                                                        | No. of poles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Three(3poles)                                                                                               |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| c)                                                                        | Rated operating time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Not > than 12sec                                                                                            |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| d)                                                                        | Control voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 220VDC                                                                                                      |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| e)                                                                        | Auxiliary contacts on isolator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Min.8NO & 8NC contacts per pole/isolator for 800kV & 145kV .                                                |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| f                                                                         | Auxiliary contacts on earth/ safety/grounding/high speed switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Min.6NO & 6NC contacts per pole/isolator .                                                                  |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| g)                                                                        | Operating mechanism of isolator and earth switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AC/DC/universal motor                                                                                       |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| h)                                                                        | Minimum creepage distance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 25mm/Kv                                                                                                     |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| i)                                                                        | Rated ambient temperature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 50 degree Celsius                                                                                           |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| j)                                                                        | Support structure height                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level         |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| k                                                                         | Temperature rise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | As per Table III of IEC 60694 for an ambient of 50 deg. C                                                   |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.07.01                                                                   | 765KV Isolator :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                             |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|                                                                           | <table><tr><td></td><td>Description</td><td>Parameter</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                             |                                                                                    |  | Description | Parameter |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameter                                                                                                   |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| SIPAT SUPER THERMAL POWER PROJECT<br>STAGE- III (1X800 MW)<br>EPC PACKAGE |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B                                                             | SUB-SECTION :<br>B-17: SWITCHYARD                                                  |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page<br>E- 22 of 98                                                                                         |                                                                                    |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |

# TECHNICAL REQUIREMENTS



|    |                                                                |                                                                                                                                                                                                       |
|----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) | Rated continuous current at 50deg.cen ambient temperature.     | Minimum 3150/2000 A at rated ambient temperature as per the Single line diagram                                                                                                                       |
| b) | Rated short time withstand current of isolator and earthswitch | 50 kA rms for One (1) second                                                                                                                                                                          |
| c) | Rated Insulation levels                                        |                                                                                                                                                                                                       |
| d) | Rated one minute power. Frequency withstand voltage            | i) 830 kV rms between live terminals and earth<br>ii) 1150 kV rms across isolating distance                                                                                                           |
| e) | Rated lightning impulse Withstand voltage                      | i) $\pm 2100$ kVp between live terminals and earth.<br>ii) $\pm 2100$ kVp impulse on one terminal and 455 kVp power freq. Voltage of opposite polarity on other terminal (across isolating distance). |
| f) | Rated switching impulse Withstand voltage                      | i) $\pm 1550$ kVp between live terminals and earth.<br>ii) $\pm 1175$ kVp impulse on one terminal and 650 kVp power freq.voltage of opposite polarity on other terminal (across isolating distance).  |
|    |                                                                |                                                                                                                                                                                                       |

## **132 KV Isolator :**

|    | Description                                                    | Parameter                                                                                            |
|----|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| a) | Rated continuous current at 50 deg.cen ambient temperature.    | Minimum 2000/1250 A at rated ambient temperature as per the Single line diagram                      |
| b) | Rated short time withstand current of isolator and earthswitch | 31.5kA rms for One (1) second                                                                        |
| c) | Rated Insulation levels                                        |                                                                                                      |
| d) | Rated one minute power. Frequency withstand voltage            | i) 275kV rms between live terminals and earth<br>ii) 315 kV rms across isolating distance            |
| e) | Rated lightning impulse Withstand voltage                      | i) $\pm 650$ kVp between live terminals and earth.<br>ii) $\pm 750$ kVp (across isolating distance). |
| f) | Rated switching impulse Withstand voltage                      | _____                                                                                                |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

### SECTION 3

#### PROJECT DETAILS AND GENERAL SPECIFICATIONS

#### 3.0 General

This section stipulates the General Technical Requirements under the Contract and will form an integral part of the Technical Specification.

The provisions under this section are intended to supplement general requirements for the materials, equipment and services covered under other sections of tender documents and are not exclusive. However in case of conflict between the requirements specified in this section and requirements specified under other sections, the requirements specified under respective sections shall prevail.

#### 3.1 Project Details

|                                                     | Particular                                            | Details                                                                                                                              |
|-----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| a)                                                  | Customer                                              | NTPC Limited                                                                                                                         |
| b)                                                  | Engineer/Consultant/ Inspector                        | ---                                                                                                                                  |
| c)                                                  | Project Title                                         | 765/400/132kV AIS Switchyard Extension at NTPC SIPAT STPP Stage-III (1 x 800 MW)                                                     |
| d)                                                  | Project Location                                      | Place: Sipat<br>District: Bilaspur<br>State: Chhattisgarh                                                                            |
| e)                                                  | Latitude & Longitude                                  | <b>Latitudes and Longitudes of the site are as follows:</b><br>22° - 05' to 22° - 09' (N) and 82° -16' to 82° - 18' (E) respectively |
| f)                                                  | Nearest Railway Station                               | Jairamnagar – At a distance of about 3 km on Nagpur-Raipur-Kolkata main line.                                                        |
| g)                                                  | Distance of project location from the Railway station | 3 km (approx.)                                                                                                                       |
| h)                                                  | Nearest Major Town                                    | Bilaspur city                                                                                                                        |
| i)                                                  | Distance of the town from the project site            | 20 km                                                                                                                                |
| j)                                                  | Nearest commercial airport                            | Bilaspur                                                                                                                             |
| k)                                                  | Distance of airport from the project site             | 20 km                                                                                                                                |
| <b><u>SITE CONDITIONS</u></b> (for design purposes) |                                                       |                                                                                                                                      |
| a)                                                  | Design ambient temperature                            | 50°C                                                                                                                                 |
| b)                                                  | Maximum Relative humidity                             | 80 %                                                                                                                                 |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

|           |                                                                  |                                       |
|-----------|------------------------------------------------------------------|---------------------------------------|
| <b>c)</b> | Height above mean sea level                                      | Less than 1000 meters                 |
| <b>d)</b> | Pollution Severity                                               | Heavily polluted                      |
| <b>e)</b> | Criteria for Wind Resistant design of structures and equipment   | Standard Applicable - IS 875 (Part 3) |
| <b>f)</b> | Basic Wind speed “Vb” at ten meters above the mean ground level. | 39 m/sec                              |
| <b>g)</b> | Category of terrain                                              | Cat-2                                 |
| <b>h)</b> | Risk Coefficient “K1”                                            | 1.06                                  |

### 3.1.1 System Parameters

| Sl.No. | Parameters                                | 765 kV       | 400 kV      | 132 kV      |
|--------|-------------------------------------------|--------------|-------------|-------------|
| 1      | Highest system voltage                    | 800 kV rms   | 420 kV rms  | 145 kV rms  |
| 2      | Lightning Impulse voltage                 | ±2100 kVp    | ±1425kVp    | ±650 kVp    |
| 3      | Switching impulse voltage                 | ±1550 kVp    | ±1050kVp    | NA          |
| 4      | Power frequency withstand for 1 min (rms) | 1150 kV(rms) | 650 kV(rms) | 275 kV(rms) |
| 5      | Max. fault level (1 sec.)                 | 50 kA        | 63 kA       | 31.5 kA     |
| 6      | Minimum creepage distance                 | 20000 mm     | 10500 mm    | 3625 mm     |

### 3.1.2 Auxiliary Power:

| Sl.No. | Nominal Connection Voltage | Variations in Voltage | Frequency    | Phase          | Neutral         |
|--------|----------------------------|-----------------------|--------------|----------------|-----------------|
| 1      | 415V                       | ±10%                  | 50 (+3% -5%) | 3Phase, 4 Wire | Solidly Earthed |
| 2      | 240V                       | ±10%                  | 50 (+3% -5%) | 1 phase        | Solidly Earthed |

Combined variation of voltage and frequency shall be + 10%. Design fault level of 415V system shall be restricted to 50kA rms for 1 second.

The operational limits for variation of DC voltage are (+) 10% to (-) 15%.

3.1.3 The various minimum heights of the AIS switchyard shall be as given below from plinth level:

| Voltage | Equipment /1st Level | 2nd Level | 3rd Level | Peak   |
|---------|----------------------|-----------|-----------|--------|
| 765kV   | 14000mm              | 27000mm   | 40000mm   | 8000mm |
| 400kV   | 8000mm               | 16000mm   | 23000mm   | 8500mm |

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|       |        |        |         |        |
|-------|--------|--------|---------|--------|
| 132kV | 4600mm | 8500mm | 12500mm | 5200mm |
|-------|--------|--------|---------|--------|

The minimum vertical distance from the bottom of the lowest porcelain part of the bushing, porcelain enclosures or support insulators to the bottom of the equipment structure, where it rests on the foundation pad shall be 2550mm.

The minimum height of intermediate gantry tower for 765 & 400kV wherever required shall be 29 m & 25 m respectively and the peak (s) shall be of 8 m & 8.5 m respectively. The gantry width for 765 & 400kV AIS shall be minimum 54 m & 27m or as required to meet the specified clearances.

3.1.4 The minimum clearances for 765, 400 & 132kV switchyards shall be as given below:

| Sl. No. | Parameters               | 765 kV   | 400 kV  | 132 kV  |
|---------|--------------------------|----------|---------|---------|
| 1       | Phase to earth clearance | 4900 mm  | 3500 mm | 1300 mm |
| 2       | Phase to phase clearance | 7600 mm  | 4000 mm | 1300 mm |
| 3       | Section clearance        | 10300 mm | 6500 mm | 4000 mm |

### 3.2 Instructions to Bidders

The bidders shall submit the technical requirements, data and information as per the technical data sheets, provided in Section-4.

The bidders shall furnish catalogues, engineering data, technical information, design documents, drawings etc fully in conformity with the technical specification.

The supplier should be approved by Employer. If not, it is the responsibility of the vendor to be assessed and approved Employer, before placement of order by BHEL. Any cost involved in vendor assessment/approval must be borne by the vendor himself.

The Bidder's proposal shall be based upon the use of equipment and material complying fully with the requirements specified herein. It is recognized that the Bidder may have standardized on the use of certain components, materials, processes or procedures different than those specified herein. Alternate proposals offering similar equipment based on the manufacturer's standard practice will also be considered, provided the base offer is in line with technical specifications and such proposals meet the specified design standards and performance requirement and are acceptable to the Purchaser. Sufficient amount of information for justifying such proposals shall be furnished to Purchaser alongwith the bid to enable the Purchaser to determine the acceptability of these proposals.

Wherever a material or article is specified or defined by the name of a particular brand, Manufacturer or Vendor, the specific name mentioned shall be understood to be indicative of the function and quality desired and not restrictive. Other manufacturer's products may be

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considered provided sufficient information is furnished to enable the Employer to determine that the products proposed are equivalent to those named.

Equipment furnished shall be complete in every respect with all mountings, fittings, fixtures and standard accessories normally provided with such equipment and/ or needed for erection, completion and safe operation of the equipment as required by applicable codes, though they may not have been specifically detailed in the Technical Specifications unless included in the list of exclusions. Materials and components not specifically stated in the specification but which are necessary for commissioning and satisfactory operation of the switchyard unless specifically excluded shall be deemed to be included in the scope of the specification and shall be supplied without any extra cost. All similar standard components/parts of similar standard equipment under supply shall be inter-changeable with one another.

The bidder shall supply type tested (including special tests as per tech. specification) equipment and materials. The test reports shall be furnished by the bidder along with equipment/ material drawings. In the event of any discrepancy in the test reports, (i.e., if any test report is not acceptable due to any design/ manufacturing changes or due to non-compliance with the Technical Specification and/ or applicable standard), the tests shall be carried out without any additional cost implication to the BHEL. BHEL reserves the right to get any or all type/tests conducted/repeated.

### 3.3 Codes & Standards

In addition to the codes and standards specifically mentioned in the relevant technical specifications for the equipment / plant / system, all equipment parts, systems and works covered under this specification shall comply with all currently applicable statutory regulations and safety codes of the Republic of India as well as of the locality where they will be installed, including the following:

- a) Indian Electricity Act
- b) Indian Electricity Rules
- c) Indian Explosives Act
- d) Indian Factories Act and State Factories Act
- e) Indian Boiler Regulations (IBR)
- f) Regulations of the Central Pollution Control Board, India
- g) Regulations of the Ministry of Environment & Forest (MoEF), Government of India
- h) Pollution Control Regulations of Department of Environment, Government of India

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- i) State Pollution Control Board.
- (j.) Rules for Electrical installation by Tariff Advisory Committee (TAC).
- (k.) Building and other construction workers (Regulation of Employment and Conditions of services) Act, 1996
- (l.) Building and other construction workers (Regulation of Employment and Conditions of services) Central Rules, 1998
- (m.) Explosive Rules, 1983
- (n.) Petroleum Act, 1984
- (o.) Petroleum Rules, 1976,
- (p.) Gas Cylinder Rules, 1981
- (q.) Static and Mobile Pressure Vessels (Unified) Rules, 1981
- (r.) Workmen's Compensation Act, 1923
- (s.) Workmen's Compensation Rules, 1924
- (t.) NTPC Safety Rules for Construction and Erection
- (u.) NTPC Safety Policy
- (v.) Any other statutory codes / standards / regulations, as may be applicable.

Unless covered otherwise in the specifications, the latest editions (as applicable as on date of bid opening: 03-March-2017), of the codes and standards given below shall also apply:

- a) Bureau of Indian standards (BIS)
- b) Japanese Industrial Standards (JIS)
- c) American National Standards Institute (ANSI)
- d) American Society of Testing and Materials (ASTM)
- e) American Society of Mechanical Engineers (ASME)
- f) American Petroleum Institute (API)

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- g) Standards of the Hydraulic Institute , U.S.A.
- h) International Organization for Standardization (ISO)
- i) Tubular Exchanger Manufacturer's Association (TEMA)
- j) American Welding Society (AWS)
- k) National Electrical Manufacturers Association (NEMA)
- l) National Fire Protection Association (NFPA)
- m) International Electro-Technical Commission (IEC)
- n) Expansion Joint Manufacturers Association (EJMA)
- o) Heat Exchange Institute (HEI)
- p) IEEE standard
- q) JEC standard

Other International/ National standards such as DIN, VDI, BS, GOST etc. shall also be accepted for only material codes and manufacturing standards, subject to the Employer's approval, for which the Bidder shall furnish, adequate information to justify that these standards are equivalent or superior to the standards mentioned above. In all such cases the Bidder shall furnish specifically the variations and deviations from the standards mentioned elsewhere in the specification together with the complete word to word translation of the standard that is normally not published in English.

As regards highly standardized equipment such as Steam Turbine and Generator, National /International standards such as JIS, DIN, VDI, ISO, SEL, SEW, VDE, IEC & VGB shall also be considered as far as applicable for Design, Manufacturing and Testing of the respective equipment. However, for those of the above equipment not covered by these National / International standards, established and proven standards of manufacturers shall also be considered.

In the event of any conflict between the codes and standards referred to in the above clauses and the requirement of this specification, the requirement of Technical Specification shall govern.

In case of any change in codes, standards & regulations between 03-March-2017 and the date when vendors proceed with fabrication, the Employer shall have the option to incorporate the changed requirements or to retain the original standard. It shall be the responsibility of the Contractor to bring to the notice of the Employer such changes and advise Employer of the resulting effect.

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### 3.4 Services to be performed by the Equipment being furnished

The 400kV system is being designed to limit the power frequency over voltage of 1.5 p.u. and the switching surge over voltage to 2.5 p.u. In 400 kV system the initial value of temporary over voltage could be 2.0 p.u. for 1-2 cycles. All the equipment/materials covered in this specification shall perform all its function satisfactorily without undue strain, restrike etc. under such over voltage conditions.

All equipment shall also perform satisfactorily under various other electrical, electromechanical and meteorological conditions of the site of installation. All equipment shall be able to withstand all external and internal mechanical, thermal and electromechanical forces due to various factors like wind load, temperature variation, ice & snow (not applicable for this project), short circuit etc for the equipment.

### 3.5 Engineering Data

#### 3.5.1 Drawings

All drawings submitted by the supplier including those submitted at the time of bid shall be in sufficient detail to indicate the type, size, arrangement, material description, Bill of Materials, weight of each component, break-up for packing and shipment, the external connections, fixing arrangement required. The dimensions required for installation and interconnections with other equipment and materials, clearances and spaces required for installation and interconnections between various portions of equipment and any other information specifically requested in the specifications.

Each drawing submitted by the bidder (including those of sub-vendors) shall bear a title block at the right-hand bottom corner with clear mention of the name of the Employer, the system designation, the specifications title, the specification number, the name of the Project, drawing number and revisions. If standard catalogue pages are submitted, the applicable items shall be indicated therein. All titles, noting, markings and writings on the drawing shall be in English. All the dimensions should be in metric units.

After the approval of the drawings, further work by the bidder shall be in strict accordance with these drawings and no deviation shall be permitted without the written approval of the Purchaser, if so required.

The review of these document/data/drawings by the purchaser will cover only general conformance of the document/data/drawings to the specification and contract, interfaces with the equipment provided under specification, external connections and of the dimensions which might affect plan layout. This review by the purchaser may not indicate a thorough review of the dimensions, quantities and details of the equipment, material, any devices or items indicated or the accuracy of the information submitted. The review and/or approval by the purchaser shall not be considered by the bidder, as limiting any of his responsibilities and

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liabilities for mistakes and deviations from the requirements, specified under these specifications and documents.

All manufacturing, fabrication and execution of work in connection with the equipment/system prior to the approval of the drawings shall be at the bidder's risk. The bidder is expected not to make any changes in the design of the equipment /system, once they are approved by the Purchaser. However, if some changes are necessitated in the design of the equipment/system at a later date, the bidder may do so, but such changes shall promptly be brought to the notice of the Purchaser indicating the reasons for the change and get the revised drawing approved again in strict conformance to the provisions of the Technical Specification. Approval of bidder's drawing or work by the Purchaser shall not relieve the bidder of any of his responsibilities and liabilities under the Contract.

All engineering data submitted by the contractor after final process including review and approval by the purchaser shall form part of the contract document and the entire work performed under these specifications shall be performed in strict conformity with technical specification, unless otherwise expressly requested by the purchaser in writing.

### 3.5.2 Bidder's Drawing Submission and Approval Procedure

The following procedure for submission and review/approval of the drawings, data reports, information, etc. shall be followed by the bidder:

- a. All data/information furnished by Vendor in the form of drawings, documents, Catalogues or in any other form for Employer's information/interface and/or review and approval are referred by the general term "drawings".
- b. The 'Master drawings list' indicating titles, Drawing Number, Date of submission and approval etc. shall be furnished by the bidder. This list shall be updated if required at suitable interval during detailed engineering.
- c. All drawings (including those of sub-vendor) shall bear at the right hand bottom corner the 'title plate' with all relevant information duly filled in. The bidder shall furnish this format to his sub-vendor along with his purchase order for sub-vendor's compliance.
- d. Contractor shall submit all the drawings in five (5) copies for review of Employer. Employer shall forward their comments within four (4) weeks of receipt of drawings.
- e. Upon review of each drawings, depending on the correctness and completeness of the drawings, the same will be categorised and approval accorded in one of the following categories:

|             |                                       |
|-------------|---------------------------------------|
| CATEGORY I  | Approved                              |
| CATEGORY II | Approved, subject to incorporation of |

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|              |                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------|
|              | comments/modification as noted. Resubmit revised drawing incorporating the comments                      |
| CATEGORY III | Not approved. Resubmit revised drawings for Approval after incorporating comments/modifications as noted |
| CATEGORY IV  | For information and records                                                                              |

- f. Bidder shall resubmit the drawings approved under Category II, III within one (1) week of receipt of comments on the drawings, incorporating all comments. Every revision of the drawing shall bear a revision index wherein such revisions shall be highlighted in the form of description or marked up in the drawing identifying the same with relevant revision number enclosed in a triangle (e.g 1.2.3. etc.).
- g. In case Bidder does not agree with any specific comment, he shall furnish the explanation for the same to Employer for consideration. In all such cases Bidder shall necessarily enclose explanations along with the revised drawing (taking care of balance comments) to avoid any delay and/or duplication in review work.
- h. It is the responsibility of the Bidder to get all the drawings approved in the Category I or IV (as the case may be) and complete engineering activities within the agreed schedule. Any delay arising out of submission and modification of drawings shall not alter the contract completion schedule.
- i. Bidder shall not make any changes in the portions of the drawing other than those commented. If changes are required to be made in the portions already approved, the Bidder shall resubmit the drawings identifying the changes (along with reasons for changes) for Employer's review and approval. **Drawings resubmitted shall show clearly the portions where the same are revised marking the relevant revision numbers and Employer shall review only such revised portion of documents.**
- j. As Built Drawings  
  
After final acceptance of individual equipment / system by the Employer, the Bidder will update all original drawings and documents for the equipment / system to "as built" conditions and submit no. of copies as per clause 3.5.5.
- k. Approval of drawings will not in any way relieve the Bidder of his obligations of furnishing the equipment in accordance with the specification and shall not prevent subsequent rejection if such equipment is later found to be defective.

### 3.5.3 Erection Drawings.

- a. Contractor shall furnish erection drawings for the guidance or commencement of erection or the first shipment, whichever is earlier. These shall generally comprise of fabrication/assembly drawings, various component/part details drawing, assembly, clearance data requirements, etc. The drawings shall contain details of components/

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equipment with identification number, match marks, bill of materials, assembly procedures etc.

- b. For all major equipment apart from above details, assembly sequence and instructions with check-lists shall be furnished in the form of erection manuals.

### 3.5.4 Instruction Manual

- a. The Contractor shall submit to the Employer preliminary instruction manuals for all the equipments for review. The final instructions manuals incorporating Employer's comments and complete in all respect shall be submitted at least sixty (60) days before the first shipment of the equipment. The instruction manuals shall contain full details and drawings of all the equipments, the transportation, storage, installation, testing, commissioning, operation and maintenance procedures, etc. separately for each component/equipment along with log record format. These instruction manuals shall be submitted in five (5) copies for approval.
- b. If after commissioning and initial operation of the plant, the instruction manuals require any modifications/additions/changes, the same shall be incorporated and the updated final instruction manuals shall be submitted.
- c. The operating and maintenance instructions together with drawings (other than shop drawings) of the equipment, as completed, shall have sufficient details to enable the Employer to maintain, dismantle, reassemble and adjust all parts of the equipment. They shall give a step by step procedure for all operations likely to be carried out during the life of the plant/equipment, including erection, testing, commissioning, operation, maintenance dismantling and repair. Each manual shall also include a complete set of approved drawings together with performance/rating curves of the equipment and test certificates, wherever applicable. The contract shall not be considered completed for purpose of taking over until such instructions and drawings have been supplied to the Employer.
- d. A separate section of the manual shall be for each size/type of equipment and shall contain a detailed description of construction and operation, together with all relevant pamphlets.
- e. The manuals shall include the following
  - a) List of spare parts along with their drawing and catalogues and procedure for ordering spares.
  - b) Lubrication Schedule including charts showing lubrication checking, testing and replacement procedure to be carried daily, weekly, monthly & at longer intervals to ensure trouble free operation.
- f. Where applicable, fault location charts shall be included to facilitate finding the cause of mal-operation or break down.

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- g. A collection of the manufacturer's standard leaflets will not accepted to be taken as a compliance of this clause. The manual shall be specifically compiled for the concerned project.

The Instruction Manuals shall comprise of the following:

### 3.5.4.1 Erection Manuals

The erection manuals shall be submitted atleast three (3) months prior to the commencement of erection activities of particular equipment/system. The erection manual should contain the following as a minimum.

- a) Erection strategy.
- b) Sequence of erection.
- c) Erection instructions.
- d) Critical checks and permissible deviation/tolerances.
- e) List of tool, tackles, heavy equipments like cranes, dozers, etc.
- f) Bill of Material
- g) Procedure for erection and General Safety procedures to followed during erection/installation.
- h) Procedure for initial checking after erection.
- i) Procedure for testing and acceptance norms.
- j) Procedure / Check list for pre-commissioning activities.
- k) Procedure / Check list for commissioning of the system.
- l) Safety precautions to be followed in electrical supply distribution during erection.

### 3.5.4.2 Operation and Maintenance Manuals

- a) The manual shall be a two rim PVC bound stiff sided binder able to withstand constant usage or where a thicker type is required it shall have locking steel pins, the size of the manual shall not be larger than international size A3. The cover shall be printed with the Project Name, Services covered and Volume / Book number Each section of the manual shall be divided by a stiff divider of the same size as the holder.

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The dividers shall clearly state the section number and title. All written instructions within the manual not provided by the manufacturers shall be typewritten with a margin on the left hand side.

b) The arrangement and contents of O & M manuals shall be as follows :

1) Chapter 1 - Plant Description : To contain the following sections specific to the equipment/system supplied

- (a) Description of operating principle of equipment / system with schematic drawing / layouts.
- (b) Functional description of associated accessories / controls. Control interlock protection write up.
- (c) Integrated operation of the equipment along-with the intended system. (This is to be given by the supplier of the Main equipment by taking into account the operating instruction given by the associated suppliers).
- (d) Exploded view of the main equipment, associated accessories and auxiliaries with description. Schematic drawing of the equipment along-with its accessories and auxiliaries.
- (e) Design data against which the plant performance will be compared.
- (f) Master list of equipment, Technical specification of the equipment/ system and approved data sheets.
- (g) Identification system adopted for the various components, (it will be of a simple process linked tagging system).
- (h) Master list of drawings (as built drawing - Drawings to be enclosed in a separate volume).

2) Chapter 2 - Plant Operation: To contain the following sections specific to the equipment supplied

- (a) Protection logics provided for the equipment along-with brief philosophy behind the logic, Drawings etc.
- (b) Limiting values of all protection settings.
- (c) Various settings of annunciation/interlocks provided.

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- (d) Start-up and shut down procedure for equipment along-with the associated systems in step mode.
  - (e) Do's and Don'ts related to operation of the equipment.
  - (f) Safety precautions to be taken during normal operation. Emergency instruction on total power failure condition/lubrication failure/any other conditions.
  - (g) Parameters to be monitored with normal value and limiting values.
  - (h) Equipment isolating procedures.
  - (i) Trouble shooting with causes and remedial measures.
  - (j) Routine testing procedure to ascertain healthiness of the safety devices along-with schedule of testing.
  - (k) Routine Operational Checks, Recommended Logs and Records
  - (l) Change over schedule if more than one auxiliary for the same purpose is given.
  - (m) Preservation procedure on long shut down.
  - (n) System/plant commissioning procedure.
- 3) Chapter 3 - Plant Maintenance: To contain the following sections specific to the equipment supplied
- (a) Exploded view of each of the equipment. Drawings along-with bill of materials including name, code no. & population.
  - (b) Exploded view of the spare parts and critical components with dimensional drawings (In case of Electronic cards, the circuit diagram to be given) and spare parts catalogue for each equipment.
  - (c) List of Special T/ P required for Overhauling /Trouble shooting including special testing equipment required for calibration etc.
  - (d) Stepwise dismantling and assembly procedure clearly specifying the tools to be used, checks to be made, records to be maintained etc. Clearance to be maintained etc.

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- (e) Preventive Maintenance schedules linked with running hours/calendar period along-with checks to be carried out.
- (f) Overhauling schedules linked with running hours/calendar period along-with checks to be done.
- (g) Long term maintenance schedules
- (h) Consumables list along-with the estimated quantity required during normal running and during maintenance like Preventive Maintenance and Overhauling.
- (i) List of lubricants with their Indian equivalent, Lubrication Schedule including charts showing lubrication checking, testing and replacement procedure to be carried daily, weekly, monthly & at longer intervals to ensure trouble free operation and quantity required for complete replacement.
- (j) Tolerance for fitment of various components.
- (k) Details of sub vendors with their part no. in case of bought out items.
- (l) List of spare parts with their Part No, total population, life expediency & their interchangeability with already supplied spares to NTPC.
- (m) List of mandatory and recommended spare list along with manufacturing drawings, material specification & quality plan for fast moving consumable spares.
- (n) Lead time required for ordering of spares from the equipment supplier, instructions for storage and preservation of spares.
- (o) General information on the equipment such as modification carried out in the equipment from its inception, equipment population in the country / foreign country and list of utilities where similar equipments have been supplied.

After finalization and approval of the Employer, the O & M Manuals shall be submitted as indicated in table below. The Contract shall not be considered to be completed for purposes of taking over until the final Instructions manuals (both erection and O & M manuals have been supplied to the Employer).

If after the commissioning and initial operation of the plant, the instruction manuals (Erection and /or O &M manuals) require modifications/additions/ changes, the same shall be incorporated and the updated final instruction manuals shall be submitted by the Contractor to the Employer for records and number of copies shall be as mentioned in table below:

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| S.No. | Description of Drgs/Docs       | No. of Prints | No. of CD ROMs/DVDs/Portable Hard Disk |
|-------|--------------------------------|---------------|----------------------------------------|
| 1     | Erection Manual                | 4 Sets        | 2                                      |
| 2     | Operation & Maintenance manual | 1 Set         | 1                                      |
|       | i) First Submission            |               |                                        |
|       | ii) Final Submission           | 4 Sets        | 2                                      |

### 3.5.5 Final Submission of drawings and documents:

The Bidder shall furnish the following after approval of all drawings /documents and test reports:

- a. List of drawings bearing the Employer's and Contractor's drawing number.
- b. Six (6) bound sets along-with two (2) sets of CD-ROMs/ DVD/Portable hard disk of all final drawings/documents.
- c. Bidder shall also furnish six (6) bound sets of all as-built drawings including the list of all as-built drawings bearing drawing numbers. The Contractor shall also furnish two (2) sets of CD-ROMs/ DVD/Portable hard disk of all as-built drawings as decided by the Employer.
- d. The Bidder shall also furnish four (4) copies and two (2) sets of CD-ROMs/ DVD/Portable hard disk of instruction/ operations & maintenance manuals (after approval) for all the equipments.

### 3.5.6 Test Reports

Two (2) copies of all test reports shall be supplied for approval before shipment of Equipment. The report shall indicate clearly the standard value specified for each test to facilitate checking of the reports. After final approval six (6) bound copies and two (2) sets of CD-ROMs/ DVD/Portable hard disk of all type and routine test reports shall be submitted to Employer.

## 3.6 Material /Workmanship

Where the specification does not contain references to workmanship, equipment, materials and components of the covered equipment, it is essential that the same must be new, of highest grade of the best quality of their kind, conforming to best engineering practice and suitable for the purpose for which they are intended and shall ensure satisfactory performance throughout the service life.

In case where the equipment, materials or components are indicated in the specification as "similar" to any special standard the purchaser shall decide upon the question of similarity.

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When required by the specification or when required by the purchaser the contractor shall submit, for approval, all the information concerning the materials or components to be used in manufacture. Machinery, equipment, materials and components supplied, installed or used without such approval shall run the risk of subsequent rejection, it being understood that the cost as well as the time delay associated with the rejection shall be borne by the Contractor.

The design of the Works shall be such that installation, future expansions, replacements and general maintenance may be undertaken with a minimum of time and expenses. Each component shall be designed to be consistent with its duty and suitable factors of safety subject to mutual agreements. All joints and fastenings shall be devised, constructed and documented so that the component parts shall be accurately positioned and restrained to fulfill their required function. In general, screw threads shall be standard metric threads. The use of other thread forms will only be permitted when prior approval has been obtained from the Purchaser.

Whenever possible, all similar part of the works shall be made to gauge and shall also be made interchangeable with similar parts. All spare parts shall also be interchangeable and shall be made of the same materials and workmanship as the corresponding parts of the equipment supplied under the specification. Where feasible, common component units shall be employed in different pieces of equipment in order to minimize spare parts stocking requirements. All equipment of the same type and rating shall be physically and electrically interchangeable.

The equipment offered in the bid only shall be accepted for supply, with the minimum modifications as agreed/accepted.

### 3.7 Provisions for Exposure to Hot and Humid Climate

Outdoor equipment supplied under the specification shall be suitable for service and storage under tropical conditions of high temperature, high humidity' heavy rainfall and environment favorable to the growth of fungi and mildew. The indoor equipment located in non-air-conditioned areas shall also be of same type.

#### Space Heaters

The heaters shall be suitable for continuous operation at 240 V as supply voltage. On –off switch and fuse shall be provided.

One or more adequately rated thermostatically connected heaters shall be supplied to prevent condensation in any compartment. The heaters shall be installed in the compartment and electrical connections shall be made sufficiently away from below the heaters to minimize deterioration of supply wire insulation. The heaters shall be suitable to maintain the compartment temperature to prevent condensation.

The heaters shall be suitably designed to prevent any contact between the heater wire and the air and shall consist of coiled resistance wire centered in a metal sheath and completely

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encased in a highly compacted powder of magnesium oxide or other material having equal heat conducting and electrical insulation properties or they shall consist of resistance wire wound on a ceramic and completely covered with a ceramic material to prevent any contact between the wire and the air. Alternatively, they shall consist of a resistance wire mounted into a tubular ceramic body built into an envelope of stainless steel or the resistance wire is wound on a tubular ceramic body and embedded in vitreous glaze. The surface temperature of the heaters shall be restricted to a value which will not shorten the life of the heater sheaths or that of insulated wire or other component in the compartments.

Control cubicles installed in air-conditioned area need not be provided with space heaters. These cubicles shall, however, have space heaters in case of storage of cubicles for long duration.

### **Fungi Static Varnish**

Besides the space heaters, special moisture and fungus resistance varnish shall be applied on parts which may be subjected or predisposed to the formation of fungi due to the presence or deposit of nutrient substances. The varnish shall not be applied to any surface of part where the treatment will interfere with the operation or performance of the equipment. Such surfaces or parts shall be protected against the application of the varnish.

### **Ventilation opening**

In order to ensure adequate ventilation, compartments shall have ventilation openings provided with fine wire mesh of brass to prevent the entry of insects and to reduce to a minimum the entry of dirt and dust. Outdoor compartment openings shall be provided with shutter type blinds.

### **Degree of Protection**

The enclosure of the Control Cabinets, Junction boxes and Marshalling Boxes, panels etc. to be installed shall provide degree of protection as detailed here under:

- a. Installed outdoor: IP- 55
- b. Installed indoor in air conditioned area: IP-32
- c. Installed in covered area: IP-52
- d. Installed indoor in non air-conditioned area where possibility of entry of water is limited: IP-41.
- e. For LT Switchgear (AC & DC distribution Boards) : IP-52

The degree of protection shall be in accordance with IS: 13947 (Part –I) / IEC-947 (Part-I) / IS 12063/IEC 529. Type test report for degree of protection test, on each type of the box shall be submitted for approval.

### **Preservative Shop Coating**

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All exposed metallic surfaces subject to corrosion shall be protected by shop application of suitable coatings. All surfaces which will not be easily accessible after the shop assembly, shall be treated beforehand and protected for the life of the equipment. All surfaces shall be thoroughly cleaned of all mill scales, oxides and other coatings and prepared in the shop. The surfaces that are to be finish-painted after installation or require corrosion protection until installation, shall be shop painted as per the requirements covered in the relevant part of the Technical Specification.

Transformers and other electrical equipments, if included shall be shop finished with one or more coats of primer and two coats of high grade resistance enamel. The finished colors shall be as per manufacturer's standards, to be selected and specified by the Employer at a later date.

Shop primer for all steel surfaces which will be exposed to operating temperature below 95 degrees Celsius shall be selected by the Bidder after obtaining specific approval of the Employer regarding the quality of primer proposed to be applied. Special high temperature primer shall be used on surfaces exposed to temperature higher than 95 degrees Celsius and such primer shall also be subject to the approval of the Employer.

### **3.8 Rating Plates, Name Plates and Labels**

- 3.8.1 Each equipment shall have permanently attached to it in a conspicuous position, a rating plate of non-corrosive material upon which shall be engraved manufacturer's name, equipment, type or serial number together with details of the ratings, service conditions under which the item of plant in question has been designed to operate, and such diagram plates as may be required by the Employer.
- 3.8.2 Such nameplates or labels shall be of white non-hygroscopic material with engraved black lettering or alternately, in the case of indoor circuit breakers, starters, etc. of transparent plastic material with suitably coloured lettering engraved on the back.
- 3.8.3 Each equipment shall be provided with nameplate or label designating the service of the particular equipment. The inscriptions shall be approved by the Employer or as detailed in appropriate section of the technical specifications.
- 3.8.4 The rated current, extended current rating and rated thermal current shall be clearly indicated in the name plate in case of current transformer.
- 3.8.5 Rated voltage, voltage factor and intermediate voltage shall be clearly indicated on the nameplate in case of capacitor voltage transformer.
- 3.8.6 Each switch shall have a clear inscription identifying its function. Switches shall also have a clear inscription of each position indication.

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3.8.7 All such plates, instruction plates, etc. shall be bilingual with Hindi inscription first, followed by English. Alternatively, two separate plates one with Hindi and the other with English inscriptions may be provided.

3.8.8 All segregated phases of conductors or bus ducts, indoor or outdoor, shall be provided with coloured phase plates to clearly identify the phase of the system.

### 3.9 Galvanising

3.9.1 All exposed ferrous parts shall be hot dip galvanised as per IS:2629 & IS:2633, Galvanising shall be uniform, clean, smooth continuous and free from acid spots. Should the galvanising of the sample be found defective, the entire batch of steel shall have to be re-galvanised at bidder's cost.

3.9.2 The amount of zinc deposit over threaded portion of the bolts, nuts and screws shall not be less than 300 gms. per sq. meter of surface area. The amount of zinc deposit on washers shall not be less than 340 gms. per sq. meter of surface area or a minimum of 30 microns. The threads shall have extra deposit of zinc which shall be removed by die cutting after the completion of galvanising. The removal of extra zinc shall be carefully done so that threads shall have the required deposits of zinc on them as specified.

### 3.10 Painting

Unless explicitly stated in relevant chapters of the specification, the painting of all electrical equipment shall be as follows:

Epoxy based with suitable additives. The thickness of finish coat shall be minimum 50 microns (minimum total DFT shall be 100 microns). However in case electrostatic process of painting is offered for any electrical equipment, minimum paint thickness of 50 microns shall be acceptable for finish coat. Paint shade shall be as per technical specification.

### 3.11 Quality Assurance Programme

3.11.1 The Bidder shall adopt suitable quality assurance programme to ensure that the equipment and services under the scope of contract whether manufactured or performed within the Bidder's works or at his subcontractor's premises or at the Employer's site or at any other place of work are in accordance with the specifications. Such programmes shall be outlined by the Contractor and shall be finally accepted by the Employer/authorised representative after discussions before the award of the contract. The QA programme shall be generally in line with ISO-9001/IS- 14001.

A quality assurance programme of the contractor shall generally cover the following:

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- i. His organisation structure for the management and implementation of the proposed quality assurance programme.
- ii. Quality System Manual
- iii. Design Control System
- iv. Documentation Data Control System
- v. Qualification data for Bidder's key Personnel.
- vi. The procedure for purchase of materials, parts, components and selection of sub-contractor's services including vendor analysis, source inspection, incoming raw-material inspection, verification of materials purchased etc.
- vii. System for shop manufacturing and site erection controls including process controls and fabrication and assembly controls.
- viii. Control of non-conforming items and system for corrective actions and resolution of deviations.
- ix. Inspection and test procedure both for manufacture and field activities.
- x. Control of calibration and testing of measuring testing equipments.
- xi. System for Quality Audits.
- xii. System for identification and appraisal of inspection status.
- xiii. System for authorising release of manufactured product to the Employer.
- xiv. System for handling storage and delivery.
- xv. System for maintenance of records, and
- xvi. Furnishing quality plans for manufacturing detailing out the specific quality control procedure adopted for controlling the quality characteristics relevant to each item of equipment/component as per format enclosed as Annexure-I.

### 3.12 General Requirements - Quality Assurance

3.12.1 All materials, components and equipment covered under this specification shall be procured, manufactured, erected, commissioned and tested at all the stages, as per a comprehensive Quality Assurance Programme. An indicative programme of inspection/tests to be carried out by the Bidder for some of the major items is given in the respective technical specification. This is, however, not intended to form a comprehensive programme as it is the Bidder's responsibility to draw up and implement such programme duly approved by the Employer. The detailed Quality Plans for manufacturing and field activities should be drawn up by the Bidder and will be submitted to Employer for approval.

3.12.2 Manufacturing Quality Plan will detail out for all the components and equipment, various tests/inspection, to be carried out as per the requirements of this specification and standards mentioned therein and quality practices and procedures followed by Bidder's/ Sub-contractor's/ sub-supplier's Quality Control Organisation, the relevant reference documents and standards, acceptance norms, inspection documents raised etc., during all stages of materials procurement, manufacture, assembly and final testing/performance testing. The Quality Plan shall be submitted on electronic media e.g. E-mail in addition to hard copy, for review. Once the same is finalised, hard copies

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shall be submitted for approval. After approval the same shall be submitted in compiled form on CD ROM.

- 3.12.3 The Bidder shall also furnish copies of the reference documents/plant standards/acceptance norms/tests and inspection procedure etc., as referred in Quality Plans along with Quality Plans. These Quality Plans and reference documents/standards etc. will be subject to Employer's approval without which manufacturer shall not proceed.
- 3.12.4 These approved documents shall form a part of the contract. In these approved Quality Plans, Employer shall identify customer hold points (CHP), i.e. test/checks which shall be carried out in presence of the Employer's Project Manager or his authorised representative and beyond which the work will not proceed without consent of Employer/Authorised representative in writing. All deviations to this specification, approved quality plans and applicable standards must be documented and referred to Employer along with technical justification for approval and dispositioning.
- 3.12.5 No material shall be despatched from the manufacturer's works before the same is accepted subsequent to pre-despatch final inspection including verification of records of all previous tests/inspections by Employer's Project Manager/Authorised representative and duly authorised for despatch by issuance of Material Dispatch Clearance Certificate (MDCC).
- 3.12.6 All material used for equipment manufacture including casting and forging etc. shall be of tested quality as per relevant codes/standards. Details of results of the tests conducted to determine the mechanical properties, chemical analysis and details of heat treatment procedure recommended and actually followed shall be recorded on certificates and time temperature chart. Tests shall be carried out as per applicable material standards and/or agreed details.
- 3.12.7 All welding and brazing shall be carried out as per procedure drawn and qualified in accordance with requirements of ASME Section IX/BS-4870 or other International equivalent standard acceptable to the Employer.
- 3.12.8 All welding/brazing procedures shall be submitted to the Employer or its authorised representative for approval prior to carrying out the welding/brazing.
- 3.12.9 All brazers, welders and welding operators employed on any part of the contract either in Bidder's/his sub-contractor's works or at site or elsewhere shall be qualified as per ASME Section-IX or BS-4871 or other equivalent International Standards acceptable to the Employer.
- 3.12.10 Test results or qualification tests and specimen testing shall be furnished to the Employer for approval. However, where required by the Employer, tests shall be conducted in presence of Employer/authorised representative.

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- 3.12.11 For all pressure parts and high pressure piping welding, the latest applicable requirements of the IBR (Indian Boiler Regulations) shall also be essentially complied with. Similarly, any other statutory requirements for the equipments/systems shall also be complied with. On all back-gauged welds MPI/LPI shall be carried before seal welding.
- 3.12.12 All the heat treatment results shall be recorded on time temperature charts and verified with recommended regimes.
- 3.12.13 No welding shall be carried out on cast iron components for repair.
- 3.12.14 Unless otherwise proven and specifically agreed with the Employer, welding of dissimilar materials and high alloy materials shall be carried out at shop only.
- 3.12.15 All non-destructive examination shall be performed in accordance with written procedures as per International Standards. The NDT operator shall be qualified as per SNT-TC-IA (of the American Society of non-destructive examination). NDT shall be recorded in a report which includes details of methods and equipment used, result/evaluation, job data and identification of personnel employed and details of correlation of the test report with the job. In general all plates of thickness greater than 40mm & for pressure parts plates of thickness equal to or greater than 25mm shall be ultrasonically tested otherwise as specified in respective equipment specification. All bar stock/Forging of diameter equal to or greater than 40mm shall be ultrasonically tested.
- 3.12.16 The Bidder shall list out all major items/ equipment/ components to be manufactured in house as well as procured from sub-contractors (BOI). All the subcontractor proposed by the Contractor for procurement of major bought out items including castings, forging, semi-finished and finished components/equipment etc., list of which shall be drawn up by the Bidder and finalised with the Employer, shall be subject to Employer's approval. The Bidder's proposal shall include vendor's facilities established at the respective works, the process capability, process stabilization, QC systems followed, experience list, etc. along with his own technical evaluation for identified subcontractors enclosed and shall be submitted to the Employer for approval within the period agreed at the time of pre-awards discussion and identified in "DR" category prior to any procurement. Such vendor approval shall not relieve the Bidder from any obligation, duty or responsibility under the contract.
- 3.12.17 For components/equipment procured by the Bidders for the purpose of the contract, after obtaining the written approval of the Employer, the Bidder's purchase specifications and inquiries shall call for quality plans to be submitted by the suppliers. The quality plans called for from the subcontractor shall set out, during the various stages of manufacture and installation, the quality practices and procedures followed

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by the vendor's quality control organisation, the relevant reference documents/standards used, acceptance level, inspection of documentation raised, etc.

3.12.18 Employer reserves the right to carry out quality audit and quality surveillance of the systems and procedures of the Bidder's or their sub-contractor's quality management and control activities. The Bidder shall provide all necessary assistance to enable the Employer carry out such audit and surveillance.

3.12.19 The Bidder shall carry out an inspection and testing programme during manufacture in his work and that of his sub-contractor's and at site to ensure the mechanical accuracy of components, compliance with drawings, conformance to functional and performance requirements, identity and acceptability of all materials parts and equipment. Bidder shall carry out all tests/inspection required to establish that the items/equipments conform to requirements of the specification and the relevant codes/standards specified in the specification, in addition to carrying out tests as per the approved quality plan.

3.12.20 Quality audit/surveillance/approval of the results of the tests and inspection will not, however, prejudice the right of the Employer to reject the equipment if it does not comply with the specification when erected or does not give complete satisfaction in service and the above shall in no way limit the liabilities and responsibilities of the Bidder in ensuring complete conformance of the materials/equipment supplied to relevant specification, standard, data sheets, drawings, etc.

3.12.21 For all spares and replacement items, the quality requirements as agreed for the main equipment supply shall be applicable.

3.12.22 Repair/rectification procedures to be adopted to make the job acceptable shall be subject to the approval of the Employer/ authorised representative.

3.12.23 Environmental Stress Screening

All solid state electronic system / equipment / sub assembly shall be free from infant mortile components. For establishing the compliance to this requirement, the Bidder / sub – contractor should meet the following.

1. The Bidder / Sub – contractor shall furnish the established procedure being followed for eliminating infant mortile components. The procedure followed by the Contractor / Sub – contractor should be substantiated along with the statistical figures to validate the procedure being followed. The necessary details as required under this clause shall be furnished at the stage of QP finalization.

**Or**

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In case the Bidder / Sub – contractor do not have any established procedure to eliminate infant mortile components then two or 10% whichever is less, most densely populated Panels shall be tested for Elevated Temperature Cycle Test as per the following procedure.

### **Elevated Temperature Test Cycle**

During the elevated temperature test which shall be for 48 hours, the ambient temperature shall be maintained at 50° C. The equipment shall be interconnected with devices and kept under energized conditions so as to repeatedly perform all operations it is expected to perform in actual service with load on various components being equal to those which will be experienced in actual service.

During the elevated temperature test the cubicle doors shall be closed (or shall be in the position same as they are supposed to be in the field) and inside temperature in the zone of highest heat dissipating components / modules shall be monitored. The temperature rise inside the cubicle should not exceed 10° C above the ambient temperature at 50° C.

In case of any failure during the test cycle, the further course of action should be mutually discussed for demonstrating the intent of the above requirement.

### **Burn In Test Cycle**

The test shall be conducted on all the panels fully assembled and wired including the panels having undergone the above mentioned elevated temperature test.

The period of Burn in Test Cycle shall be 120 hrs and process shall be similar to the elevated temperature test as above except that the temperature shall be reduced to the ambient temperature prevalent at that time.

During the above tests, the process I/O and other load on the system shall be simulated by simulated inputs and in the case of control systems, the process which is to be controlled shall also be simulated. Testing of individual components or modules shall not be acceptable.

During the Burn in Test the cubicle doors shall be closed (or shall be in the position same as they are supposed to be in the field) and inside temperature in the zone of highest heat dissipating components / modules shall be monitored. The temperature rise inside the cubicle should not exceed 10° C above the ambient temperature.

The Bidder / Sub-contractor shall carry out routine test on 100% item at Bidder's / sub-contractor's works. The quantum of check / test for routine & acceptance test by employer shall be generally as per criteria / sampling plan defined in referred

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standards. Wherever standards have not been mentioned quantum of check / test for routine / acceptance test shall be as agreed during detailed engineering stage.

### 3.13 Quality Assurance Documents

The Contractor shall be required to submit two hard copies and two sets on CDROM of the following Quality Assurance Documents as identified in respective quality plan with tick (✓) mark.

Each QA Documentation shall have a project specific Cover Sheet bearing name & identification number of equipment and including an index of its contents with page control on each document.

The QA Documentation file shall be progressively completed by the Supplier's sub-supplier to allow regular reviews by all parties during the manufacturing.

The final quality document will be compiled and issued at the final assembly place of equipment before dispatch. However, CD-Rom may be issued not later than three weeks.

#### 3.13.1 Typical contents of Quality Assurance Document are as below: -

- i) Quality Plan,
- ii) Material mill test reports on components as specified by the specification and approved Quality Plans.
- iii) Manufacturer / works test reports/results for testing required as per applicable codes and standard referred in the specification and approved Quality Plans.
- iv) Non-destructive examination results /reports including radiography interpretation reports. Sketches/drawings used for indicating the method of traceability of the radiographs to the location on the equipment.
- v) Heat Treatment Certificate/Record (Time- temperature Chart)
- vi) All the accepted Non-conformance Reports (Major/Minor) / deviation, including complete technical details / repair procedure).
- vii) CHP / Inspection reports duly signed by the Inspector of the Employer and Contractor for the agreed Customer Hold Points.
- viii) Certificate of Conformance (COC) whoever applicable.
- ix) MDCC

#### 3.13.2 Before dispatch/ commissioning of any equipment, the Supplier shall make sure that the corresponding quality document or in the case of protracted phased deliveries, the applicable section of the quality document file is completed. The supplier will then notify the Inspector regarding the readiness of the quality document (or applicable section) for review.

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- i) If the result of the review carried out by the Inspector of the Quality document (or applicable section) is satisfactory, the Inspector shall stamp the quality document (or applicable section) for release.
- ii) If the quality document is unsatisfactory, the Supplier shall endeavour to correct the incompleteness, thus allowing to finalize the quality document (or applicable section) by time compatible with the requirements as per contract documents. When it is done, the quality document (or applicable section) is stamped by the Inspector.
- iii) If a decision is made for dispatch, whereas all outstanding actions cannot be readily cleared for the release of the quality document by that time, the supplier shall immediately, upon shipment of the equipment, send a copy of the quality document Review Status signed by the Supplier Representative to the Inspector and notify of the committed date for the completion of all outstanding actions & submission. The Inspector shall stamp the quality document for applicable section when it is effectively completed. The submission of QA documentation package shall not be later than 3 weeks after the dispatch of equipment.

### 3.14 Transmission of Quality Documents

As a general rule, two hard copies of the quality document and Two CD ROMs shall be issued to the Employer on release of QA Documentation by Inspector. One set of quality document shall be forwarded to Corporate Quality Assurance Department and other set to respective Site.

For the particular case of phased deliveries, the complete quality document to the Employer shall be issued not later than 3 weeks after the date of the last delivery similarly as stated above.

### 3.15 Inspection, Testing & Inspection Certificate

- 3.15.1 The word 'Inspector' shall mean the Project Manager and/or his authorised representative and/or an outside inspection agency acting on behalf of the Employer to inspect and examine the materials and workmanship of the works during its manufacture or erection.
- 3.15.2 The Project Manager or his duly authorised representative and/or an outside inspection agency acting on behalf of the Employer shall have access at all reasonable times to inspect and examine the materials and workmanship of the works during its manufacture or erection and if part of the works is being manufactured or assembled on other premises or works, the Bidder shall obtain for the Project Manager and for his duly authorised representative permission to inspect as if the works were manufactured or assembled on the Bidder's own premises or works.

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- 3.15.3 The Bidder shall give the Project Manager/Inspector fifteen (15) days written notice of any material being ready for testing. Such tests shall be to the Bidder's account except for the expenses of the Inspector's. The Project Manager/Inspector, unless the witnessing of the tests is virtually waived and confirmed in writing, will attend such tests within fifteen (15) days of the date on which the equipment is noticed as being ready for test/inspection failing which the Bidder may proceed with test which shall be deemed to have been made in the inspector's presence and he shall forthwith forward to the inspector duly certified copies of test reports in two (2) copies.
- 3.15.4 The Project Manager or Inspector shall within fifteen (15) days from the date of inspection as defined herein give notice in writing to the Bidder, or any objection to any drawings and all or any equipment and workmanship which is in his opinion not in accordance with the contract. The Bidder shall give due consideration to such objections and shall either make modifications that may be necessary to meet the said objections or shall inform in writing to the Project Manager/Inspector giving reasons therein, that no modifications are necessary to comply with the contract.
- 3.15.5 When the factory tests have been completed at the Bidder's or subcontractor's works, the Project Manager /Inspector shall issue a certificate to this effect fifteen (15) days after completion of tests but if the tests are not witnessed by the Project Manager /Inspectors, the certificate shall be issued within fifteen (15) days of the receipt of the Contractor's test certificate by the Project Manager /Inspector. Project Manager /Inspector to issue such a certificate shall not prevent the Bidder from proceeding with the works. The completion of these tests or the issue of the certificates shall not bind the Employer to accept the equipment should it, on further tests after erection be found not to comply with the contract.
- 3.15.6 In all cases where the contract provides for tests whether at the premises or works of the Bidder or any sub-contractor, the Bidder, except where otherwise specified shall provide free of charge such items as labour, material, electricity, fuel, water, stores, apparatus and instruments as may be reasonably demanded by the Project Manager /Inspector or his authorised representatives to carry out effectively such tests on the equipment in accordance with the Bidder and shall give facilities to the Project Manager/Inspector or to his authorised representative to accomplish testing.
- 3.15.7 The inspection by Project Manager / Inspector and issue of Inspection Certificate thereon shall in no way limit the liabilities and responsibilities of the Contractor in respect of the agreed Quality Assurance Programme forming a part of the contract.
- 3.15.8 To facilitate advance planning of inspection in addition to giving inspection notice, the Bidder shall furnish quarterly inspection programme indicating schedule dates of inspection at Customer Hold Point and final inspection stages. Updated quarterly inspection plans will be made for each three consecutive months and shall be furnished before beginning of each calendar month.

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- 3.15.9 All inspection, measuring and test equipments used by contractor shall be calibrated periodically depending on its use and criticality of the test/measurement to be done. The Bidder shall maintain all the relevant records of periodic calibration and instrument identification, and shall produce the same for inspection by NTPC. Wherever asked specifically, the contractor shall re-calibrate the measuring/test equipments in the presence of Project Manager / Inspector.

### 3.16 Packaging & Transportation

All the equipment shall be suitably protected, coated, covered or boxed and crated to prevent damage or deterioration during transit, handling and storage at Site till the time of erection. While packing all the materials, the limitation from the point of view of the sizes of railway wagons available in India should be taken account of. The Bidder shall be responsible for any loss or damage during transportation, handling and storage due to improper packing. The Bidder shall ascertain the availability of Railway wagon sizes from the Indian Railways or any other agency concerned in India well before effecting despatch of equipment. Before despatch it shall be ensured that complete processing and manufacturing of the components is carried out at shop, only restricted by transport limitation, in order to ensure that site works like grinding, welding, cutting & preassembly to bare minimum. The Employer's Inspector shall have right to insist for completion of works in shops before despatch of materials for transportation.

### 3.17 Clamps & Connectors including Terminal Connectors

- 3.17.1 The material of clamps and connectors shall be Aluminium alloy casting conforming to designation A6 of IS:617 for connecting to equipment terminals and conductors of aluminium. In case the terminals are of copper, the same clamps/connectors shall be used with 2mm thick bimetallic liner.
- 3.17.2 The material of clamps and connectors shall be Galvanised mild steel for connecting to shield wire.
- 3.17.3 Bolts, nuts and plain washers shall be hot dip galvanised mild steel for sizes M12 and above. For sizes below M12, they shall be electro-galvanised mild steel. The spring washers shall be electro-galvanised mild steel.
- 3.17.4 All castings shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be rounded off to meet specified corona and radio interference requirements.
- 3.17.5 They shall have same current rating as that of the connected equipment. All current carrying parts shall be at least 10 mm thick. The connectors shall be manufactured to have minimum contact resistance.

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- 3.17.6 Flexible connectors, braids or laminated strips shall be made up of copper/aluminium.
- 3.17.7 Current rating and size of terminal/conductor for which connector is suitable shall be put on a suitable sticker on each component which should last atleast till erection time.

### **3.18 Spacers**

- 3.18.1 Spacers shall conform to IS: 10162. They shall be of non-magnetic material except nuts and bolts, which shall be of hot dip galvanised mild steel.
- 3.18.2 Spacers shall generally meet the requirements of clamps and connectors as specified above. Its design shall take care of fixing and removing during installation and maintenance.
- 3.18.3 In addition to the type tests as per IS: 10162, clamp slip test should have been conducted. In this test the sample shall be installed on test span of twin/quad bundle string at a tension of 44.2kN (4500 kg). One of the clamps when subjected to a longitudinal pull of 2.5kN (250 kg) parallel to the axis of conductor shall not slip, i.e. permanent displacement between conductor and clamp after test shall not exceed 1.0 mm. This test should have been performed on all other clamps of the sample.

### **3.19 Bushings, Hollow Column Insulators, Support Insulators, and Disc Insulators**

- 3.19.1 Bushings shall be manufactured and tested in accordance with IS: 2099 & IEC: 60137 while hollow column insulators shall be manufactured and tested in accordance with IEC62155/IS 5284. The support insulators shall be manufactured and tested as per IS: 2544/IEC 60168/IEC 60273. The insulators shall also conform to IEC 60815 as applicable having alternate long and short sheds.  
Support insulators/ bushings/ hollow column insulators shall be designed to have ample insulation, mechanical strength and rigidity for the conditions under which they will be used.
- 3.19.2 Porcelain used shall be homogenous, free from laminations, cavities and other flaws or imperfections that might affect the mechanical or dielectric quality and shall be thoroughly vitrified, tough and impervious to moisture.
- 3.19.3 Glazing of the porcelain shall be uniform brown in colour, free from blisters, burns and other similar defects.

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- 3.19.4 The design of the insulator shall be such that stresses due to expansion and contraction in any part of the insulator shall be lead to deterioration. All ferrous parts shall be hot dip galvanised.
- 3.19.5 Post type insulators shall consist of a porcelain part permanently secured in metal base to be mounted on supporting structures. They shall be capable of being mounted upright. They shall be designed to withstand all shocks to which they may be subjected to during operation of the associated equipment.
- 3.19.6 Bushing porcelain shall be robust and capable of withstanding the internal pressures likely to occur in service. The design and location of clamps, the shape and the strength of the porcelain flange securing the bushing to the tank shall be such that there is no risk of fracture. All portions of the assembled porcelain enclosures and supports other than gaskets, which may in any way be exposed to the atmosphere shall be composed of completely non-hygrosopic material such as metal or glazed porcelain.
- 3.19.7 All iron parts shall be hot dip galvanised and all joints shall be air tight. Surface of joints shall be trued, porcelain parts by grinding and metal parts by machining. Insulator/ bushing design shall be such as to ensure a uniform compressive pressure on the joints.
- 3.19.8 In accordance with the requirement stipulated elsewhere, bushing, hollow column insulators and support insulators shall conform to type tests and shall be subjected to routine tests and acceptance test/sample test in accordance with relevant standards.

### **3.20 Control Cabinets, Junction Boxes, Terminal Boxes & Marshalling Boxes For Outdoor Equipment.**

- 3.20.1 All types of control cabinets, junction boxes, marshaling boxes, lighting panels, terminal boxes, operating mechanism boxes, Kiosks etc. shall generally conform to IS:5039, IS:8623 and IEC:60439 as applicable.
- 3.20.2 They shall be of Stainless steel or Aluminium. The thickness of Stainless steel shall be minimum 1 mm. The thickness of aluminium shall be minimum 3 mm and shall provide rigidity. Top of the boxes shall be sloped towards the rear of the box.

#### **3.20.3 Bay Marshalling Box**

Bay Marshaling Box located at a convenient location to receive and distribute cables shall be provided as required. It shall meet all the requirements as specified for cabinets/boxes.

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It shall have three separate distinct compartments for following purposes:

- To receive two incoming 415V, three phase, AC supplies controlled by 100A four pole MCBs with auto changeover provision, and to distribute five (5) three phase ac supplies controlled by 32A four pole MCBs. It shall also be provided with 63A, 3 phase 4 pin industrial grade receptacle with rotary switch.

- To receive three phase incoming from first compartment and to distribute ten (10) single phase ac supplies controlled by 16A two pole MCBs.

- 150 nos. terminal blocks in vertical formation for interlocking facility.

### 3.20.4 Auxiliary Switch

**The auxiliary switch shall conform of following type tests:**

- a) Electrical endurance test - A minimum of 1000 operations for 2A. D.C. with a time constant greater than or equal to 20 milliseconds with a subsequent examination of mV drop/ visual defects/ temperature rise test.
- b) Mechanical endurance test - A minimum of 5000 operations with a subsequent checking of contact pressure test/ visual examination
- c) Heat run test on contacts
- d) IR/HV test, etc.

### 3.21 Cable Glands & Lugs/ Ferrules

3.21.1 Cable shall be terminated using double compression type cable glands. Testing requirements of Cable glands shall conform to BS:6121 and gland shall be of robust construction capable of clamping cable and cable armour (for armoured cables) firmly without injury to insulation. Cable glands shall be made of heavy duty brass machine finished and nickel chrome plated. Thickness of plating shall not be less than 10 micron. All washers and hardware shall also be made of brass with nickel chrome plating Rubber components shall be of neoprene or better synthetic material and of tested quality. Cable glands shall be suitable for the sizes of cable supplied/erected.

3.21.2 Cable lugs/ferrules for power cables shall be tinned copper solderless crimping type suitable for aluminium compacted conductor cables. Cable lugs and ferrules for control cables shall be tinned copper type. The cable lugs for control cables shall be provided with insulating sleeve and shall suit the type of terminals provided on the equipment. Cable lugs and ferrule shall conform to DIN standards.

### 3.22 Conduits, Pipes & Accessories

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- 3.22.1 The bidder shall supply and install all rigid conduits, mild steel pipes, flexible conduits, hume pipes, etc. including all necessary sundry materials, such as tees, elbows, check nuts, bushing reduces, enlargers, wooden plugs, coupling caps, nipples, gland sealing fittings, pull boxes, etc.
- 3.22.2 The size of the conduit/pipe shall be selected to limit the fill to a maximum of 40%. All conduits/pipes shall have their ends closed by caps until cables are pulled. After cables are pulled, the ends of conduits/pipes shall be sealed in an approved manner to prevent damage to threaded portions and entrance of moisture and foreign materials.
- 3.22.3 PVC conduits shall be of high impact, heavy gauge (at least class 2) conduit conforming to BS-4607.
- 3.22.4 The outer surface of the steel conduits shall be coated with hot-dip zinc and chromate conversion coatings. The inner surface shall have silicone epoxy ester coating for easy cable pulling. Mild steel pipes shall be hot-dip galvanized. All rigid conduits/pipes shall be of a reputed make.
- 3.22.5 The hume pipes and accessories shall be of reinforced concrete conforming to class NP2 of IS-458. All tests on hume pipes shall be conducted as per IS-458.
- 3.22.6 Flexible conduits shall be of heat-resistant lead coated steel, water-leak, fire and rust proof.

### 3.23 Motors

The voltage level for motors shall be as follows:

- |                                  |                                       |
|----------------------------------|---------------------------------------|
| a) Upto 0.2 KW                   | : Single phase 240V AC / 3 phase 415V |
| AC                               |                                       |
| b) Above 0.2 KW and upto 200 KW  | : 3 phase, 415V AC                    |
| c) Above 200 KW and upto 1500 KW | : 3 phase, 3.3 kV AC                  |
| d) Above 1500 KW                 | : 11 kV                               |

The bidder may adopt 415V/3.3 KV for the drives rated in the range of 160-210 KW.

The voltage rating of the drives indicated above is for basic guideline.

- 3.23.1 All motors shall conform to IEC-60034-5 / IS Standard and with principal dimensions in accordance with IEC 60072-1 (1991), IEC 60072-2 (1990) and IEC 60072-3 (1994).

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3.23.2 All equipment shall be suitable for rated frequency of 50 Hz with a variation of +3% & -5%, and 10% combined variation of voltage and frequency unless specifically brought out in the specification

3.23.3 Paint shade shall be as per RAL 5012 (Blue) for indoor and outdoor equipment.

3.23.4 Degree of Protection

Degree of protection for various enclosures as per IEC60034-05 shall be as follows:

Indoor motors - IP 54

Outdoor motors - IP 55

Cable box-indoor area - IP 54

Cable box-Outdoor area - IP 55

3.23.5 Type:

### **AC Motors:**

- a) Squirrel cage induction motor suitable for direct-on-line starting.
- b) Continuous duty LT motors upto 200 KW Output rating (at 50 deg.C ambient temperature), shall be Premium Efficiency class-IE3, conforming to IS 12615, or IEC:60034-30.
- c) Crane duty motors shall be squirrel cage Induction motor as per the requirement.
- d) Motor operating through variable frequency drives shall be suitable for inverter duty. Also these motors shall comply the requirements stipulated in IEC: 60034-18-41 and IEC: 60034-18-42 as applicable.

**DC Motors**      Shunt wound

## **3.24      Auxiliary Switch**

The auxiliary switch shall conform of following type tests:

- a) Electrical endurance test - A minimum of 1000 operations for 2A. D.C. with a time constant greater than or equal to 20 milliseconds with a subsequent examination of mV drop/ visual defects/ temperature rise test.
- b) Mechanical endurance test - A minimum of 5000 operations with a subsequent checking of contact pressure test/ visual examination
- c) Heat run test on contacts
- d) IR/HV test, etc.

## **3.25      Lamps & Sockets**

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### 3.25.1 Lamps:

All incandescent lamps shall use a socket base as per IS-1258, except in the case of signal lamps.

### 3.25.2 Sockets

All sockets (convenience outlets) shall be suitable to accept both 5 Amp & 15 Amp pin round Standard Indian plugs. They shall be switched sockets with shutters.

### 3.25.3 Hand Lamp:

A 240 Volts, single Phase, 50 Hz AC plug point shall be provided in the interior of each cubicle with ON-OFF Switch for connection of hand lamps.

## 3.26 Switches & Fuses

Each control panel shall be provided with necessary arrangements for receiving, distributing, isolating and fusing of DC and AC supplies for various control, signaling, lighting and space heater circuits. The incoming and sub-circuits shall be separately provided with switch-fuse units. Selection of the main and sub-circuit fuse ratings shall be such as to ensure selective clearance of sub-circuit faults. Potential circuits for relaying and metering shall be protected by HRC fuses.

All fuses shall be of HRC cartridge type conforming to IS 9228 mounted on plug-in type fuse bases. Miniature circuit breakers with thermal Protection and alarm contacts will also be accepted. All accessible live connection to fuse bases shall be adequately shrouded. Fuses shall have operation indicators for indicating blown fuse condition. Fuse carrier base shall have imprints of the fuse rating and voltage.

All control switches shall be of rotary type. Toggle/piano switches shall not be accepted.

## 3.27 Type, Routine & Acceptance Tests

### 3.27.1 Type Test Requirements for Equipment other than Circuit Breaker

a) All equipment to be supplied shall be of type tested design. During detail engineering, the contractor shall submit for Owner's approval the reports of all the type tests as listed in this specification. The validity period (counted from 02 May 2024) of reports shall be as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical Equipment in power Transmission- May2020 (with latest amendments). These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a Client.

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b) However if contractor is not able to submit report of the type test(s) conducted as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical equipment in power Transmission-May2020 (with latest amendments) from 02 May 2024, or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor shall conduct all such tests under this contract at no additional cost to the owner either at third party lab or in presence of client/ owners representative and submit the reports for approval.

c) All acceptance and routine tests as per the specification and relevant standards shall be carried out. Charges for these shall be deemed to be included in the equipment price.

### 3.27.2 Type Test Requirements for Circuit Breaker

a) The Contractor shall carry out the type tests as listed in this specification on the equipment to be supplied under this contract. The Bidder shall indicate the charges for each of these type tests separately in the relevant schedule of BPS and the same shall be considered for the evaluation of bids. The type test charges shall be paid as per the charges quoted for each of these type tests separately in the relevant schedule of BPS (Bid Proposal Sheet) & no qty variation is allowed. only for the test(s) conducted successfully under the contract and upon certification by the Employer's engineer.

b) The type tests shall be carried out in the presence of the Employer's representative, for which minimum 60 days' notice shall be given by the Contractor. The Contractor shall obtain the Employer's approval for the type test procedure before conducting the type test. The type test procedure shall clearly specify the test set up, instrument to be used, procedure, acceptance norms, recording of various parameters, interval of recording, precautions to be taken etc. for the type test(s) to be carried out.

c) In case the Contractor has conducted such specified type test(s) according to the relevant standard and / or specification as per CEA Guidelines for the validity period (counted from 02 May 2024) of Type test(s) conducted on Major Electrical equipment in power Transmission-May2020 & with latest amendments, submit the type test reports to the Employer for waiver of conductance of such type test(s). These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a Client. The Employer reserves the right to waive conducting of any or all the specified type tests(s) under this contract. In case the type tests are waived, the type test charges shall not be payable to the Contractor.

However, if any type test report is found not meeting the specification requirements, bidder shall conduct all such type tests successfully according to relevant standards without any cost and delivery implication to BHEL.

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### 3.27.2.1 Type tests to be conducted on AIS Circuit Breaker

- a) Di-electric tests
- b) Radio interference voltage test
- c) Temperature rise test
- d) short time withstand current, peak withstand test, short circuit test duties, short line fault test
- e) Mechanical endurance
- f) Out of phase making & breaking test
- g) Line charging current breaking test
- h) Corona test for 765kV only
- i) IP: 55 test on each type of box
- j) Seismic withstand test with structure for 765kV only
- k) Test for reactor switching duty for 765kV CB only (applicable for Bus reactor CB)

### 3.28 Corona, RIV Tests & Seismic Withstand Test:

- a) The corona and RIV tests shall conform to the requirements as per Annexure A.
- b) The seismic withstand test shall conform to requirements as per Annexure B.

### 3.29 Enclosures:

1. ANNEXURE- A - CORONA AND RADIO INTERFERENCE VOLTAGE (RIV) TEST
2. ANNEXURE- B - SEISMIC WITHSTAND TEST
3. ANNEXURE- I – MQP (NTPC format)
4. ANNEXURE- II – QUALITY ASSURANCE FOR SWITCHYARD

ANNEXURE – A

## **CORONA AND RADIO INTERFERENCE VOLTAGE (RIV) TEST (For 765 & 400kv AIS Only)**

### **1.0 General:**

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Unless otherwise stipulated, all equipment together with its associated connectors where applicable shall be tested for external corona both by observing the voltage level for the extinction of visible corona under falling power frequency voltage and measurement of radio interference voltage (RIV).

### 2.0 Test Methods for RIV:

2.1 RIV tests shall be made according to measuring circuit as per International Special-committee on Radio Interference (CISPR) Publication 16 -1 (1993) Part – I. The measuring circuit shall preferably be tuned to frequency with 10 % of 0.5 MHz but other frequencies in the range of 0.5 MHz to 2 MHz may be used, the measuring frequency being recorded. The result shall be in microvolts. Alternatively, RIV tests shall be in accordance with NEMA standard Publication No. 107 – 1964 except otherwise noted herein. In measurement of RIV only standard fittings of identical type supplied with the equipment and a simulation of the connections as used in the actual installation will be permitted in the vicinity within 3.5 meters of terminals.

2.2 Ambient noise shall be measured before and after each series of tests to ensure that there is no variation in ambient noise level. If variation is present, the lowest ambient noise level will form basis for the measurements. RIV levels shall be measured at increasing and decreasing voltages of 85%, 100%, 115% and 130% for the specified RIV test voltage for all equipment unless otherwise specified. The specified RIV test voltage for 765 & 400KV is listed in the detailed specification together with maximum permissible RIV level in microvolts.

### 3.0 Test Methods for visible Corona:

The purpose of this test is to determine the corona extinction voltage of the apparatus, connectors etc. The test shall be carried out in the same manner as RIV test described above with the exception that RIV measurements are not required during test and a search technique shall be used near the onset and extinction voltage, when the test voltage is raised and lowered to determine their precise values. The test voltage shall be raised to 130 % of RIV test voltage and maintained there for five minutes. In case corona inception does not take place at 130 %, the voltage level shall be raised till inception of corona or rated voltage whichever is lower. The voltage will then be decreased slowly until all visible corona disappears. The test procedure shall be repeated at least 4 times with corona inception and extinction voltage recorded each time. The corona extinction voltage for purposes of determining compliance with the specification shall be the lowest of the four values at which the visible corona (negative or positive polarity) disappears.

ANNEXURE – B

### SEISMIC WITHSTAND TEST (for 765 & 400 kV AIS only)

a) The seismic withstand test on the complete equipment (except BPI) shall be carried out along with supporting structure.

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b) The seismic level specified shall be applied at the base of the structure. The accelerometers shall be provided at the terminal pad of the equipment and at any other point as agreed by the owner. The seismic test shall be carried out in all possible combinations of the equipment. The seismic test procedure shall be furnished for approval of the purchaser.

ANNEXURE-I

|                 |                                    |                            |                   |                |
|-----------------|------------------------------------|----------------------------|-------------------|----------------|
| MFGR.'s<br>LOGO | MANUFACTURER'S NAME AND<br>ADDRESS | MANUFACTURING QUALITY PLAN |                   | PROJECT :      |
|                 |                                    | ITEM :                     | QP NO.:           | PACKAGE :      |
|                 |                                    | SUB-SYSTEM:                | REV.NO.:          | CONTRACT NO. : |
|                 |                                    |                            | DATE:             | MAIN-SUPPLIER: |
|                 |                                    |                            | PAGE: .... OF.... |                |

| SL.<br>NO                     | COMPONENT &<br>OPERATIONS | CHARACTERISTICS | CLASS                                                                                                                                                                                                                                                                                             | TYPE OF<br>CHECK | QUANTUM<br>OF CHECK |       | REFERENCE<br>DOCUMENT | ACCEPTANCE<br>NORMS                                                                                             | FORMAT OF<br>RECORD |    | AGENCY            |   |                  | REMARKS |
|-------------------------------|---------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-------|-----------------------|-----------------------------------------------------------------------------------------------------------------|---------------------|----|-------------------|---|------------------|---------|
|                               |                           |                 |                                                                                                                                                                                                                                                                                                   |                  | M                   | C / N |                       |                                                                                                                 |                     |    | M                 | C | N                |         |
| 1.                            | 2.                        | 3.              | 4.                                                                                                                                                                                                                                                                                                | 5.               | 6.                  |       | 7.                    | 8.                                                                                                              | 9.                  | D* | ** 10.            |   |                  | 11.     |
|                               |                           |                 |                                                                                                                                                                                                                                                                                                   |                  |                     |       |                       |                                                                                                                 |                     |    |                   |   |                  |         |
|                               |                           |                 | <b>LEGEND:</b> * RECORDS, IDENTIFIED WITH “TICK” ( ✓ ) SHALL BE ESSENTIALLY INCLUDED BY SUPPLIER IN QA DOCUMENTATION.<br>** M: MANUFACTURER/SUB-SUPPLIER C: MAIN SUPPLIER, N: NTPC<br>P: PERFORM W: WITNESS AND V: VERIFICATION. AS APPROPRIATE,<br>CHP: NTPC SHALL IDENTIFY IN COLUM “N” AS ‘ W” |                  |                     |       |                       | <br><br>FOR<br>NTPC<br>USE | DOC. NO.:           |    | REV..... CAT..... |   |                  |         |
| MANUFACTURER/<br>SUB-SUPPLIER |                           | MAIN-SUPPLIER   |                                                                                                                                                                                                                                                                                                   |                  |                     |       |                       |                                                                                                                 |                     |    |                   |   |                  |         |
| SIGNATURE                     |                           |                 |                                                                                                                                                                                                                                                                                                   |                  |                     |       |                       |                                                                                                                 | REVIEWED BY         |    | APPROVED BY       |   | APPROVAL<br>SEAL |         |

FORMAT NO.: QS-01-QAI-P-09/F1-R1

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## ANNEXURE-II

| ENDORSEMENT SHEET FOR QP<br>REFERENCE / STANDARD / FIELD QUALITY PLAN (RQP / SQP/RFQP/SFQP)                                                                                                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| TO BE FILLED IN BY SUPPLIER AT TIME OF SUBMISSION                                                                                                                                          |                                    |                                                                                                                                                                                                                                                                                                                                                             | To be filled in by NTPC |
| PROJECT NAME                                                                                                                                                                               |                                    | <b>REVIEW &amp; ENDORSEMENT BY NTPC PROJECT</b><br><b>SPECIFIC QP NUMBER ALLOTTED</b><br><b>QP NO.:</b><br><br><b>REV. NO.:      DATE:</b><br><b>** The RQP/SQP/RFQP/SFQP once endorsed for a particular contract shall remain valid even though the original QP may have expired or revised, unless / otherwise mutually agreed with the supplier. ①</b>                                                                                      |                         |
| CONTRACT NO.:                                                                                                                                                                              |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| MAIN SUPPLIER                                                                                                                                                                              |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| MANUFACTURER WORKS & ADDRESS                                                                                                                                                               | M/S                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| ITEM /EQUIPMENT / SYSTEM/<br>SUB-SYSTEM DETAILS i.e. MODEL TYPE /<br>SIZE /RATING etc.                                                                                                     |                                    | <b>REV. NO.:      DATE:</b><br><b>** The RQP/SQP/RFQP/SFQP once endorsed for a particular contract shall remain valid even though the original QP may have expired or revised, unless / otherwise mutually agreed with the supplier. ①</b>                                                                                                                                                                                                     |                         |
| APPROVED QP NO.: RQP/SQP/RFQP/SFQP                                                                                                                                                         | 0000-999-QV - - REV. NO.: DATED**: |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| Confirmation by Main Supplier (TICK WHICHEVER APPLICABLE)                                                                                                                                  |                                    | (TICK APPLICABLE)                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
| I. That the item/ component is identical to that considered for QP approval. OR.                                                                                                           |                                    | The QP is endorsed for this project without any change                                                                                                                                                                                                                                                                                                                                                                                         |                         |
| II. That there are minor changes in the item/ component with respect to that considered for QP approval, however the same do not affect the contents of QP. OR                             |                                    | The QP is endorsed for this project with changes as indicated.                                                                                                                                                                                                                                                                                                                                                                                 |                         |
| III. That there are minor changes in the item/ component with respect to that considered for QP approval, however the same affect the QP slightly, as indicated below / in attached sheet. |                                    | <b><u>DISTRIBUTION OF ENDORSEMENT OF</u></b><br><b>A) RQP/SQP:</b><br><b>1. MAIN SUPPLIER (WITH A COPY OF QP)</b><br><b>2. MANUFACTURER</b><br><b>3. RIO</b><br><b>4. CQA-SPL</b><br><b>5. CQA-O/C</b><br><br><b>B) RFQP/SFQP:</b><br><b>1. MAIN SUPPLIER (with a copy of QP)</b><br><b>2. MANUFACTURER</b><br><b>3. NTPC FQA (with a copy of QP)</b><br><b>4. NTPC Erection (with a copy of QP)</b><br><b>5. CQA-SPL</b><br><b>6. CQA-O/C</b> |                         |
| SIGN.: (Main Supplier)                                                                                                                                                                     | DATE                               | SIGN.: (Manufacturer)                                                                                                                                                                                                                                                                                                                                                                                                                          | DATE:                   |
|                                                                                                                                                                                            |                                    | NTPC (Reviewed /Approved by/ Date & Seal)                                                                                                                                                                                                                                                                                                                                                                                                      |                         |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

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| ANNEXURE-B: | Deviation/ Change Request of Technical Specification |
| ANNEXURE-C: | Technical Checklist                                  |
| ANNEXURE-D: | Format for Guaranteed Technical Particulars          |

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### **ANNEXURE-A: Compliance Certificate of Technical Specification**

The bidder shall confirm compliance to the following by signing and stamping this compliance certificate and furnishing same with the offer.

1. The scope of supply, technical details, construction features, design parameters etc. shall be as per technical specification & there are no exclusion/ deviation with regard to same.
2. There are no deviation(s) with respect to specification other than those furnished in the schedule of deviations.
3. Only those technical submittals which are specifically asked for in Notice Inviting Tender (NIT) to be submitted at tender stage shall be considered as part of offer. Any other submission, even if made, shall not be considered as part of technical offer.
4. Any comments/ clarifications on technical/ inspection requirements furnished as part of bidder's covering letter shall not be considered by BHEL, and bidder's offer shall be construed to be in conformance with the specification.
5. Any changes made by the bidder in the price schedule with respect to the description/ quantities from those given in 'BOQ' of the specification shall not be considered (i.e., technical description & quantities as per the specification shall prevail).

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

### ANNEXURE-B: Deviation/ Change Request of Technical Specification

Bidder shall list out all technical potential deviation/ change request (s) along with clause with respect to technical specifications.

| Sl. No. | Page No. | Clause No. | Deviation | Reason/ Justification(s) |
|---------|----------|------------|-----------|--------------------------|
|---------|----------|------------|-----------|--------------------------|

Any deviation not specifically brought out in this section shall not be admissible for any commercial implication at later stage. Except to the technical deviations listed in this schedule, bidder's offer shall be considered in full compliance to the tender specifications irrespective of any such deviation indicated / taken elsewhere in the submitted offer.

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

### ANNEXURE-C: Technical Checklist

| Sl. No.  | Particulars                                                                                                                                                                                                                                                  | Confirmation by Bidder |         |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|
| <b>1</b> | <b>Technical Qualifying Requirement</b>                                                                                                                                                                                                                      |                        |         |
| 1.1      | The bidder to furnish relevant documents for meeting the qualifying requirement. Performance certificates shall be submitted in English. Translated pages should be attested by the ultimate customer, if attested only by the bidder it shall be notarized. | Confirmed              | Yes/ No |
| 1.2      | The bidder's scope includes supply and services such as Supervision of installation, Testing and commissioning.                                                                                                                                              | Confirmed              | Yes/ No |
| <b>2</b> | <b>Un-priced BOQ</b>                                                                                                                                                                                                                                         |                        |         |
| 2.1      | Confirm that all items have been quoted separately. (If any item has not been quoted, the same shall be specifically brought out with technical reasons thereof. Record the same in schedule of technical deviations.                                        | Confirmed              | Yes/ No |
| <b>3</b> | <b>Technical</b>                                                                                                                                                                                                                                             |                        |         |
| 3.1      | Catalogues, indicative OGA of the offered equipment is attached.                                                                                                                                                                                             | Confirmed              | Yes/ No |
| 3.2      | Minimum Number of auxiliary contacts on each isolator - besides requirement of this specification/ Equipment Schematics. The bidder shall wire up 12 NO +12 NC+ 4 MBB to Terminal Block (reversible) for purchaser 's use.                                   | Confirmed              | Yes/ No |
| 3.3      | Minimum Number of auxiliary contacts on each earthing switch - besides requirement of this specification/ Equipment Schematics. The bidder shall wire up 12 NO +12 NC + 4 MBB to Terminal Block (reversible) for purchaser 's use.                           | Confirmed              | Yes/ No |
| 3.4      | Bidder to provide flexible Copper Braid for Earthing System for Isolator and earth switch MOM Box (2 number per box), size equivalent to 75X12 mm GS flat.                                                                                                   | Confirmed              | Yes/ No |
| <b>4</b> | <b>Technical Deviations</b>                                                                                                                                                                                                                                  |                        |         |
| 4.1      | Confirm that the Complete systems have been offered as per the requirements of Technical Specification and Technical Deviation sheet has been submitted. Deviations mentioned elsewhere in the bid shall not be considered.                                  | Confirmed              | Yes/ No |
| <b>5</b> | <b>Guaranteed Technical Particulars</b>                                                                                                                                                                                                                      |                        |         |
| 5.1      | All equipment being supplied shall conform to Guaranteed Technical Particulars as per technical specification and                                                                                                                                            | Confirmed              | Yes/ No |

## Bharat Heavy Electricals Limited

**Project:** 765/400/132kV AIS Switchyard Extn. at NTPC SIPAT STPP Stage-III (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolator with/ without Earth Switches

**Document No.** TB-435-316-TS-004B Rev 00

|          |                                                                                                                                                                                                                                                                                                                                                                                                            |           |         |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|
|          | applicable IS/ IEC.                                                                                                                                                                                                                                                                                                                                                                                        |           |         |
| <b>6</b> | <b>Type Tests Requirements</b>                                                                                                                                                                                                                                                                                                                                                                             |           |         |
| 6.1      | All equipment being supplied shall conform to type tests as per technical specification and shall be subject to routine tests in accordance with requirements stipulated under respective sections.                                                                                                                                                                                                        | Confirmed | Yes/ No |
| 6.2      | The validity of type test reports shall be as per the latest CEA guidelines (amended time to time) as date of bid opening (i.e. <b>10.07.2024</b> ). In case, where type test certificates are older than period as per latest CEA guidelines (amended time to time), bidder/ manufacturer shall carry out the type tests prior to dispatch of equipment without any commercial implication on BHEL/ NTPC. | Confirmed | Yes/ No |

Date:

Bidder's Stamp & Signature

|                       |                                                                |                                                                                     |
|-----------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>CLAUSE<br/>NO.</b> | <b>Bidder's Name .....</b>                                     |  |
|                       | <b>DB:12B</b>                                                  |                                                                                     |
|                       | <b>EHV DISCONNECTORS/ISOLATORS</b>                             |                                                                                     |
|                       | (Bidder shall furnish the data for each type of DS separately) |                                                                                     |
|                       | 1. General .....                                               |                                                                                     |
|                       | a) Name & country of the Manufacturer .....                    |                                                                                     |
|                       | b) Manufacturer's type designation .....                       |                                                                                     |
|                       | c) Standard Applicable .....                                   |                                                                                     |
|                       | d) Rated Voltage (KV) .....                                    |                                                                                     |
|                       | e) Rated Current (Amps) .....                                  |                                                                                     |
|                       | i) Under normal .....                                          |                                                                                     |
|                       | ii) Under site conditions at 50 deg C ambient .....            |                                                                                     |
|                       | f) Number of poles .....                                       |                                                                                     |
|                       | g) Whether all 3 poles are ganged mechanically .....           |                                                                                     |
|                       | h) Pole to pole spacing .....                                  |                                                                                     |
| 2.                    | Guaranteed Ratings .....                                       |                                                                                     |
|                       | a) Rated short time current of Isolator .....                  |                                                                                     |
|                       | i) for 1 sec. (kA rms) .....                                   |                                                                                     |
|                       | ii) for 3 sec. (kA rms) .....                                  |                                                                                     |
|                       | iii) Dynamic current .....                                     |                                                                                     |
|                       | b) Opening Time of .....                                       |                                                                                     |
|                       | i) Isolator (sec) .....                                        |                                                                                     |
|                       | ii) Earth switch (sec) .....                                   |                                                                                     |
| <b>Section-4</b>      |                                                                |                                                                                     |

| <div>CLAUSE<br/>NO.</div> | <div>Bidder's Name .....</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <div>एनटीपीसी<br/>NTPC</div> |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <div>3.</div>             | <div> <div>c) Closing time of</div> <div> <div>i) Isolator (sec)</div> <div>ii) Earth switch (sec)</div> </div> <div>Dielectric with-stand capacity of Completely assembled Isolator and earth switch</div> <div> <div>a) One minute power frequency withstand test voltage (kV) rms</div> <div> <div>i) against ground (kV peak)</div> <div>- dry (kV rms)</div> <div>- Wet (kV rms)</div> </div> <div>b) 1.2/50 micro second impulse withstand test voltage</div> <div> <div>i) against ground (kV peak)</div> <div>ii) across open contacts (kV peak)</div> </div> <div>c) 250/2500 micro second switching surge withstand test voltage (dry &amp; wet)</div> <div> <div>i) against ground (kV peak)</div> <div>ii) across open contacts (kV peak)</div> </div> <div>d) Corona extinction voltage (kV rms)</div> <div>e) Radio interference level at <math>1.1 \times U_m/\sqrt{3}</math> (in micro volts) for frequency between 0.5 MHz to 2.0 MHz</div> <div>f) Total creepage distance to ground (mm)</div> </div> </div> |                              |

| CLAUSE<br>NO. | Bidder's Name .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 4.            | Operating Mechanism .....<br><br>a) Type and rating (KW) of motor .....<br><br>b) Rated voltage of motor .....                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                     |
| 4.1           | Interlockings .....<br><br>a) Whether mechanical/ constructional interlock between Isolator and Earth switch provided .....<br><br>b) Whether interlock coil is continuously rated .....                                                                                                                                                                                                                                                                                                                                                                        | Yes/No .....                                                                        |
| 5             | Constructional Features<br><br>a) Main contacts .....<br>i) Type of contacts .....<br><br>ii) Contact area (cm <sup>2</sup> ) .....<br><br>iii) Material of contacts .....<br><br>iv) Contact pressure (Kg/cm <sup>2</sup> ) .....<br><br>v) Maximum current density under normal current carrying capacity (Amp/cm <sup>2</sup> ) .....<br><br>vi) Thickness of silver plating .....<br><br>b) Auxiliary contacts on Isolator .....<br><br>i) Total number .....<br>i) NO .....<br>iii) NC .....<br><br>iv) Adjustable .....<br><br>v) Make before break ..... |                                                                                     |

| <b>CLAUSE<br/>NO.</b> | <b>Bidder's Name .....</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                       | <div> <div>vi)</div> <div>Rated voltage (volts)</div> <div>.....</div> </div> <div> <div>vii)</div> <div>Rated continuous current (Amps)</div> <div>.....</div> </div> <div> <div>i)</div> <div>Rated DC breaking current with 20 ms time constant (A)</div> <div>.....</div> </div> <div> <div>c)</div> <div>Auxiliary contacts on earth switch</div> <div>.....</div> </div> <div> <div>i)</div> <div>Total number</div> <div>.....</div> </div> <div> <div>i)</div> <div>NO</div> <div>.....</div> </div> <div> <div>iii)</div> <div>NC</div> <div>.....</div> </div> <div> <div>iv)</div> <div>Adjustable</div> <div>.....</div> </div> <div> <div>v)</div> <div>Rated Voltage (volts)</div> <div>.....</div> </div> <div> <div>vi)</div> <div>Rated continuous current (Amps)</div> <div>.....</div> </div> <div> <div>vii)</div> <div>Rated DC breaking current with 20 ms time constant (A)</div> <div>.....</div> </div> <div> <div>e)</div> <div>Whether details of constructional interlock enclosed</div> <div>Yes/No</div> </div> |                                                                                     |

| <b>CLAUSE<br/>NO.</b> | <b>Bidder's Name .....</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>A.</b>             | <p style="text-align: center;"><b>DB:12G</b></p> <p style="text-align: center;"><b>EHV SWITCHYARD<br/>AUXILIARY ITEMS</b></p> <p><b>CONTROL CABINETS</b></p> <p>(Bidder shall furnish this data for each equipment marshalling box/operating mechanism box separately i.e.Bay Marshalling Box, cabinets of breakers, disconnecting switches, instrument transformers, lighting panels, etc.)</p> <ol style="list-style-type: none"> <li>1. Manufacturer's Name and country of origin .....</li> <li>2. Design ambient air temperature (°C) .....</li> <li>3. Standards Applicable .....</li> <li>4. Thickness of sheet steel (mm) and whether cold rolled or hot rolled .....</li> <li>5. Degree of protection provided .....</li> <li>6. Shede of finish paint as per RAL .....</li> <li>a) Outside .....</li> <li>b) Inside .....</li> </ol> |                                                                                     |
|                       | <p><b>B. TERMINAL CLAMPS AND CONNECTORS, AND SPACERS</b></p> <p>(Bidder shall furnish this data for Terminal clamps and connector separately for each type and equipment)</p> <ol style="list-style-type: none"> <li>1. Manufacturer's Name and address .....</li> <li>2. Applicable Standards .....</li> </ol> <p><b>C. INSULATOR HARDWARE</b></p> <p>(Bidder shall furnish this data separtely for each type and rating)</p> <ol style="list-style-type: none"> <li>1. Manufacturer .....</li> </ol>                                                                                                                                                                                                                                                                                                                                         |                                                                                     |

### TECHNICAL PRE-QUALIFICATION REQUIREMENT

**Name of Project:** 765/132kV AIS Switchyard Extension at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Name of Customer:** NTPC Limited

**Name of Item:** 132kV Isolator/ Disconnectors

#### **TECHNICAL PRE-QUALIFICATION REQUIREMENT**

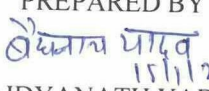
(a) The Bidder should have manufactured and supplied minimum five (05) nos. of three phase Disconnectors/ Isolators suitable for Air Insulated Substation/ Switchyard of 132kV or above class which should have been in successful operation for minimum two (2) years prior to the date of Techno-Commercial bid opening (i.e. 10.07.2024).

#### **SUPPORTING DOCUMENTS TO BE ATTACHED**

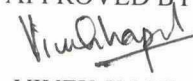
| Sr | Required Criteria    | Supporting Documents to be submitted by bidder along with technical bid                                                                                             |
|----|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Manufacturing        | Approved Drawings/ GTP/ Approved Quality Plan/ Factory Inspection Test Report etc. of offered item.                                                                 |
| 2  | Supply               | Purchase Order (PO)/ Dispatch clearance/ Lorry Receipt (LR)/ Material Receipt certificate at site / installation or commissioning certificate etc. of offered item. |
| 3  | Successful operation | Successful operation means certificate issued by the Employer/Utility certifying the operation without any adverse remark.                                          |

Notes (General points):

1. Consideration of offer shall be subject to customer's approval of bidder's, if applicable.
2. Bidder to submit all supporting documents in English. If documents submitted by bidder are in language other than English, a self- attested English translated document should also be submitted.
3. Notwithstanding anything stated above, BHEL reserves the right to assess the capabilities and capacity of the bidder to perform the contract, should the circumstances warrant such assessment in the overall interest of BHEL.
4. After satisfactory fulfilment of all the above criteria/ requirement, offer shall be considered for further evaluation as per NIT and all the other terms of the tender.
5. Bidder to submit complete documents complying above Pre-qualification Criterion (PQR) along with duly filled 3K document (format attached).

PREPARED BY  
  
 BAIDYANATH YADAV  
 MANAGER-TBEM

REVIEWED BY  
  
 JAI K SONI  
 DGM-TBEM

APPROVED BY  
  
 VIVEK KAPIL  
 AGM-TBEM

**Sub : Sub-Qualifying Requirements for the 132 KV Disconnectors**

**5.14.2.B.III. For 132 KV Disconnectors :** In order to satisfy the requirement as indicated in Technical Specifications, Clause No. 5.14.2.B.III, of Sub-Section-IA, Part-A, Section-VI, We declare that M/s ..... is a manufacturer who have manufactured and supplied atleast five (05) nos. of Three phase Disconnectdor of 132 KV or above Voltage Class suitable for Air Insulated Substation/Switchyard which are in successful operation for minimum two (2) years. The details are given below :

**For 132 kV or above class Disconectors**

| Sl. No. | Item Description                                                                                           | Details |
|---------|------------------------------------------------------------------------------------------------------------|---------|
| 1.      | Name and Address of Manufacturer and Name of contact person with email ID, Telephone & Fax Nos.            |         |
| 2       | Location of manufacturer plant & its address, Telephone & Fax Nos.                                         |         |
| 3.      | Client Name and its Address, including Tel. No. and Fax no.                                                |         |
| 4       | Name of the plant / project & its location for which the manufacturer has supplied the 132KV Disconnectors |         |
| 5.      | Name and designation of the responsible person in Clients organisation.                                    |         |
| 6       | Name & locatin of the substation / switchyard                                                              |         |
| 7.      | i)Contract No. & Date                                                                                      |         |
|         | ii)Brief scope of work                                                                                     |         |
|         | iii) Value of Order                                                                                        |         |
| 08      | Scope of work for the aforesaid contract includes the following                                            |         |
|         | (i) Manufactured                                                                                           | Yes/No  |

|     |                                                                                                                                                                                                                     |            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|     | (ii) Supplied                                                                                                                                                                                                       | Yes/No     |
|     | iii)Type test                                                                                                                                                                                                       | Yes/No     |
| 09  | i)No. of Three phase disconnectors supplied                                                                                                                                                                         |            |
|     | ii)Voltage Level (in KV)                                                                                                                                                                                            |            |
| 10  | Date of Commissioning                                                                                                                                                                                               |            |
| 11  | Date of commencement of successful Operation & No of years of service                                                                                                                                               |            |
| 12  | Whether Three phase disconnectors supplied which are in successful operation for at least two (2) years as on the date of consideration for approval but not later than six months after award date of the package. | Yes / No   |
| 13. | Client(s) certificate(s) enclosed in support of stated experience above at Annexure.....to this Attachment-3K.(Use separate sheet for each experience/Contract)                                                     | YES* / NO* |

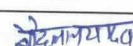
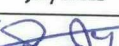
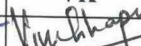
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Date : (Signature).....

Place : (Printed Name).....

(Designation).....

(Common Seal).....

|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                         | <p align="center"><b>BHARAT HEAVY ELECTRICALS LIMITED</b><br/> <b>TRANSMISSION BUSINESS ENGINEERING MANAGEMENT</b><br/> <b>NOIDA</b></p> |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
| <p align="center"><b>Copyright &amp; Confidential</b><br/> The information in this document is the property of BHARAT HEAVY ELECTRICALS LIMITED<br/> This must not be used directly or indirectly in anyway detrimental to the interest of the company.</p>                              | DOCUMENT NO.                                                                                                                             | TB-436-316-TS-004B                                                                       | REV 00  | Prepared                                                                           | Checked                                                                             | Approved                                                                            |
|                                                                                                                                                                                                                                                                                          | TYPE OF DOC.                                                                                                                             | TECHNICAL SPECIFICATION                                                                  | NAME    | BY                                                                                 | JK/DKS                                                                              | VK                                                                                  |
|                                                                                                                                                                                                                                                                                          | <b>Title:</b><br><b>132kV Isolators with/ without Earth Switches</b>                                                                     |                                                                                          | SIGN    |  |  |  |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          | DATE    | 15.01.26                                                                           | 15.01.26                                                                            | 15.01.26                                                                            |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          | GROUP   | TBEM                                                                               |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          | WO No.  | CS-9551-001-2-FC/SC/TC-NOA-7912/7913/7914                                          |                                                                                     |                                                                                     |
| CUSTOMER                                                                                                                                                                                                                                                                                 | NTPC Limited                                                                                                                             |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
| <p align="center"><b>Copyright &amp; Confidential</b><br/> The information in this document is the property of BHARAT HEAVY ELECTRICALS LIMITED<br/> This must not be used directly or indirectly in anyway detrimental to the interest of the company.</p>                              | PROJECT                                                                                                                                  | 765/132kV AIS Switchyard Extension at NTPC DARLIPALI STPP Stage-II (1 x 800 MW)          |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | <b>Contents</b>                                                                                                                          |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Section No.                                                                                                                              | Description                                                                              |         |                                                                                    |                                                                                     | No of Pages                                                                         |
|                                                                                                                                                                                                                                                                                          | Section-1                                                                                                                                | Scope, Technical Requirements and Quantities                                             |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Section-2                                                                                                                                | Equipment Specification under Scope of Supplies                                          |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Section-3                                                                                                                                | Project details and general technical requirements (For all equipment under the Project) |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Section-4                                                                                                                                | Annexures                                                                                |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          | Annexure-A: Compliance Certificate to Technical Specification                            |         |                                                                                    |                                                                                     |                                                                                     |
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|                                                                                                                                                                                                                                                                                          | Annexure-D: Guaranteed Technical Particulars                                                                                             |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
| <p><b>Remarks:</b> Bidder to note that data and details of guaranteed technical particulars shall not be reviewed during technical evaluation/ scrutiny, hence compliance of guaranteed technical particulars in line with technical specification shall be bidder's responsibility.</p> |                                                                                                                                          |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          |         |                                                                                    |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Rev. No.                                                                                                                                 | Date                                                                                     | Altered | Checked                                                                            | Approved                                                                            |                                                                                     |
|                                                                                                                                                                                                                                                                                          | Distribution                                                                                                                             |                                                                                          |         |                                                                                    | To                                                                                  |                                                                                     |
|                                                                                                                                                                                                                                                                                          |                                                                                                                                          |                                                                                          |         |                                                                                    | Copies                                                                              |                                                                                     |

### CHECKLIST FOR TECHNICAL EVALUATION

Along with the technical offer/ bids, the bidder should submit this checklist confirming the inclusion of the enclosures as listed below,

| Sl. No. | Documents to be enclosed                                                                                   | Bidder to confirm<br>(Please tick "Confirmed") |
|---------|------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1.      | Supporting documents for compliance of Technical Qualifying Requirement (as per 3K format, as applicable). | Confirmed                                      |
| 2.      | Unpriced BOQ duly mentioning "Quoted" for all the items, signed and sealed.                                | Confirmed                                      |
| 3.      | Annexure- A duly signed and sealed & Annexure- B duly filled, signed and sealed.                           | Confirmed                                      |

Note: Any bidder not meeting the above requirement shall be liable for non-evaluation.

The above checklist is reviewed and verified for,

NIT Reference No.:

Name of Bidder:

Name of Project: **765/132kV AIS Switchyard Extension at NTPC DARLIPALI STPP Stage-II (1x800 MW)**

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

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**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 132kV Isolators with/ without Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

### SECTION 1: SCOPE, PROJECT SPECIFIC TECHNICAL REQUIREMENTS & BILL OF QUANTITIES

#### 1. Scope

This technical specification covers the requirements of design, manufacture, inspection including third party inspection and testing at manufacturer's work before supply, proper packing and delivery to project of equipment (**132kV Isolators with/ without Earth Switches**) complete with all fittings, accessories, mandatory spares, as applicable of the equipment complete in all respects for efficient & trouble-free working mentioned under this specification.

This section covers the specific technical requirements of equipment. This constitutes minimum technical parameters for the above item as specified by the BHEL/ NTPC. The offered equipment shall also comply with the Section-3 (Project Details and General technical requirements for all equipment under the Project) of this specification.

The specification comprises of following sections:

- Section-1 : Scope, Project Specific Technical Requirements & Bill of Quantities
- Section-2 : Equipment Specification under scope of Supplies/ Service
- Section-3 : Project Details & General Technical Requirements (For all equipment under the Project)
- Section-4 : Annexures
  - Annexure-A: Compliance Certificate to Technical Specification
  - Annexure-B: Deviation/ Change Request to Technical Specification
  - Annexure C: Technical Checklist
  - Annexure-D: Guaranteed Technical Particulars

The following order of priority/ precedence shall be followed. In case of conflict between requirements specified in various documents, the more stringent one shall be followed. BHEL/ NTPC concurrence shall, however, be obtained before taking a final decision in such matters.

#### 1. Statutory Regulations

In particular, the latest version of the following statutory regulations, as applicable, shall be followed for system,

- o Indian Electricity Act
- o CEA regulations
- o The Factory Act
- o Requirements of other statutory bodies as applicable, e.g. CEA etc.

#### 2. Section-1

#### 3. Section-2

#### 4. Section-3

#### 5. Codes & Standards

Bidder shall furnish list of conflicts/ ambiguities/ deviations, if any, along with their technical offer and also furnish the basis that is considered for submitting technical offer. BHEL/ NTPC will resolve listed conflicts prior to award. In case of ambiguity, bidder shall inform BHEL/ NTPC of

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their interpretation. In case bidder fails to convey the same prior to award, BHEL/ NTPC decision on interpretation shall be considered final if need arises during the execution. No additional cost or extra time on account of conflicts/ ambiguities/ deviations shall be admissible.

In general, no deviation from the requirements specified in various clauses of this specification shall be allowed and hence, a certificate to this effect shall have to be furnished along with the offer (Annexure-A), however bidder shall furnish list of conflicts/ ambiguities/ deviations (Annexure-B), if any. Any conflicts/ ambiguities/ deviations mentioned elsewhere in technical offer shall not be reviewed.

The equipment is required for the following project:

|                         |   |                                                                                  |
|-------------------------|---|----------------------------------------------------------------------------------|
| Name of the Customer    | : | <b>NTPC Limited (NTPC Ltd.)</b>                                                  |
| Name of Main Contractor | : | <b>Bharat Heavy Electricals Limited (BHEL)</b>                                   |
| Name of the Project     | : | <b>765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)</b> |

**The terms used in this specification namely, “Employer” refers to “NTPC Ltd.”, “PURCHASHER” refers to BHEL, “Contractor” refers to successful Bidder and “EQUIPMENT” refers to “132kV Isolators with/ without Earth Switches”.**

The scope of supplies shall be as per commercial terms and conditions enclosed separately with the notice inviting tender/ enquiry.

## 2. Codes & Standards

1. The Isolator shall comply with the latest editions and amendments of the following standards as applicable, unless otherwise specified elsewhere in this specification,
 

|                      |                                                                      |
|----------------------|----------------------------------------------------------------------|
| IEC 62271-102        | Alternating current isolators (dis-connectors) and earthing switches |
| IS:9921              | Alternating current isolators (dis-connectors) and earthing switches |
| IS:2544              | Insulators                                                           |
| IS 13947             | Degree of protection provided by enclosure                           |
| IS/IEC60034-5 (2000) | Degree of protection provided by enclosure                           |
| IS:325               | Three phase induction motors                                         |
| IS:4722              | Rotating electrical machines                                         |
| IS:262               | Recommended practice for hot dip galvanizing of iron & steel         |
| IS:4759              | Hot dip galvanization coating on structural steel                    |
| IS:2633              | Method of testing weight, thickness and uniformity                   |
| IS:1573              | Electroplated coating of zinc on iron and steel                      |
| IS:6735              | Spring washers                                                       |
| IS:2016              | Plain washers                                                        |
2. For the purpose of this specification, all technical terms used hereinafter shall have the meaning as per IEC/ ISS specification.
3. The equipment meeting with the requirements of other authoritative standards, which ensure equal or better quality than the standards mentioned above shall also be acceptable. Where the equipment offered by the bidder confirms to other standards, salient points of difference between the standards adopted and the specified standards shall be clearly brought out in the

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offer.

4. In case of imported equipment, standards of the country of origin shall be applicable, if these standards are equivalent or stringent than the applicable Indian standards.
5. The equipment shall also conform to the provisions of Indian Electricity Rules, 1956 and other statutory regulations currently in force in the country.
6. In case Indian standards are not available for any equipment, standards issued by IEC/ BS/ VDE/ IEEE/ NEMA or equivalent agency shall be applicable.

### 3. Specific Technical Requirements

1. Isolator shall perform satisfactorily under various other electrical, electromechanical and meteorological conditions of the site of installation.
2. Equipment shall be able to withstand all external and internal mechanical, thermal and electromechanical forces due to various factors like wind load, temperature variation, ice & snow, (wherever applicable) short circuit etc. for the equipment.
3. The equipment shall also comply to facilitate erection of equipment, all items to be assembled at site shall be “match marked”.

| Sl. No. | Description                                                             | Unit                                                                               | Technical Parameters<br>(132kV system)                                                                                               |
|---------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Rated voltage                                                           | kVrms                                                                              | 132                                                                                                                                  |
| 2       | Rated Maximum Voltage                                                   | kVrms                                                                              | 145                                                                                                                                  |
| 3       | Rated frequency                                                         | Hz                                                                                 | 31.5                                                                                                                                 |
| 4       | No. of poles                                                            | Nos.                                                                               | 3                                                                                                                                    |
| 5       | Design ambient temperature                                              | °C                                                                                 | 50                                                                                                                                   |
| 6       | Type                                                                    |                                                                                    | Outdoor, off-load type<br>Isolator: Horizontal Centre Break (HCB)<br>Earth Switch: Single/ double earth switch/ without Earth Switch |
| 6       | Rated current at 50°C ambient temperature                               | A                                                                                  | 1250                                                                                                                                 |
| 7       | Rated short time withstand current of isolator and earth switch         | kA                                                                                 | 31.5 for 1sec                                                                                                                        |
| 8       | Rated dynamic short time withstand current of isolator and earth switch | kAp                                                                                | 80                                                                                                                                   |
| 9       | Temperature rise over design ambient temperature                        | As per Table-3 of IEC 62271-1/ IEC 60694 for an ambient design temperature of 50°C |                                                                                                                                      |
| 10      | Rated mechanical terminal load                                          | N                                                                                  | As per table III of IEC-62271-102                                                                                                    |
| 11      | Mechanical Endurance Class                                              | Isolator-M2 as per IEC<br>E/S-M0/ M1 as per IEC                                    |                                                                                                                                      |
| 12      | Operating mechanism of isolator/earth switch                            | AC/DC/universal motor                                                              |                                                                                                                                      |
| 13      | Minimum no. of auxiliary contacts on each isolator                      | Isolator: In addition to requirements of specification/                            |                                                                                                                                      |

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|     |                                                                                          |                                                                                                                   |                                                                                                                                                                                                              |
|-----|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                          |                                                                                                                   | Equipment Schematics (Min.8NO+8NC contacts per pole/isolator), 12NO +12NC+4MBB contacts wired to Terminal Blocks (Reversible) for future use.                                                                |
| 14  | Minimum no. of auxiliary contacts on each earthing switch                                |                                                                                                                   | Earth switch: In addition to requirements of specification/ Equipment Schematics (Min.6NO+6NC contacts per pole/isolator), 12 NO+ 12 NC+4 MBB contacts wired to Terminal Blocks (Reversible) for future use. |
| 15  | Max. Operating time                                                                      | secs                                                                                                              | Not more than 12sec                                                                                                                                                                                          |
| 16  | Number of terminals in control cabinet                                                   | All contacts & control circuits are to be wired up to control cabinet plus 24 spare terminals evenly distributed. |                                                                                                                                                                                                              |
| 17  | Rated Insulation levels                                                                  |                                                                                                                   |                                                                                                                                                                                                              |
| a)  | Full wave impulse withstand voltage (1.2/50 microsec.)                                   |                                                                                                                   |                                                                                                                                                                                                              |
| i)  | between line terminals and ground                                                        | kVpeak                                                                                                            | ±650                                                                                                                                                                                                         |
| ii) | between terminals with isolator open                                                     | kVpeak                                                                                                            | ±750 (across isolating distance)                                                                                                                                                                             |
| b)  | Switching impulse withstand voltage (250/2500 micro-second) dry and wet                  |                                                                                                                   |                                                                                                                                                                                                              |
| i)  | between line terminals and ground                                                        | kVpeak                                                                                                            | Not Applicable                                                                                                                                                                                               |
| ii) | between terminals with Isolator open                                                     | kVpeak                                                                                                            | Not Applicable                                                                                                                                                                                               |
| c)  | One-minute power frequency dry withstand voltage                                         |                                                                                                                   |                                                                                                                                                                                                              |
| i)  | between line terminals and ground                                                        | kV rms                                                                                                            | 275                                                                                                                                                                                                          |
| ii) | between terminals with isolator open                                                     | kV rms                                                                                                            | 315 (across isolating distance)                                                                                                                                                                              |
| 18  | Minimum Corona extinction voltage with Isolator in all positions                         | kV rms                                                                                                            | 92                                                                                                                                                                                                           |
| 19  | Max. radio interference voltage for frequency between 0.5 MHz and 2 MHz in all positions | µV                                                                                                                | 1000µV at 92kVrms                                                                                                                                                                                            |
| 20  | Minimum Creepage distance                                                                | mm                                                                                                                | 3625mm (i.e. 25mm/kV)                                                                                                                                                                                        |
| 21  | Seismic acceleration                                                                     |                                                                                                                   | 0.3g/ As per IS 1893                                                                                                                                                                                         |
| 22  | Thermal Rating of Auxiliary Contacts                                                     | A                                                                                                                 | 10A at 220 V DC                                                                                                                                                                                              |
| 23  | Breaking Capacity of auxiliary contacts                                                  |                                                                                                                   | 2A DC with circuit time constant not less than 20 ms                                                                                                                                                         |

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|    |                                                                     |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|---------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | Distance between support structures foundations (within same phase) | m  | As per OEM Design                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 25 | System neutral earthing                                             |    | Effectively Earthed                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 26 | Local remote switch indication of Isolator for SAS/ SCADA           |    | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 27 | Phase to phase spacing (mm) (min)                                   | mm | 2100                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 28 | Rated mechanical Terminal load                                      |    | As per IEC – 62271-102 or value calculated                                                                                                                                                                                                                                                                                                                                                                                          |
| 29 | Number of terminals in Control cabinet                              |    | (1) All contacts and control circuits are to be wired up to control cabinet including potential free auxiliary contacts of Isolator/Earth switch.<br>(2) Additional 24 nos. evenly distributed spare TBs in Master to be provided for interpole cabling of Aux contacts (employer's use).<br>(3) Sufficient TBs shall be provided in control cabinet for looping of AC supply from master to follower of Isolator and earth switch. |
| 30 | DC operated Electrical Interlocking coil                            |    | To be included for Isolator and Earth Switch                                                                                                                                                                                                                                                                                                                                                                                        |
| 31 | TB size & Type                                                      |    | (1) TBs shall be stud type<br>(2) Power Cable- Each TB Should be suitable for terminating two cores of 10 Sq. mm size on each side<br>(3) Control cable- Each TB shall be suitable for terminating two wires of 2.5 Sq. mm size on each side                                                                                                                                                                                        |

**4. Pitch Circle Diameter (PCD) details for 132kV, 6kN Insulator: (For Isolator)-BHEL scope of supply**

|              |         |
|--------------|---------|
| Top PCD      | 127     |
| No. of holes | 4 x Ø16 |
| Bottom PCD   | 254     |
| No. of holes | 8 x Ø18 |

*All dimensions are in mm*

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5. The earthing switches shall be capable of discharging trapped charges of the associated lines. Isolator and earth switches shall be able to bear on the terminals the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.
6. The isolator shall be capable for making / breaking normal currents when no significant change in voltage occurs across the terminals of each pole of the isolator on account of making /breaking operation.
7. Disconnecting Switches with Single Earth switch shall be so constructed that Earth Switch can be installed on either side of the disconnecter switch at site.
8. All the hardware (Nut, bolt & washer etc.) required for the fixing the isolator & its accessories to structure shall be included in the bidder's scope.
9. Any other applicable parameters, not mentioned here, please refer section-2 of this specification.

#### 4. General Technical Requirements

The general technical requirements for Equipment shall be as follows,

1. Bidder shall provide adequate power distribution circuit & terminal block in Isolator/ earth switch marshalling box. Bidder shall provide Power circuit diagram for all isolator/ earth switch during detailed engineering stage. Duplicated wiring on TB shall not be allowed.
2. Flexible Cu braided earthing for Isolator/ earth switch marshalling box (2 number/ box), size equivalent to 75X12 mm GS flat. Bidder to provide minimum cross section of 270Sq mm flexible copper braid on each side of marshaling box. Hole to hole distance for flexible copper braid shall be minimum 200mm. Hardwires for installation of flexible Cu braid for fixing on marshalling box shall be under bidder's scope.
3. Earth switch shall have grounded connection with minimum 2 number's 155sqmm Cu flexible braid.
4. The commencement of erection, testing and commissioning activities of Isolators shall be communicated to manufacturers from time to time as per front readiness/ availability at site.
5. Any special tools and tackle, which are specifically required for the equipment and are proprietary in nature shall be included in offer. List of such special tools and tackle should be clearly listed along with the technical offer. Any special tool which is not listed in the list but required during the erection/ testing/ commissioning of equipment shall also be deemed in bidder's scope.
6. Supervision of erection, testing and commissioning of all supplied Isolator with earth switch shall be carried out under the supervision of the Isolator manufacturer's representative. The commissioning report shall be prepared and signed by the manufacturer's representative. However, BHEL shall extend all the support including manpower, general tools & plants required for erection work.
7. The bidder must fill up all the details required for offered item/s. Instead of indicating "refer drawing, or as per IS/IEC", the exact value/s must be filled in.

#### 5. Bill of Quantities

1. Bill of quantities shall be as per follows, however, any item not appearing herein but required for completeness of the work and mentioned elsewhere in technical specification is deemed to be included in bidder's scope.

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| Sl. No.   | Item Description                                                                                                                                                                                                                                                                                                         | Unit | Quantity |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|
| <b>A.</b> | <b>Supply: Main Items</b>                                                                                                                                                                                                                                                                                                |      |          |
| 1         | Supply: 132kV, 1250, 3-Ø, HCB type isolator (mechanically ganged motor operated) line isolators/ disconnectors with 2 earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors         | Set  | 02       |
| 2         | Supply: 132kV, 1250, 3-Ø, HCB type isolator (mechanically ganged motor operated) line isolators/ disconnectors with 1 earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors         | Set  | 02       |
| 3         | Supply: 132kV, 1250, 3-Ø, HCB type Tandem isolator (mechanically ganged motor operated) line isolators/ disconnectors without earth switches (locally motor operated & mechanically ganged) complete with operating mechanism boxes and accessories etc. excluding support insulators, structure and terminal connectors | Set  | 02       |
| <b>B.</b> | <b>Service</b>                                                                                                                                                                                                                                                                                                           |      |          |
| 1         | Services: 132kv, 1250, 3-Ø, HCB type Isolator-Supervision of Erection, Testing And Commissioning including Alignment Check                                                                                                                                                                                               | Nos. | 06       |

2. Vendor to quote individual spare items against the above BOQ separately, even if the items are covered in any other spare items of this BOQ. Vendor to mention, spares which are not applicable as per the design the of the offered Isolator. However, in latter stage, if found that the spares are applicable for the offered circuit breakers, the same should be provided without any price implication.
3. Flexible Braided Copper (2 Nos. of 300 mm Length, 75sq.mm Cross Section for each MOM Box) along with Lugs for earthing of each Isolator & Earth Switch Shall Be Included in Equipment Prices.
4. Prices for all applicable accessories, fixing hardware of Isolators, Post Insulators, Operating Rod Insulators and MOM box on structure of isolators and earth switches, as applicable shall be included in the equipment (against above BOQ) prices.
5. Erection, testing and commissioning of all 132kV Isolators and earth switches shall be carried out under the supervision of the Isolator manufacturer's representative. However, required unskilled men power/labour, tools (other than special tools and tackles which shall be in

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- 
- bidder's scope) shall be provided by BHEL.
6. Respective dates for the commencement of erection, testing and commissioning activities of Isolators shall be communicated to manufacturers from time to time as per the readiness of respective sites. Bidder to note that erection, testing and commissioning of all of the supplied breakers shall not be done in one go. Multiple visits may be required.
  7. The quantities in BOQ may vary up to  $\pm 10\%$  in line with quantity variation clause. However, individual quantities may be deleted or vary up to any extent.

### 6. Drawings / Documents required for Technical Clearance for Manufacturing

The engineering drawings/ documents, shall be used for providing technical clearance for manufacturing of the equipment, which shall be used for delay analysis, if applicable for respective group.

|   |                                                                       |
|---|-----------------------------------------------------------------------|
| 1 | 132kV Isolator- Outline General Arrangement, BOM & Control Schematics |
| 2 | 132kV Isolator- Guaranteed Technical Particulars/ Datasheet           |
| 3 | 132kV Isolator- Type Test Reports                                     |
| 4 | 132kV Isolator- Quality Assurance Plan                                |

Technical clearance for manufacturing shall be issued after approval of drawings in category-I (approval without any comments)/ category-II (approval with comments) from customer/ BHEL. In case drawing/ document are not duly stamped in category-1/ category-2 by customer, BHEL stamp in Category-1 & 2 shall be treated final to proceed further.

The successful bidder shall have to extend all possible supports like timely submission/ re-submission of drawings, visit to end customer to facilitate documents approval without any commercial implications to BHEL. Acceptance of bidder's documents shall be subject to end customer/ NTPC approval.

### 7. Type Testing

All equipment to be supplied shall be of type tested design. During detail engineering, the contractor/ bidder shall submit for Owner/ NTPC approval the reports of all the type tests as listed in this specification. The validity period of reports shall be as per CEA Guidelines for the validity period of Type test (s) conducted on Major Electrical Equipment in power Transmission-May2020 & with latest amendments from the date of bid opening (i.e. **10.07.2024**). These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test (s) should have been either conducted at an independent laboratory or should have been witnessed by a Client/ NTPC.

However, if contractor/ bidder is not able to submit report of the type test(s) conducted as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical equipment in power Transmission-May'2020 & with latest amendments from the date of bid opening (i.e. **10.07.2024**), or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor/ bidder shall conduct all such tests under this contract at no additional cost to the owner either at third party lab or in presence of client/ owner/ customer representative

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and submit the reports for approval.

All acceptance and routine tests as per the specification and relevant standards shall be carried out. Charges for these shall be deemed to be included in the equipment price.

### 8. Quality Plan

The successful bidder shall submit Quality Assurance Plan with in-process inspection methods, tests, records, etc. for BHEL/ NTPC approval. Customer/ NTPC hold points will also be included in the plan, which shall be mutually agreed by the BHEL/ NTPC. In case bidder has reference, Quality Assurance Plan agreed with BHEL/ NTPC same shall be submitted for specific project to BHEL/ NTPC approval. There shall be no commercial implication to BHEL/ NTPC on account of Quality Plan approval.

Superior quality control system shall be adopted to assure high product quality. Raw materials of the best commercial grade quality and high reliability shall be used in the manufacture of the equipment. All materials shall be procured, manufactured, inspected and tested by vendor/ sub-vendor as per approved quality plan. The supplier shall perform all tests necessary to ensure that the material and workmanship conform to the relevant standards and comply with the requirements of the specification. Charges for all tests for the equipment shall be deemed to be included in bidder's scope.

### 9. Inspection & Testing

1. The equipment shall be subject to inspection by BHEL/ NTPC or authorized representative at bidder/ manufacturers' works. Hence, Bidder shall furnish all necessary information concerning the supply to BHEL/ NTPC.
2. Routine and acceptance tests as listed in relevant standard and section-2, technical specifications shall be complied.
3. Bidder shall also furnish factory acceptance test (FAT) from manufacturers for BHEL/ NTPC approval in line with specific requirements mentioned in section-2, technical specification.

### 10. Packing and Dispatch

1. The equipment shall be properly packed for selected mode of transportation i.e. sea, rail and road in such a manner that it is protected against the climatic conditions and for any damage during transportation, transit and storage. The panels shall be wrapped in polyethylene sheets before being placed in wooden crates/ cases to prevent damage to the finish. Crates/ cases shall have skid bottoms for handling. Special notations such as 'Fragile', 'This side up', 'Weight', 'Owner's particulars\ 'PO nos.' etc., shall be clearly marked on the package together with other details as per purchase order.
2. All equipment shall be suitably protected, coated, covered or boxed and crated to prevent damage or deterioration during transit, handling and outdoor storage (for a minimum period of 1 years) at site till the time of erection. While packing all the materials, the limitations from the point of view of availability of transportation facilities in India should be considered. The Bidder shall be responsible for any loss or damage during transportation, handling and storage.
3. The Bidder shall include and provide for security, protection and packing the equipment so as to avoid loss or damage during transport by any mode.

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4. All packing shall allow for easy removal and checking at site. Wherever necessary, proper arrangement for attaching slings for lifting shall be provided. All packages shall be clearly marked for with signs showing 'UP' and 'DOWN' side of boxes, and handling and unpacking instructions as considered necessary. Special precautions shall be taken to prevent rusting of steel and iron parts during transit and storage. Gas seals or other methods proposed to be adopted for protection against moisture during transit shall be to the satisfaction of the purchaser.
5. Each package delivered under the contract shall be marked by the Bidder at his expense and such marking must be distinct (all previous irrelevant marking being carefully obliterated). Such marking shall show the description and quantity of contents, the name of consignee and address, the gross and net weights of the package, the name of Bidder with a distinctive number of mark sufficient for purpose of identification. All markings shall be carried out with such materials as to ensure quickness of drying, fastness and legibility.
6. Each Package shall contain a note quoting specifically the name of the Bidder, the number and date of contract or order and the name of office placing the contract, nomenclature of the stores and include a schedule of parts for each complete equipment giving the parts number with reference to the General Arrangement/ Assembly drawing and the quantity of each part, drawing number and tag numbers.
7. All equipment/ material shall be suitably packed for transport, carriage at site and outdoor storage during transit. The Bidder shall be responsible for any damage to the equipment during transit. The contents of each package shall bear marking that can be readily identified from the package list and packing shall provide complete protection from moisture, termites and mechanical shocks etc.
8. Any material found short inside the packing cases shall be supplied by the Bidder without any extra cost.
9. Notwithstanding anything stated in this clause the Bidder shall be entirely responsible for any loss, damage or depreciation to the stores.

### 11. Exceptions

Followings are not in bidder's scope of supply (BHEL supplied items)

- a. Equipment support structure
- b. Terminal Connector

### 12. Spares

All spares supplied under this contract shall be strictly interchangeable with the parts for which they are intended for replacement. The spares shall be treated and packed for long term storage in the climatic conditions prevailing at the site. Small items shall be packed in sealed transparent plastic covers with desiccant bags as necessary. Each spare part shall be clearly marked and labeled on the outside of the packing together with the description when more than one spare part is packed in single case. A general description of the contents shall be shown on outside of the case and detailed list enclosed. All cases, containers and other packages must be suitably marked and numbered for the purpose of identification.

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### 13. Sub-Suppliers

Bidder should consider NTPC/NTPC approved make of components and fitments. In case the offered make is not approved by NTPC/NTPC, bidder has to provide alternate make components without any commercial/ time of delivery implication to BHEL.

**Bharat Heavy Electricals Limited**

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)  
**Customer:** NTPC Limited  
**Technical Specification:** 132kV Isolators with/ without Earth Switches  
**Document No.** TB-436-316-TS-004B Rev 00

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SECTION 2:  
EQUIPMENT SPECIFICATION FOR UNDER SCOPE OF SUPPLY/SERVICES .....2

|                                                        |                                                                                                                                 |                                   |                  |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------|
|                                                        | <div>TECHNICAL REQUIREMENTS</div> <div></div> |                                   |                  |
|                                                        |                                                                                                                                 |                                   |                  |
| 3.00.00                                                | DISCONNECTOR : GENERAL                                                                                                          |                                   |                  |
| EPC PACKAGE FOR DARLIPALI STPP,<br>STAGE-II (1X800 MW) | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B<br>BID DOC. NO. CS-9551-001-2                                                   | SUB-SECTION :<br>B-17: SWITCHYARD | Page<br>19 of 98 |

|                                                        | <b>TECHNICAL REQUIREMENTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                               |                                          |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| 3.01.01                                                | The isolators and accessories shall conform in general to relevant IEC 62271-102 (or equivalent Indian Standard) except to the extent explicitly modified in specification. Earth switches shall be provided on isolators as marked on SLD. Isolators shall be horizontal centre break type.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                               |                                          |
| 3.01.02                                                | The isolators and earth switches shall be A. C / D.C. motor operated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                               |                                          |
| 3.02.00                                                | <b>DUTY REQUIREMENTS</b><br><br>Isolators and earth switches shall be capable of withstanding the dynamic and thermal effects of the maximum possible short circuit current of the system in their closed position. They shall be constructed such that they do not open under influence of short circuit current and wind pressure together. The earth switches wherever provided shall be constructional interlocked so that the earth switches can be operated only when the isolator is open and vice-versa.                                                                                                                                                                                                                                           |                                                                               |                                          |
| 3.02.01                                                | The earth switches wherever provided shall be constructional interlocked so that the earth switches can be operated only when the isolator is open and vice-versa. Mechanical Endurance : M2 type of duty as per IEC for 765kV and earthing switches shall be of class M0 / M1 duty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                               |                                          |
| 3.02.02                                                | In addition to the constructional interlock, isolator and earth switches shall have provision to prevent their electrical and manual operation unless the associated and other interlocking conditions are met. All these interlocks shall be of failsafe type. Suitable individual interlocking coil arrangements shall be provided. The interlocking coil shall be suitable for continuous operation from DC supply and within a variation range as stipulated in relevant section. The interlock coil shall be provided with adequate contacts for facilitating permissive logic for 'DC' control scheme of the isolator as well as for AC circuit of the motor to prevent opening or closing of isolators when the interlocking coil is not energised. |                                                                               |                                          |
| 3.02.03                                                | The earthing switches shall be capable of discharging trapped charges of the associated lines. Isolator and earth switches shall be able to bear on the terminals the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                               |                                          |
| 3.02.04                                                | The isolator shall be capable for making / breaking normal currents when no significant change in voltage occurs across the terminals of each pole of the isolator on account of making / breaking operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                               |                                          |
| 3.03.00                                                | <b>CONSTRUCTIONAL FEATURES :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                               |                                          |
| 3.03.01                                                | The isolators shall be provided with high pressure current carrying contacts on the hinge/ jaw ends and all contact surfaces shall be silver plated. The thickness of silverplating should not be less than 25 microns. The contacts shall be accurately machined and self aligned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                               |                                          |
| 3.03.02                                                | The isolator shall be provided with a galvanised steel base provided with holes and designed for mounting on a lattice/pipe support structure. The base shall be rigid and self supporting. The position of movable contact system (main blades) of each of the isolator and earthing switch shall be indicated by a mechanical indicator at the lower end of the vertical rod of shaft for the isolator and earthing switch. The indicator shall be of metal and shall be visible from operating level.                                                                                                                                                                                                                                                   |                                                                               |                                          |
| EPC PACKAGE FOR DARLIPALI STPP,<br>STAGE-II (1X800 MW) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B<br>BID DOC. NO. CS-9551-001-2 | SUB-SECTION :<br><b>B-17: SWITCHYARD</b> |
|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                               | Page<br>20 of 98                         |

|                                                                | <b>TECHNICAL REQUIREMENTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |                                           |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------|
| <b>3.03.03</b>                                                 | All metal parts shall be of non-rusting and non-corroding metal. Current carrying parts shall be from high conductivity electrolytic copper/aluminium. Bolts, screws and pins shall be provided with lock washers. Keys or equivalent locking facilities, if provided on current carrying parts, shall be made of copper silicon alloy or equivalent. The live parts shall be designed to eliminate sharp joints, edges and other corona producing surfaces.                                                                                                                                                                                                                                                                                                                           |                                                                                        |                                           |
| 3.03.04                                                        | The isolators shall be so constructed that the switch blade will not fall to the closed position if the operating shaft gets disconnected. Isolators and earthing switches including their operating parts shall be such that they cannot be dislodged from their open or closed positions by gravity, wind pressure, vibrations shocks or accidental touching of the connecting rods of the operating mechanism. The switch shall be designed such that no lubrication of any part is required except at very infrequent intervals.                                                                                                                                                                                                                                                   |                                                                                        |                                           |
| 3.03.05                                                        | The insulator of the isolator shall conform to the requirements stipulated under clause no: 08.00.00 as specified in the specification and shall have a min. cantilever strength of 800 kg for 765kV and 600kg for 132kV. Pressure due to the contact shall not be transferred to the insulators after the main blades full close. The insulators shall be so arranged that leakage current will pass to earth and not between terminals of the same pole or between phases. The terminal connectors shall conform to requirements stipulated under clause no:08.00.00 as specified else where in the specification.                                                                                                                                                                   |                                                                                        |                                           |
| 3.04.00                                                        | <b>EARTHING SWITCHES:</b><br><br>Where earthing switches are specified these shall include the complete operating mechanism and auxiliary contacts. The earthing switches shall form an integral part of the isolator and shall be mounted on the base frame of the isolator. Earthing switches shall be suitable for local operation only. The earthing switches shall be constructional interlocked with the isolator so that the earthing switches can be operated only when isolator is open and vice versa.                                                                                                                                                                                                                                                                       |                                                                                        |                                           |
| 3.05.00                                                        | <b>OPERATING MECHANISM AND CONTROL</b><br><br>The Contractor shall offer, motor operated switches having padlock arrangement on both 'ON' and 'OFF' positions. Limit switches for control shall be fitted on the isolator/ earth switch shaft, within the cabinet to sense the open and close positions of the isolators and earth switches. It shall not be possible, after final adjustment has been made for any part of the mechanism to be displaced at any point in the travel sufficient enough to allow improper functioning of the isolator when the isolator is opened or closed at any speed. Control cabinet/operating mech. box shall conform to requirements stipulated in clause no: 08.00.00 else where in the specification and IS:5039/IS 8623/IEC 439 as applicable |                                                                                        |                                           |
| <b>3.05.01</b>                                                 | <b>OPERATION</b><br><br>Isolator shall be gang operated for main blades and earth switches. The operation of the three poles shall be well synchronised and interlocked.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                        |                                           |
| 3.05.02                                                        | The design shall be such as to provide maximum reliability under all service conditions. All operating linkages carrying mechanical loads shall be designed for negligible deflection. The length of inter insulator and interpole operating rods shall be capable of adjustments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                        |                                           |
| 3.05.03                                                        | The design of linkages and gears be such so as to allow one man to operate the handle with ease for isolator and earth switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                        |                                           |
| <b>3.06.00</b>                                                 | <b>TESTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                        |                                           |
| <b>EPC PACKAGE FOR DARLIPALI STPP,<br/>STAGE-II (1X800 MW)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>TECHNICAL SPECIFICATION<br/>SECTION – VI, PART-B<br/>BID DOC. NO. CS-9551-001-2</b> | <b>SUB-SECTION :<br/>B-17: SWITCHYARD</b> |
|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        | <b>Page<br/>21 of 98</b>                  |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                             |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--|-------------|-----------|----|------------------|-------------------------------------------------------------------------------------------------------------|----|--------------|---------------|----|----------------------|------------------|----|-----------------|--------|----|--------------------------------|--------------------------------------------------------------|---|-----------------------------------------------------------------|--------------------------------------------|----|--------------------------------------------------|-----------------------|----|---------------------------|---------|----|---------------------------|-------------------|----|--------------------------|-----------------------------------------------------------------------------------------------------|---|------------------|-----------------------------------------------------------|
|         | <div>TECHNICAL REQUIREMENTS</div> <div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                             |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.07.01 | In continuation to the requirements stipulated under Part-I the isolator alongwith operating mechanism shall conform to the type tests and shall be subjected to routine tests and acceptance tests in accordance with IEC 62271-102. Minimum 50 nos. mechanical operations will be carried out on 1 (one) isolator assembled completely with all accessories as acceptance test. During final testing of isolator sequential closing/ opening of earth switch shall also be checked only after isolator is fully open/close. Acceptance test shall be carried out with operating box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                             |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.06.02 | <p>The insulator shall conform to all the type tests as per IEC60168. In addition to all type, routine and acceptance tests, as per IEC-60168, the following additional routine/ acceptance tests shall also be carried out:</p> <p>a)Bending load test in four directions at 50% min. bending load guaranteed in all insulators.<br/>b) Bending load test in four directions at 100% min. bending load guaranteed on sample insulators in a lot. c) Torsional test on sample insulator of a lot.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                             |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.7.00  | <p>General PARAMETERS:</p> <table><tr><td></td><td>Description</td><td>Parameter</td></tr><tr><td>a)</td><td>Type of isolator</td><td>Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV</td></tr><tr><td>b)</td><td>No. of poles</td><td>Three(3poles)</td></tr><tr><td>c)</td><td>Rated operating time</td><td>Not &gt; than 12sec</td></tr><tr><td>d)</td><td>Control voltage</td><td>220VDC</td></tr><tr><td>e)</td><td>Auxiliary contacts on isolator</td><td>Min.8NO &amp; 8NC contacts per pole/isolator for 800kV &amp; 145kV .</td></tr><tr><td>f</td><td>Auxiliary contacts on earth/ safety/grounding/high speed switch</td><td>Min.6NO &amp; 6NC contacts per pole/isolator .</td></tr><tr><td>g)</td><td>Operating mechanism of isolator and earth switch</td><td>AC/DC/universal motor</td></tr><tr><td>h)</td><td>Minimum creepage distance</td><td>25mm/Kv</td></tr><tr><td>i)</td><td>Rated ambient temperature</td><td>50 degree Celsius</td></tr><tr><td>j)</td><td>Support structure height</td><td>Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level</td></tr><tr><td>k</td><td>Temperature rise</td><td>As per Table III of IEC 60694 for an ambient of 50 deg. C</td></tr></table> |                                                                                                             |  | Description | Parameter | a) | Type of isolator | Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV | b) | No. of poles | Three(3poles) | c) | Rated operating time | Not > than 12sec | d) | Control voltage | 220VDC | e) | Auxiliary contacts on isolator | Min.8NO & 8NC contacts per pole/isolator for 800kV & 145kV . | f | Auxiliary contacts on earth/ safety/grounding/high speed switch | Min.6NO & 6NC contacts per pole/isolator . | g) | Operating mechanism of isolator and earth switch | AC/DC/universal motor | h) | Minimum creepage distance | 25mm/Kv | i) | Rated ambient temperature | 50 degree Celsius | j) | Support structure height | Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level | k | Temperature rise | As per Table III of IEC 60694 for an ambient of 50 deg. C |
|         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter                                                                                                   |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| a)      | Type of isolator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Out door type , 50Hz,Vertical Break /Double break/knee type ( for 800kV). Horizontal Center Break for 145kV |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| b)      | No. of poles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Three(3poles)                                                                                               |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| c)      | Rated operating time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Not > than 12sec                                                                                            |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| d)      | Control voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 220VDC                                                                                                      |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| e)      | Auxiliary contacts on isolator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Min.8NO & 8NC contacts per pole/isolator for 800kV & 145kV .                                                |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| f       | Auxiliary contacts on earth/ safety/grounding/high speed switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Min.6NO & 6NC contacts per pole/isolator .                                                                  |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| g)      | Operating mechanism of isolator and earth switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AC/DC/universal motor                                                                                       |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| h)      | Minimum creepage distance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 25mm/Kv                                                                                                     |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| i)      | Rated ambient temperature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 50 degree Celsius                                                                                           |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| j)      | Support structure height                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Adequate so that lowest part of support insulator of equipment is minimum 2550 mm from plinth level         |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| k       | Temperature rise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | As per Table III of IEC 60694 for an ambient of 50 deg. C                                                   |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
| 3.07.01 | <p>765KV Isolator :</p> <table><tr><td></td><td>Description</td><td>Parameter</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                             |  | Description | Parameter |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |
|         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter                                                                                                   |  |             |           |    |                  |                                                                                                             |    |              |               |    |                      |                  |    |                 |        |    |                                |                                                              |   |                                                                 |                                            |    |                                                  |                       |    |                           |         |    |                           |                   |    |                          |                                                                                                     |   |                  |                                                           |

|                                                     |                                                                               |                                   |                  |
|-----------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------|------------------|
| EPC PACKAGE FOR DARLIPALI STPP, STAGE-II (1X800 MW) | TECHNICAL SPECIFICATION<br>SECTION – VI, PART-B<br>BID DOC. NO. CS-9551-001-2 | SUB-SECTION :<br>B-17: SWITCHYARD | Page<br>22 of 98 |
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# TECHNICAL REQUIREMENTS



|    |                                                                |                                                                                                                                                                                                       |
|----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) | Rated continuous current at 50deg.cen ambient temperature.     | Minimum 3150/2000 A at rated ambient temperature as per the Single line diagram                                                                                                                       |
| b) | Rated short time withstand current of isolator and earthswitch | 50 kA rms for One (1) second                                                                                                                                                                          |
| c) | Rated Insulation levels                                        |                                                                                                                                                                                                       |
| d) | Rated one minute power. Frequency withstand voltage            | i) 830 kV rms between live terminals and earth<br>ii) 1150 kV rms across isolating distance                                                                                                           |
| e) | Rated lightning impulse Withstand voltage                      | i) $\pm 2100$ kVp between live terminals and earth.<br>ii) $\pm 2100$ kVp impulse on one terminal and 455 kVp power freq. Voltage of opposite polarity on other terminal (across isolating distance). |
| f) | Rated switching impulse Withstand voltage                      | i) $\pm 1550$ kVp between live terminals and earth.<br>ii) $\pm 1175$ kVp impulse on one terminal and 650 kVp power freq.voltage of opposite polarity on other terminal (across isolating distance).  |
|    |                                                                |                                                                                                                                                                                                       |

## **132 KV Isolator :**

|    | Description                                                    | Parameter                                                                                            |
|----|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| a) | Rated continuous current at 50 deg.cen ambient temperature.    | Minimum 2000/1250 A at rated ambient temperature as per the Single line diagram                      |
| b) | Rated short time withstand current of isolator and earthswitch | 31.5kA rms for One (1) second                                                                        |
| c) | Rated Insulation levels                                        |                                                                                                      |
| d) | Rated one minute power. Frequency withstand voltage            | i) 275kV rms between live terminals and earth<br>ii) 315 kV rms across isolating distance            |
| e) | Rated lightning impulse Withstand voltage                      | i) $\pm 650$ kVp between live terminals and earth.<br>ii) $\pm 750$ kVp (across isolating distance). |
| f) | Rated switching impulse Withstand voltage                      | _____                                                                                                |

EPC PACKAGE FOR DARLIPALI STPP,  
STAGE-II (1X800 MW)

TECHNICAL SPECIFICATION  
SECTION – VI, PART-B  
BID DOC. NO. CS-9551-001-2

SUB-SECTION :  
**B-17: SWITCHYARD**

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**PROJECT: 765/132kV AIS Switchyard Extension at NTPC DARLIPALI STPP Stage-II (1 x 800 MW)****CUSTOMER: NTPC Limited****TECHNICAL SPECIFICATION: TB-436-316-TS-004B Rev 00**

### **SECTION- 3**

#### **PROJECT DETAILS AND GENERAL SPECIFICATIONS**

### **3.0 GENERAL**

This section stipulates the General Technical Requirements under the Contract and will form an integral part of the Technical Specification.

The provisions under this section are intended to supplement general requirements for the materials, equipment and services covered under other sections of tender documents and are not exclusive. However, in case of conflict between the requirements specified in this section and requirements specified under other sections, the requirements specified under respective sections shall prevail.

### **3.1 PROJECT DETAILS**

|                                                     | <b>Particular</b>                                              | <b>Details</b>                                                                            |
|-----------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| a)                                                  | Customer                                                       | NTPC Limited                                                                              |
| b)                                                  | Engineer/Consultant/ Inspector                                 | ---                                                                                       |
| c)                                                  | Project Title                                                  | 765/132kV AIS Switchyard Extension at NTPC DARLIPALI STPP Stage-II (1x800 MW)             |
| d)                                                  | Project Location                                               | Place: Jharsuguda,<br>District: Sundargarh<br>State: Odisha                               |
| e)                                                  | Latitude & Longitude                                           | The latitude of the project: 21°55'00" (N)<br>The longitude of the project: 83°53'35" (E) |
| f)                                                  | Nearest Railway Station                                        | Brajarajnagar at 20Km on SEC Railway.                                                     |
| g)                                                  | Distance of project location from the Railway station          | 20km (approx.)                                                                            |
| h)                                                  | Nearest Major Town                                             | Jharsuguda                                                                                |
| i)                                                  | Distance of the town from the project site                     | 25km                                                                                      |
| j)                                                  | Nearest commercial airport                                     | Veer Surendra Sai Airport Jharsuguda                                                      |
| k)                                                  | Distance of airport from the project site                      | 40km                                                                                      |
| <b><u>SITE CONDITIONS</u></b> (for design purposes) |                                                                |                                                                                           |
| a)                                                  | Design ambient temperature                                     | 50°C                                                                                      |
| b)                                                  | Maximum Relative humidity                                      | 95%                                                                                       |
| c)                                                  | Height above mean sea level                                    | Less than 230 meters                                                                      |
| d)                                                  | Pollution Severity                                             | Heavily polluted                                                                          |
| e)                                                  | Criteria for Wind Resistant design of structures and equipment | Standard Applicable - IS 875 (Part 3)                                                     |

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|           |                                                                  |          |
|-----------|------------------------------------------------------------------|----------|
| <b>f)</b> | Basic Wind speed “Vb” at ten meters above the mean ground level. | 39m/ sec |
| <b>g)</b> | Category of terrain                                              | Cat –2   |
| <b>h)</b> | Risk Coefficient “K1”                                            | 1.06     |

**3.1.1 SYSTEM PARAMETERS:**

| Sl. No. | Parameters                                | 765kV       | 132kV      |
|---------|-------------------------------------------|-------------|------------|
| 1       | Highest system voltage                    | 800kV rms   | 145kV rms  |
|         | Rated / nominal system voltage            | 765kV       | 132kV      |
| 2       | Lightning Impulse voltage                 | ±2100 kVp   | ±650 kVp   |
| 3       | Switching impulse voltage                 | ±1550 kVp   | NA         |
| 4       | Power frequency withstand for 1 min (rms) | 1150kV(rms) | 275kV(rms) |
| 5       | Max. fault level (1 sec.)                 | 50kA        | 31.5kA     |
| 6       | Minimum creepage distance                 | 20000mm     | 3625mm     |

**3.1.2 AUXILIARY POWER:**

| Sl. No. | Nominal Connection Voltage | Variations in Voltage | Frequency    | Phase          | Neutral         |
|---------|----------------------------|-----------------------|--------------|----------------|-----------------|
| 1       | 415V                       | ±10%                  | 50 (+3% -5%) | 3Phase, 4 Wire | Solidly Earthed |
| 2       | 240V                       | ±10%                  | 50 (+3% -5%) | 1 phase        | Solidly Earthed |

Combined variation of voltage and frequency shall be +10%. Design fault level of 415V system shall be restricted to 50kA rms for 1 second.

The operational limits for variation of DC voltage are +10% to -15%.

3.1.3 The various minimum heights of the AIS switchyard shall be as given below from plinth level:

| Voltage | Equipment /1st Level | 2nd Level | 3rd Level |
|---------|----------------------|-----------|-----------|
| 765kV   | 14000mm              | 27000mm   | 40000mm   |
| 132kV   | 4600mm               | 8500mm    | 12500mm   |

The peak of towers for 765kV shall be 8000mm. The Bay width (Beam Span) for 765kV gantry structures should be 54m. All ‘T’ off connection associated with lines/ transformers shall be provided with a bye pass utilizing two PG clamps for each ‘T’ off. As far as possible, the conductor shall pass without cut/ joint unless otherwise necessary for planned shutdown/ maintenance.

3.1.4 The minimum clearances for 765 & 132kV switchyards shall be as given below:

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| Sl. No. | Parameters                                           | 765 kV  | 132 kV |
|---------|------------------------------------------------------|---------|--------|
| 1       | Phase to earth clearance<br>(Conductor to conductor) | 4900mm  | 1300mm |
|         | Phase to earth clearance<br>(rod to structure)       | 6400mm  |        |
| 2       | Phase to earth clearance<br>(Conductor to conductor) | 7600mm  | 1300mm |
|         | Phase to earth clearance<br>(rod to structure)       | 9400mm  |        |
| 3       | Section clearance                                    | 10300mm | 4000mm |

### 3.2 INSTRUCTION TO BIDDERS:

The bidders shall submit the technical requirements, data and information as per the technical data sheets, provided in Section-4.

The bidders shall furnish catalogues, engineering data, technical information, design documents, drawings etc fully in conformity with the technical specification.

The supplier should be approved by Employer. If not, it is the responsibility of the vendor to be assessed and approved Employer, before placement of order by BHEL. Any cost involved in vendor assessment/approval must be borne by the vendor himself.

The Bidder's proposal shall be based upon the use of equipment and material complying fully with the requirements specified herein. It is recognized that the Bidder may have standardized on the use of certain components, materials, processes or procedures different than those specified herein. Alternate proposals offering similar equipment based on the manufacturer's standard practice will also be considered, provided the base offer is in line with technical specifications and such proposals meet the specified design standards and performance requirement and are acceptable to the Purchaser. Sufficient amount of information for justifying such proposals shall be furnished to Purchaser along with the bid to enable the Purchaser to determine the acceptability of these proposals.

Wherever a material or article is specified or defined by the name of a particular brand, Manufacturer or Vendor, the specific name mentioned shall be understood to be indicative of the function and quality desired and not restrictive. Other manufacturer's products may be considered provided sufficient information is furnished to enable the Employer to determine that the products proposed are equivalent to those named.

Equipment furnished shall be complete in every respect with all mountings, fittings, fixtures and standard accessories normally provided with such equipment and/ or needed for erection, completion and safe operation of the equipment as required by applicable codes, though they may not have been specifically detailed in the Technical Specifications unless included in the list of exclusions. Materials and components not specifically stated in the specification but which are necessary for commissioning and satisfactory operation of the switchyard unless specifically excluded shall be deemed to be included in the scope of the specification and

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shall be supplied without any extra cost. All similar standard components/parts of similar standard equipment under supply shall be inter-changeable with one another.

The bidder shall supply type tested (including special tests as per tech. specification) equipment and materials. The test reports shall be furnished by the bidder along with equipment/ material drawings. In the event of any discrepancy in the test reports, (i.e., if any test report is not acceptable due to any design/ manufacturing changes or due to non-compliance with the Technical Specification and/ or applicable standard), the tests shall be carried out without any additional cost implication to the BHEL. BHEL reserves the right to get any or all type/tests conducted/repeated.

### 3.3 CODES AND STANDARDS

In addition to the codes and standards specifically mentioned in the relevant technical specifications for the equipment / plant / system, all equipment parts, systems and works covered under this specification shall comply with all currently applicable statutory regulations and safety codes of the Republic of India as well as of the locality where they will be installed, including the following,

- a) Indian Electricity Act
- b) Indian Electricity Rules
- c) Indian Explosives Act
- d) Indian Factories Act and State Factories Act
- e) Indian Boiler Regulations (IBR)
- f) Regulations of the Central Pollution Control Board, India
- g) Regulations of the Ministry of Environment & Forest (MoEF), Government of India
- h) Pollution Control Regulations of Department of Environment, Government of India
- i) State Pollution Control Board.
- (j.) Rules for Electrical installation by Tariff Advisory Committee (TAC).
- (k.) Building and other construction workers (Regulation of Employment and Conditions of services) Act, 1996
- (l.) Building and other construction workers (Regulation of Employment and Conditions of services) Central Rules, 1998
- (m.) Explosive Rules, 1983
- (n.) Petroleum Act, 1984

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(o.) Petroleum Rules, 1976,

(p.) Gas Cylinder Rules, 1981

(q.) Static and Mobile Pressure Vessels (Unified) Rules, 1981

(r.) Workmen's Compensation Act, 1923

(s.) Workmen's Compensation Rules, 1924

(t.) NTPC Safety Rules for Construction and Erection

(u.) NTPC Safety Policy

(v.) Any other statutory codes / standards / regulations, as may be applicable.

Unless covered otherwise in the specifications, the latest editions (as applicable as on date of bid opening: 03-March-2017), of the codes and standards given below shall also apply:

a) Bureau of Indian standards (BIS)

b) Japanese Industrial Standards (JIS)

c) American National Standards Institute (ANSI)

d) American Society of Testing and Materials (ASTM)

e) American Society of Mechanical Engineers (ASME)

f) American Petroleum Institute (API)

g) Standards of the Hydraulic Institute , U.S.A.

h) International Organization for Standardization (ISO)

i) Tubular Exchanger Manufacturer's Association (TEMA)

j) American Welding Society (AWS)

k) National Electrical Manufacturers Association (NEMA)

l) National Fire Protection Association (NFPA)

m) International Electro-Technical Commission (IEC)

n) Expansion Joint Manufacturers Association (EJMA)

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o) Heat Exchange Institute (HEI)

p) IEEE standard

q) JEC standard

Other International/ National standards such as DIN, VDI, BS, GOST etc. shall also be accepted for only material codes and manufacturing standards, subject to the Employer's approval, for which the Bidder shall furnish, adequate information to justify that these standards are equivalent or superior to the standards mentioned above. In all such cases the Bidder shall furnish specifically the variations and deviations from the standards mentioned elsewhere in the specification together with the complete word to word translation of the standard that is normally not published in English.

As regards highly standardized equipment such as Steam Turbine and Generator, National /International standards such as JIS, DIN, VDI, ISO, SEL, SEW, VDE, IEC & VGB shall also be considered as far as applicable for Design, Manufacturing and Testing of the respective equipment. However, for those of the above equipment not covered by these National / International standards, established and proven standards of manufacturers shall also be considered.

In the event of any conflict between the codes and standards referred to in the above clauses and the requirement of this specification, the requirement of Technical Specification shall govern.

In case of any change in codes, standards & regulations between 03-March-2017 and the date when vendors proceed with fabrication, the Employer shall have the option to incorporate the changed requirements or to retain the original standard. It shall be the responsibility of the Contractor to bring to the notice of the Employer such changes and advise Employer of the resulting effect.

### **3.4 SERVICES TO BE PERFORMED BY THE EQUIPMENT BEING FURNISHED**

The 400kV system is being designed to limit the power frequency over voltage of 1.5 p.u. and the switching surge over voltage to 2.5 p.u. In 400 kV system the initial value of temporary over voltage could be 2.0 p.u. for 1-2 cycles. All the equipment/materials covered in this specification shall perform all its function satisfactorily without undue strain, restrike etc. under such over voltage conditions.

All equipment shall also perform satisfactorily under various other electrical, electromechanical and meteorological conditions of the site of installation. All equipment shall be able to withstand all external and internal mechanical, thermal and electromechanical forces due to various factors like wind load, temperature variation, ice & snow (not applicable for this project), short circuit etc for the equipment.

### **3.5 ENGINEERING DATA**

|                                                                                                 |
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### 3.5.1 Drawings

All drawings submitted by the supplier including those submitted at the time of bid shall be in sufficient detail to indicate the type, size, arrangement, material description, Bill of Materials, weight of each component, break-up for packing and shipment, the external connections, fixing arrangement required. The dimensions required for installation and interconnections with other equipment and materials, clearances and spaces required for installation and interconnections between various portions of equipment and any other information specifically requested in the specifications.

Each drawing submitted by the bidder (including those of sub-vendors) shall bear a title block at the right-hand bottom corner with clear mention of the name of the Employer, the system designation, the specifications title, the specification number, the name of the Project, drawing number and revisions. If standard catalogue pages are submitted, the applicable items shall be indicated therein. All titles, noting, markings and writings on the drawing shall be in English. All the dimensions should be in metric units.

After the approval of the drawings, further work by the bidder shall be in strict accordance with these drawings and no deviation shall be permitted without the written approval of the Purchaser, if so required.

The review of these document/data/drawings by the purchaser will cover only general conformance of the document/data/drawings to the specification and contract, interfaces with the equipment provided under specification, external connections and of the dimensions which might affect plan layout. This review by the purchaser may not indicate a thorough review of the dimensions, quantities and details of the equipment, material, any devices or items indicated or the accuracy of the information submitted. The review and/or approval by the purchaser shall not be considered by the bidder, as limiting any of his responsibilities and liabilities for mistakes and deviations from the requirements, specified under these specifications and documents.

All manufacturing, fabrication and execution of work in connection with the equipment/system prior to the approval of the drawings shall be at the bidder's risk. The bidder is expected not to make any changes in the design of the equipment /system, once they are approved by the Purchaser. However, if some changes are necessitated in the design of the equipment/system at a later date, the bidder may do so, but such changes shall promptly be brought to the notice of the Purchaser indicating the reasons for the change and get the revised drawing approved again in strict conformance to the provisions of the Technical Specification. Approval of bidder's drawing or work by the Purchaser shall not relieve the bidder of any of his responsibilities and liabilities under the Contract.

All engineering data submitted by the contractor after final process including review and approval by the purchaser shall form part of the contract document and the entire work performed under these specifications shall be performed in strict conformity with technical specification, unless otherwise expressly requested by the purchaser in writing.

### 3.5.2 Bidder's Drawing Submission and Approval Procedure

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The following procedure for submission and review/approval of the drawings, data reports, information, etc. shall be followed by the bidder:

- a. All data/information furnished by Vendor in the form of drawings, documents, Catalogues or in any other form for Employer's information/interface and/or review and approval are referred by the general term "drawings".
- b. The 'Master drawings list' indicating titles, Drawing Number, Date of submission and approval etc. shall be furnished by the bidder. This list shall be updated if required at suitable interval during detailed engineering.
- c. All drawings (including those of sub-vendor) shall bear at the right hand bottom corner the 'title plate' with all relevant information duly filled in. The bidder shall furnish this format to his sub-vendor along with his purchase order for sub-vendor's compliance.
- d. Contractor shall submit all the drawings in five (5) copies for review of Employer. Employer shall forward their comments within four (4) weeks of receipt of drawings.
- e. Upon review of each drawings, depending on the correctness and completeness of the drawings, the same will be categorised and approval accorded in one of the following categories:

|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| CATEGORY I   | Approved                                                                                                                  |
| CATEGORY II  | Approved, subject to incorporation of comments/modification as noted. Resubmit revised drawing incorporating the comments |
| CATEGORY III | Not approved. Resubmit revised drawings for Approval after incorporating comments/modifications as noted                  |
| CATEGORY IV  | For information and records                                                                                               |

- f. Bidder shall resubmit the drawings approved under Category II, III within one (1) week of receipt of comments on the drawings, incorporating all comments. Every revision of the drawing shall bear a revision index wherein such revisions shall be highlighted in the form of description or marked up in the drawing identifying the same with relevant revision number enclosed in a triangle (e.g 1.2.3. etc.).
- g. In case Bidder does not agree with any specific comment, he shall furnish the explanation for the same to Employer for consideration. In all such cases Bidder shall necessarily enclose explanations along with the revised drawing (taking care of balance comments) to avoid any delay and/or duplication in review work.
- h. It is the responsibility of the Bidder to get all the drawings approved in the Category I or IV (as the case may be) and complete engineering activities within the agreed schedule. Any delay arising out of submission and modification of drawings shall not alter the contract completion schedule.
- i. Bidder shall not make any changes in the portions of the drawing other than those commented. If changes are required to be made in the portions already approved, the

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Bidder shall resubmit the drawings identifying the changes (along with reasons for changes) for Employer's review and approval. **Drawings resubmitted shall show clearly the portions where the same are revised marking the relevant revision numbers and Employer shall review only such revised portion of documents.**

j. As Built Drawings

After final acceptance of individual equipment / system by the Employer, the Bidder will update all original drawings and documents for the equipment / system to "as built" conditions and submit no. of copies as per clause 3.5.5.

- k. Approval of drawings will not in any way relieve the Bidder of his obligations of furnishing the equipment in accordance with the specification and shall not prevent subsequent rejection if such equipment is later found to be defective.

### 3.5.3 Erection Drawings.

- a. Contractor shall furnish erection drawings for the guidance or commencement of erection or the first shipment, whichever is earlier. These shall generally comprise of fabrication/assembly drawings, various component/part details drawing, assembly, clearance data requirements, etc. The drawings shall contain details of components/equipment with identification number, match marks, bill of materials, assembly procedures etc.
- b. For all major equipment apart from above details, assembly sequence and instructions with check-lists shall be furnished in the form of erection manuals.

### 3.5.4 Instruction Manual

- a. The Contractor shall submit to the Employer preliminary instruction manuals for all the equipments for review. The final instructions manuals incorporating Employer's comments and complete in all respect shall be submitted at least sixty (60) days before the first shipment of the equipment. The instruction manuals shall contain full details and drawings of all the equipments, the transportation, storage, installation, testing, commissioning, operation and maintenance procedures, etc. separately for each component/equipment along with log record format. These instruction manuals shall be submitted in five (5) copies for approval.
- b. If after commissioning and initial operation of the plant, the instruction manuals require any modifications/additions/changes, the same shall be incorporated and the updated final instruction manuals shall be submitted.
- c. The operating and maintenance instructions together with drawings (other than shop drawings) of the equipment, as completed, shall have sufficient details to enable the Employer to maintain, dismantle, reassemble and adjust all parts of the equipment. They shall give a step by step procedure for all operations likely to be carried out during the life of the plant/equipment, including erection, testing, commissioning, operation, maintenance dismantling and repair. Each manual shall also include a complete set of

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approved drawings together with performance/rating curves of the equipment and test certificates, wherever applicable. The contract shall not be considered completed for purpose of taking over until such instructions and drawings have been supplied to the Employer.

- d. A separate section of the manual shall be for each size/type of equipment and shall contain a detailed description of construction and operation, together with all relevant pamphlets.
- e. The manuals shall include the following
  - a) List of spare parts along with their drawing and catalogues and procedure for ordering spares.
  - b) Lubrication Schedule including charts showing lubrication checking, testing and replacement procedure to be carried daily, weekly, monthly & at longer intervals to ensure trouble free operation.
- f. Where applicable, fault location charts shall be included to facilitate finding the cause of mal-operation or break down.
- g. A collection of the manufacturer's standard leaflets will not accepted to be taken as a compliance of this clause. The manual shall be specifically compiled for the concerned project.

The Instruction Manuals shall comprise of the following:

#### 3.5.4.1 Erection Manuals

The erection manuals shall be submitted atleast three (3) months prior to the commencement of erection activities of particular equipment/system. The erection manual should contain the following as a minimum.

- a) Erection strategy.
- b) Sequence of erection.
- c) Erection instructions.
- d) Critical checks and permissible deviation/tolerances.
- e) List of tool, tackles, heavy equipment like cranes, dozers, etc.
- f) Bill of Material
- g) Procedure for erection and General Safety procedures to followed during erection/installation.
- h) Procedure for initial checking after erection.

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- i) Procedure for testing and acceptance norms.
- j) Procedure / Check list for pre-commissioning activities.
- k) Procedure / Check list for commissioning of the system.
- l) Safety precautions to be followed in electrical supply distribution during erection.

#### 3.5.4.2 Operation and Maintenance Manuals

- a) The manual shall be a two rim PVC bound stiff sided binder able to withstand constant usage or where a thicker type is required it shall have locking steel pins, the size of the manual shall not be larger than international size A3. The cover shall be printed with the Project Name, Services covered and Volume / Book number Each section of the manual shall be divided by a stiff divider of the same size as the holder. The dividers shall clearly state the section number and title. All written instructions within the manual not provided by the manufacturers shall be typewritten with a margin on the left-hand side.
- b) The arrangement and contents of O & M manuals shall be as follows,
  - 1) Chapter 1 - Plant Description: To contain the following sections specific to the equipment/system supplied
    - (a) Description of operating principle of equipment / system with schematic drawing / layouts.
    - (b) Functional description of associated accessories / controls. Control interlock protection write up.
    - (c) Integrated operation of the equipment along-with the intended system. (This is to be given by the supplier of the Main equipment by taking into account the operating instruction given by the associated suppliers).
    - (d) Exploded view of the main equipment, associated accessories and auxiliaries with description. Schematic drawing of the equipment along-with its accessories and auxiliaries.
    - (e) Design data against which the plant performance will be compared.
    - (f) Master list of equipment, Technical specification of the equipment/ system and approved data sheets.
    - (g) Identification system adopted for the various components, (it will be of a simple process linked tagging system).

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- (h) Master list of drawings (as built drawing - Drawings to be enclosed in a separate volume).
- 2) Chapter 2 - Plant Operation: To contain the following sections specific to the equipment supplied
- (a) Protection logics provided for the equipment along-with brief philosophy behind the logic, Drawings etc.
  - (b) Limiting values of all protection settings.
  - (c) Various settings of annunciation/interlocks provided.
  - (d) Start-up and shut down procedure for equipment along-with the associated systems in step mode.
  - (e) Do's and Don'ts related to operation of the equipment.
  - (f) Safety precautions to be taken during normal operation. Emergency instruction on total power failure condition/lubrication failure/any other conditions.
  - (g) Parameters to be monitored with normal value and limiting values.
  - (h) Equipment isolating procedures.
  - (i) Trouble shooting with causes and remedial measures.
  - (j) Routine testing procedure to ascertain healthiness of the safety devices along-with schedule of testing.
  - (k) Routine Operational Checks, Recommended Logs and Records
  - (l) Change over schedule if more than one auxiliary for the same purpose is given.
  - (m) Preservation procedure on long shut down.
  - (n) System/plant commissioning procedure.
- 3) Chapter 3 - Plant Maintenance: To contain the following sections specific to the equipment supplied
- (a) Exploded view of each of the equipment. Drawings along-with bill of materials including name, code no. & population.

|                                                                                                 |
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- (b) Exploded view of the spare parts and critical components with dimensional drawings (In case of Electronic cards, the circuit diagram to be given) and spare parts catalogue for each equipment.
- (c) List of Special T/ P required for Overhauling /Trouble shooting including special testing equipment required for calibration etc.
- (d) Stepwise dismantling and assembly procedure clearly specifying the tools to be used, checks to be made, records to be maintained etc. Clearance to be maintained etc.
- (e) Preventive Maintenance schedules linked with running hours/calendar period along-with checks to be carried out.
- (f) Overhauling schedules linked with running hours/calendar period along-with checks to be done.
- (g) Long term maintenance schedules
- (h) Consumables list along-with the estimated quantity required during normal running and during maintenance like Preventive Maintenance and Overhauling.
- (i) List of lubricants with their Indian equivalent, Lubrication Schedule including charts showing lubrication checking, testing and replacement procedure to be carried daily, weekly, monthly & at longer intervals to ensure trouble free operation and quantity required for complete replacement.
- (j) Tolerance for fitment of various components.
- (k) Details of sub vendors with their part no. in case of bought out items.
- (l) List of spare parts with their Part No, total population, life expediency & their interchangeability with already supplied spares to NTPC.
- (m) List of mandatory and recommended spare list along with manufacturing drawings, material specification & quality plan for fast moving consumable spares.
- (n) Lead time required for ordering of spares from the equipment supplier, instructions for storage and preservation of spares.
- (o) General information on the equipment such as modification carried out in the equipment from its inception, equipment population in the country / foreign country and list of utilities where similar equipments have been supplied.

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After finalization and approval of the Employer, the O & M Manuals shall be submitted as indicated in table below. The Contract shall not be considered to be completed for purposes of taking over until the final Instructions manuals (both erection and O & M manuals have been supplied to the Employer.

If after the commissioning and initial operation of the plant, the instruction manuals (Erection and /or O & M manuals) require modifications/additions/ changes, the same shall be incorporated and the updated final instruction manuals shall be submitted by the Contractor to the Employer for records and number of copies shall be as mentioned in table below:

| S.No. | Description of Drgs/Docs                                 | No. of Prints | No. of CD ROMs/<br>DVDs/ Portable Hard<br>Disk |
|-------|----------------------------------------------------------|---------------|------------------------------------------------|
| 1     | Erection Manual                                          | 4 Sets        | 2                                              |
| 2     | Operation & Maintenance<br>manual<br>i) First Submission | 1 Set         | 1                                              |
|       | ii) Final Submission                                     | 4 Sets        | 2                                              |

### 3.5.5 Final Submission of drawings and documents:

The Bidder shall furnish the following after approval of all drawings /documents and test reports:

- List of drawings bearing the Employer's and Contractor's drawing number.
- Six (6) bound sets along-with two (2) sets of CD-ROMs/ DVD/Portable hard disk of all final drawings/documents.
- Bidder shall also furnish six (6) bound sets of all as-built drawings including the list of all as-built drawings bearing drawing numbers. The Contractor shall also furnish two (2) sets of CD-ROMs/ DVD/Portable hard disk of all as-built drawings as decided by the Employer.
- The Bidder shall also furnish four (4) copies and two (2) sets of CD-ROMs/ DVD/Portable hard disk of instruction/ operations & maintenance manuals (after approval) for all the equipment.

### 3.5.6 TEST REPORTS

Two (2) copies of all test reports shall be supplied for approval before shipment of Equipment. The report shall indicate clearly the standard value specified for each test to facilitate checking of the reports. After final approval six (6) bound copies and two (2) sets of CD-ROMs/ DVD/Portable hard disk of all type and routine test reports shall be submitted to Employer.

## 3.6 MATERIAL/ WORKMANSHIP

Where the specification does not contain references to workmanship, equipment, materials and components of the covered equipment, it is essential that the same must be new, of highest grade of the best quality of their kind, conforming to best engineering practice and

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suitable for the purpose for which they are intended and shall ensure satisfactory performance throughout the service life.

In case where the equipment, materials or components are indicated in the specification as “similar” to any special standard the purchaser shall decide upon the question of similarity. When required by the specification or when required by the purchaser the contractor shall submit, for approval, all the information concerning the materials or components to be used in manufacture. Machinery, equipment, materials and components supplied, installed or used without such approval shall run the risk of subsequent rejection, it being understood that the cost as well as the time delay associated with the rejection shall be borne by the Contractor.

The design of the Works shall be such that installation, future expansions, replacements and general maintenance may be undertaken with a minimum of time and expenses. Each component shall be designed to be consistent with its duty and suitable factors of safety subject to mutual agreements. All joints and fastenings shall be devised, constructed and documented so that the component parts shall be accurately positioned and restrained to fulfill their required function. In general, screw threads shall be standard metric threads. The use of other thread forms will only be permitted when prior approval has been obtained from the Purchaser.

Whenever possible, all similar part of the works shall be made to gauge and shall also be made interchangeable with similar parts. All spare parts shall also be interchangeable and shall be made of the same materials and workmanship as the corresponding parts of the equipment supplied under the specification. Where feasible, common component units shall be employed in different pieces of equipment in order to minimize spare parts stocking requirements. All equipment of the same type and rating shall be physically and electrically interchangeable.

The equipment offered in the bid only shall be accepted for supply, with the minimum modifications as agreed/accepted.

### **3.7 PROVISIONS FOR EXPOSURE TO HOT AND HUMID CLIMATE**

Outdoor equipment supplied under the specification shall be suitable for service and storage under tropical conditions of high temperature, high humidity' heavy rainfall and environment favorable to the growth of fungi and mildew. The indoor equipment located in non-air-conditioned areas shall also be of same type.

#### **SPACE HEATERS**

The heaters shall be suitable for continuous operation at 240 V as supply voltage. On –off switch and fuse shall be provided.

One or more adequately rated thermostatically connected heaters shall be supplied to prevent condensation in any compartment. The heaters shall be installed in the compartment and electrical connections shall be made sufficiently away from below the heaters to minimize deterioration of supply wire insulation. The heaters shall be suitable to maintain the compartment temperature to prevent condensation.

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The heaters shall be suitably designed to prevent any contact between the heater wire and the air and shall consist of coiled resistance wire centered in a metal sheath and completely encased in a highly compacted powder of magnesium oxide or other material having equal heat conducting and electrical insulation properties or they shall consist of resistance wire wound on a ceramic and completely covered with a ceramic material to prevent any contact between the wire and the air. Alternatively, they shall consist of a resistance wire mounted into a tubular ceramic body built into an envelope of stainless steel or the resistance wire is wound on a tubular ceramic body and embedded in vitreous glaze. The surface temperature of the heaters shall be restricted to a value which will not shorten the life of the heater sheaths or that of insulated wire or other component in the compartments.

Control cubicles installed in air-conditioned area need not be provided with space heaters. These cubicles shall, however, have space heaters in case of storage of cubicles for long duration.

### **FUNGI STATIC VARNISH**

Besides the space heaters, special moisture and fungus resistance varnish shall be applied on parts which may be subjected or predisposed to the formation of fungi due to the presence or deposit of nutrient substances. The varnish shall not be applied to any surface of part where the treatment will interfere with the operation or performance of the equipment. Such surfaces or parts shall be protected against the application of the varnish.

### **Ventilation opening**

In order to ensure adequate ventilation, compartments shall have ventilation openings provided with fine wire mesh of brass to prevent the entry of insects and to reduce to a minimum the entry of dirt and dust. Outdoor compartment openings shall be provided with shutter type blinds.

### **Degree of Protection**

The enclosure of the Control Cabinets, Junction boxes and Marshalling Boxes, panels etc. to be installed shall provide degree of protection as detailed here under:

- a. Installed outdoor: IP- 55
- b. Installed indoor in air-conditioned area: IP-32
- c. Installed in covered area: IP-52
- d. Installed indoor in non-air-conditioned area where possibility of entry of water is limited: IP-41.
- e. For LT Switchgear (AC & DC distribution Boards): IP-52

The degree of protection shall be in accordance with IS: 13947 (Part –I) / IEC-947 (Part-I) / IS 12063/IEC 529. Type test report for degree of protection test, on each type of the box shall be submitted for approval.

### **PRESERVATIVE SHOP COATING**

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All exposed metallic surfaces subject to corrosion shall be protected by shop application of suitable coatings. All surfaces which will not be easily accessible after the shop assembly, shall be treated beforehand and protected for the life of the equipment. All surfaces shall be thoroughly cleaned of all mill scales, oxides and other coatings and prepared in the shop. The surfaces that are to be finish-painted after installation or require corrosion protection until installation, shall be shop painted as per the requirements covered in the relevant part of the Technical Specification.

Transformers and other electrical equipments, if included shall be shop finished with one or more coats of primer and two coats of high grade resistance enamel. The finished colors shall be as per manufacturer's standards, to be selected and specified by the Employer at a later date.

Shop primer for all steel surfaces which will be exposed to operating temperature below 95 degrees Celsius shall be selected by the Bidder after obtaining specific approval of the Employer regarding the quality of primer proposed to be applied. Special high temperature primer shall be used on surfaces exposed to temperature higher than 95 degrees Celsius and such primer shall also be subject to the approval of the Employer.

### **3.8 RATING PLATES, NAME PLATES AND LABELS**

- 3.8.1 Each equipment shall have permanently attached to it in a conspicuous position, a rating plate of non-corrosive material upon which shall be engraved manufacturer's name, equipment, type or serial number together with details of the ratings, service conditions under which the item of plant in question has been designed to operate, and such diagram plates as may be required by the Employer.
- 3.8.2 Such nameplates or labels shall be of white non-hygroscopic material with engraved black lettering or alternately, in the case of indoor circuit breakers, starters, etc. of transparent plastic material with suitably coloured lettering engraved on the back.
- 3.8.3 Each equipment shall be provided with nameplate or label designating the service of the particular equipment. The inscriptions shall be approved by the Employer or as detailed in appropriate section of the technical specifications.
- 3.8.4 The rated current, extended current rating and rated thermal current shall be clearly indicated in the name plate in case of current transformer.
- 3.8.5 Rated voltage, voltage factor and intermediate voltage shall be clearly indicated on the nameplate in case of capacitor voltage transformer.
- 3.8.6 Each switch shall have a clear inscription identifying its function. Switches shall also have a clear inscription of each position indication.
- 3.8.7 All such plates, instruction plates, etc. shall be bilingual with Hindi inscription first, followed by English. Alternatively, two separate plates one with Hindi and the other with English inscriptions may be provided.

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- 3.8.8 All segregated phases of conductors or bus ducts, indoor or outdoor, shall be provided with coloured phase plates to clearly identify the phase of the system.

### **3.9 GALVANISING:**

- 3.9.1 All exposed ferrous parts shall be hot dip galvanised as per IS:2629 & IS:2633, Galvanising shall be uniform, clean, smooth continuous and free from acid spots. Should the galvanising of the sample be found defective, the entire batch of steel shall have to be re-galvanised at bidder's cost.

- 3.9.2 The amount of zinc deposit over threaded portion of the bolts, nuts and screws shall not be less than 300 gms. per sq. meter of surface area. The amount of zinc deposit on washers shall not be less than 340 gms. per sq. meter of surface area or a minimum of 30 microns. The threads shall have extra deposit of zinc which shall be removed by die cutting after the completion of galvanising. The removal of extra zinc shall be carefully done so that threads shall have the required deposits of zinc on them as specified.

### **3.10 PAINTING**

Unless explicitly stated in relevant chapters of the specification, the painting of all electrical

equipment shall be as follows:

Epoxy based with suitable additives. The thickness of finish coat shall be minimum 50 microns (minimum total DFT shall be 100 microns). However in case electrostatic process of painting is offered for any electrical equipment, minimum paint thickness of 50 microns shall be acceptable for finish coat. Paint shade shall be as per technical specification.

### **3.11 QUALITY ASSURANCE PROGRAMME**

- 3.11.1 The Bidder shall adopt suitable quality assurance programme to ensure that the equipment and services under the scope of contract whether manufactured or performed within the Bidder's works or at his subcontractor's premises or at the Employer's site or at any other place of work are in accordance with the specifications. Such programmes shall be outlined by the Contractor and shall be finally accepted by the Employer/authorised representative after discussions before the award of the contract. The QA programme shall be generally in line with ISO-9001/IS- 14001.

A quality assurance programme of the contractor shall generally cover the following:

- i. His organisation structure for the management and implementation of the proposed quality assurance programme.
- ii. Quality System Manual
- iii. Design Control System
- iv. Documentation Data Control System
- v. Qualification data for Bidder's key Personnel.

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- vi. The procedure for purchase of materials, parts, components and selection of sub-contractor's services including vendor analysis, source inspection, incoming raw-material inspection, verification of materials purchased etc.
- vii. System for shop manufacturing and site erection controls including process controls and fabrication and assembly controls.
- viii. Control of non-conforming items and system for corrective actions and resolution of deviations.
- ix. Inspection and test procedure both for manufacture and field activities.
- x. Control of calibration and testing of measuring testing equipments.
- xi. System for Quality Audits.
- xii. System for identification and appraisal of inspection status.
- xiii. System for authorising release of manufactured product to the Employer.
- xiv. System for handling storage and delivery.
- xv. System for maintenance of records, and
- xvi. Furnishing quality plans for manufacturing detailing out the specific quality control procedure adopted for controlling the quality characteristics relevant to each item of equipment/component as per format enclosed as Annexure-I.

### 3.12 GENERAL REQUIREMENTS - QUALITY ASSURANCE

3.12.1 All materials, components and equipment covered under this specification shall be procured, manufactured, erected, commissioned and tested at all the stages, as per a comprehensive Quality Assurance Programme. An indicative programme of inspection/tests to be carried out by the Bidder for some of the major items is given in the respective technical specification as per Annexure-III. This is, however, not intended to form a comprehensive programme as it is the Bidder's responsibility to draw up and implement such programme duly approved by the Employer. The detailed Quality Plans for manufacturing and field activities should be drawn up by the Bidder and will be submitted to Employer for approval.

3.12.2 Manufacturing Quality Plan will detail out for all the components and equipment, various tests/inspection, to be carried out as per the requirements of this specification and standards mentioned therein and quality practices and procedures followed by Bidder's/ Sub-contractor's/ sub-supplier's Quality Control Organisation, the relevant reference documents and standards, acceptance norms, inspection documents raised etc., during all stages of materials procurement, manufacture, assembly and final testing/performance testing. The Quality Plan shall be submitted on electronic media e.g. E-mail in addition to hard copy, for review. Once the same is finalised, hard copies shall be submitted for approval. After approval the same shall be submitted in compiled form on CD ROM.

3.12.3 The Bidder shall also furnish copies of the reference documents/plant standards/acceptance norms/tests and inspection procedure etc., as referred in Quality Plans along with Quality Plans. These Quality Plans and reference documents/standards etc. will be subject to Employer's approval without which manufacturer shall not proceed.

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- 3.12.4 These approved documents shall form a part of the contract. In these approved Quality Plans, Employer shall identify customer hold points (CHP), i.e. test/checks which shall be carried out in presence of the Employer's Project Manager or his authorised representative and beyond which the work will not proceed without consent of Employer/Authorised representative in writing. All deviations to this specification, approved quality plans and applicable standards must be documented and referred to Employer along with technical justification for approval and dispositioning.
- 3.12.5 No material shall be despatched from the manufacturer's works before the same is accepted subsequent to pre-despatch final inspection including verification of records of all previous tests/inspections by Employer's Project Manager/Authorised representative and duly authorised for despatch by issuance of Material Dispatch Clearance Certificate (MDCC).
- 3.12.6 All material used for equipment manufacture including casting and forging etc. shall be of tested quality as per relevant codes/standards. Details of results of the tests conducted to determine the mechanical properties, chemical analysis and details of heat treatment procedure recommended and actually followed shall be recorded on certificates and time temperature chart. Tests shall be carried out as per applicable material standards and/or agreed details.
- 3.12.7 All welding and brazing shall be carried out as per procedure drawn and qualified in accordance with requirements of ASME Section IX/BS-4870 or other International equivalent standard acceptable to the Employer.
- 3.12.8 All welding/brazing procedures shall be submitted to the Employer or its authorised representative for approval prior to carrying out the welding/brazing.
- 3.12.9 All brazers, welders and welding operators employed on any part of the contract either in Bidder's/his sub-contractor's works or at site or elsewhere shall be qualified as per ASME Section-IX or BS-4871 or other equivalent International Standards acceptable to the Employer.
- 3.12.10 Test results or qualification tests and specimen testing shall be furnished to the Employer for approval. However, where required by the Employer, tests shall be conducted in presence of Employer/authorised representative.
- 3.12.11 For all pressure parts and high pressure piping welding, the latest applicable requirements of the IBR (Indian Boiler Regulations) shall also be essentially complied with. Similarly, any other statutory requirements for the equipments/systems shall also be complied with. On all back-gauged welds MPI/LPI shall be carried before seal welding.
- 3.12.12 All the heat treatment results shall be recorded on time temperature charts and verified with recommended regimes.
- 3.12.13 No welding shall be carried out on cast iron components for repair.

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- 3.12.14 Unless otherwise proven and specifically agreed with the Employer, welding of dissimilar materials and high alloy materials shall be carried out at shop only.
- 3.12.15 All non-destructive examination shall be performed in accordance with written procedures as per International Standards. The NDT operator shall be qualified as per SNT-TC-IA (of the American Society of non-destructive examination). NDT shall be recorded in a report which includes details of methods and equipment used, result/evaluation, job data and identification of personnel employed and details of correlation of the test report with the job. In general all plates of thickness greater than 40mm & for pressure parts plates of thickness equal to or greater than 25mm shall be ultrasonically tested otherwise as specified in respective equipment specification. All bar stock/Forging of diameter equal to or greater than 40mm shall be ultrasonically tested.
- 3.12.16 The Bidder shall list out all major items/ equipment/ components to be manufactured in house as well as procured from sub-contractors (BOI). All the subcontractor proposed by the Contractor for procurement of major bought out items including castings, forging, semi-finished and finished components/equipment etc., list of which shall be drawn up by the Bidder and finalised with the Employer, shall be subject to Employer's approval. The Bidder's proposal shall include vendor's facilities established at the respective works, the process capability, process stabilization, QC systems followed, experience list, etc. along with his own technical evaluation for identified subcontractors enclosed and shall be submitted to the Employer for approval within the period agreed at the time of pre-awards discussion and identified in "DR" category prior to any procurement. Such vendor approval shall not relieve the Bidder from any obligation, duty or responsibility under the contract.
- 3.12.17 For components/equipment procured by the Bidders for the purpose of the contract, after obtaining the written approval of the Employer, the Bidder's purchase specifications and inquiries shall call for quality plans to be submitted by the suppliers. The quality plans called for from the subcontractor shall set out, during the various stages of manufacture and installation, the quality practices and procedures followed by the vendor's quality control organisation, the relevant reference documents/standards used, acceptance level, inspection of documentation raised, etc.
- 3.12.18 Employer reserves the right to carry out quality audit and quality surveillance of the systems and procedures of the Bidder's or their sub-contractor's quality management and control activities. The Bidder shall provide all necessary assistance to enable the Employer carry out such audit and surveillance.
- 3.12.19 The Bidder shall carry out an inspection and testing programme during manufacture in his work and that of his sub-contractor's and at site to ensure the mechanical accuracy of components, compliance with drawings, conformance to functional and performance requirements, identity and acceptability of all materials parts and equipment. Bidder shall carry out all tests/inspection required to establish that the items/equipments conform to requirements of the specification and the relevant

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codes/standards specified in the specification, in addition to carrying out tests as per the approved quality plan.

3.12.20 Quality audit/surveillance/approval of the results of the tests and inspection will not, however, prejudice the right of the Employer to reject the equipment if it does not comply with the specification when erected or does not give complete satisfaction in service and the above shall in no way limit the liabilities and responsibilities of the Bidder in ensuring complete conformance of the materials/equipment supplied to relevant specification, standard, data sheets, drawings, etc.

3.12.21 For all spares and replacement items, the quality requirements as agreed for the main equipment supply shall be applicable.

3.12.22 Repair/rectification procedures to be adopted to make the job acceptable shall be subject to the approval of the Employer/ authorised representative.

3.12.23 Environmental Stress Screening

All solid state electronic system / equipment / sub assembly shall be free from infant mortile components. For establishing the compliance to this requirement, the Bidder / sub – contractor should meet the following.

1. The Bidder / Sub – contractor shall furnish the established procedure being followed for eliminating infant mortile components. The procedure followed by the Contractor / Sub – contractor should be substantiated along with the statistical figures to validate the procedure being followed. The necessary details as required under this clause shall be furnished at the stage of QP finalization.

**Or**

In case the Bidder / Sub – contractor do not have any established procedure to eliminate infant mortile components then two or 10% whichever is less, most densely populated Panels shall be tested for Elevated Temperature Cycle Test as per the following procedure.

### **Elevated Temperature Test Cycle**

During the elevated temperature test which shall be for 48 hours, the ambient temperature shall be maintained at 50° C. The equipment shall be interconnected with devices and kept under energized conditions so as to repeatedly perform all operations it is expected to perform in actual service with load on various components being equal to those which will be experienced in actual service.

During the elevated temperature test the cubicle doors shall be closed (or shall be in the position same as they are supposed to be in the field) and inside temperature in the zone of highest heat dissipating components / modules shall be monitored. The temperature rise inside the cubicle should not exceed 10° C above the ambient temperature at 50° C.

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In case of any failure during the test cycle, the further course of action should be mutually discussed for demonstrating the intent of the above requirement.

### **Burn In Test Cycle**

The test shall be conducted on all the panels fully assembled and wired including the panels having undergone the above mentioned elevated temperature test.

The period of Burn in Test Cycle shall be 120 hrs and process shall be similar to the elevated temperature test as above except that the temperature shall be reduced to the ambient temperature prevalent at that time.

During the above tests, the process I/O and other load on the system shall be simulated by simulated inputs and in the case of control systems, the process which is to be controlled shall also be simulated. Testing of individual components or modules shall not be acceptable.

During the Burn in Test the cubicle doors shall be closed (or shall be in the position same as they are supposed to be in the field) and inside temperature in the zone of highest heat dissipating components / modules shall be monitored. The temperature rise inside the cubicle should not exceed 10° C above the ambient temperature.

The Bidder / Sub-contractor shall carry out routine test on 100% item at Bidder's / sub-contractor's works. The quantum of check / test for routine & acceptance test by employer shall be generally as per criteria / sampling plan defined in referred standards. Wherever standards have not been mentioned quantum of check / test for routine / acceptance test shall be as agreed during detailed engineering stage.

## **3.13 QUALITY ASSURANCE DOCUMENTS**

The Contractor shall be required to submit two hard copies and two sets on CDROM of the following Quality Assurance Documents as identified in respective quality plan with tick (✓) mark.

Each QA Documentation shall have a project specific Cover Sheet bearing name & identification number of equipment and including an index of its contents with page control on each document.

The QA Documentation file shall be progressively completed by the Supplier's sub-supplier to allow regular reviews by all parties during the manufacturing.

The final quality document will be compiled and issued at the final assembly place of equipment before dispatch. However CD-Rom may be issued not later than three weeks.

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3.13.1 Typical contents of Quality Assurance Document are as below: -

- i) Quality Plan,
- ii) Material mill test reports on components as specified by the specification and approved Quality Plans.
- iii) Manufacturer / works test reports/results for testing required as per applicable codes and standard referred in the specification and approved Quality Plans.
- iv) Non-destructive examination results /reports including radiography interpretation reports. Sketches/drawings used for indicating the method of traceability of the radiographs to the location on the equipment.
- v) Heat Treatment Certificate/Record (Time- temperature Chart)
- vi) All the accepted Non-conformance Reports (Major/Minor) / deviation, including complete technical details / repair procedure).
- vii) CHP / Inspection reports duly signed by the Inspector of the Employer and Contractor for the agreed Customer Hold Points.
- viii) Certificate of Conformance (COC) whoever applicable.
- ix) MDCC

3.13.2 Before dispatch/ commissioning of any equipment, the Supplier shall make sure that the corresponding quality document or in the case of protracted phased deliveries, the applicable section of the quality document file is completed. The supplier will then notify the Inspector regarding the readiness of the quality document (or applicable section) for review.

- i) If the result of the review carried out by the Inspector of the Quality document (or applicable section) is satisfactory, the Inspector shall stamp the quality document (or applicable section) for release.
- ii) If the quality document is unsatisfactory, the Supplier shall endeavour to correct the incompleteness, thus allowing to finalize the quality document (or applicable section) by time compatible with the requirements as per contract documents. When it is done, the quality document (or applicable section) is stamped by the Inspector.
- iii) If a decision is made for dispatch, whereas all outstanding actions cannot be readily cleared for the release of the quality document by that time, the supplier shall immediately, upon shipment of the equipment, send a copy of the quality document Review Status signed by the Supplier Representative to the Inspector and notify of the committed date for the completion of all outstanding actions & submission. The Inspector shall stamp the quality document for applicable section when it is effectively completed. The submission of QA documentation package shall not be later than 3 weeks after the dispatch of equipment.

### 3.14 TRANSMISSION OF QUALITY DOCUMENTS

As a general rule, two hard copies of the quality document and Two CD ROMs shall be issued to the Employer on release of QA Documentation by Inspector. One set of

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quality document shall be forwarded to Corporate Quality Assurance Department and other set to respective Site.

For the particular case of phased deliveries, the complete quality document to the Employer shall be issued not later than 3 weeks after the date of the last delivery similarly as stated above.

### **3.15 INSPECTION, TESTING & INSPECTION CERTIFICATE**

3.15.1 The word ‘Inspector’ shall mean the Project Manager and/or his authorised representative and/or an outside inspection agency acting on behalf of the Employer to inspect and examine the materials and workmanship of the works during its manufacture or erection.

3.15.2 The Project Manager or his duly authorised representative and/or an outside inspection agency acting on behalf of the Employer shall have access at all reasonable times to inspect and examine the materials and workmanship of the works during its manufacture or erection and if part of the works is being manufactured or assembled on other premises or works, the Bidder shall obtain for the Project Manager and for his duly authorised representative permission to inspect as if the works were manufactured or assembled on the Bidder’s own premises or works.

3.15.3 The Bidder shall give the Project Manager/Inspector fifteen (15) days written notice of any material being ready for testing. Such tests shall be to the Bidder’s account except for the expenses of the Inspector’s. The Project Manager/Inspector, unless the witnessing of the tests is virtually waived and confirmed in writing, will attend such tests within fifteen (15) days of the date on which the equipment is noticed as being ready for test/inspection failing which the Bidder may proceed with test which shall be deemed to have been made in the inspector’s presence and he shall forthwith forward to the inspector duly certified copies of test reports in two (2) copies.

3.15.4 The Project Manager or Inspector shall within fifteen (15) days from the date of inspection as defined herein give notice in writing to the Bidder, or any objection to any drawings and all or any equipment and workmanship which is in his opinion not in accordance with the contract. The Bidder shall give due consideration to such objections and shall either make modifications that may be necessary to meet the said objections or shall inform in writing to the Project Manager/Inspector giving reasons therein, that no modifications are necessary to comply with the contract.

3.15.5 When the factory tests have been completed at the Bidder’s or subcontractor’s works, the Project Manager /Inspector shall issue a certificate to this effect fifteen (15) days after completion of tests but if the tests are not witnessed by the Project Manager /Inspectors, the certificate shall be issued within fifteen (15) days of the receipt of the Contractor’s test certificate by the Project Manager /Inspector. Project Manager /Inspector to issue such a certificate shall not prevent the Bidder from proceeding with the works. The completion of these tests or the issue of the

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certificates shall not bind the Employer to accept the equipment should it, on further tests after erection be found not to comply with the contract.

- 3.15.6 In all cases where the contract provides for tests whether at the premises or works of the Bidder or any sub-contractor, the Bidder, except where otherwise specified shall provide free of charge such items as labour, material, electricity, fuel, water, stores, apparatus and instruments as may be reasonably demanded by the Project Manager /Inspector or his authorised representatives to carry out effectively such tests on the equipment in accordance with the Bidder and shall give facilities to the Project Manager/Inspector or to his authorised representative to accomplish testing.
- 3.15.7 The inspection by Project Manager / Inspector and issue of Inspection Certificate thereon shall in no way limit the liabilities and responsibilities of the Contractor in respect of the agreed Quality Assurance Programme forming a part of the contract.
- 3.15.8 To facilitate advance planning of inspection in addition to giving inspection notice, the Bidder shall furnish quarterly inspection programme indicating schedule dates of inspection at Customer Hold Point and final inspection stages. Updated quarterly inspection plans will be made for each three consecutive months and shall be furnished before beginning of each calendar month.
- 3.15.9 All inspection, measuring and test equipments used by contractor shall be calibrated periodically depending on its use and criticality of the test/measurement to be done. The Bidder shall maintain all the relevant records of periodic calibration and instrument identification, and shall produce the same for inspection by NTPC. Wherever asked specifically, the contractor shall re-calibrate the measuring/test equipments in the presence of Project Manager / Inspector.

### **3.16 PACKAGING & TRANSPORTATION**

All the equipment shall be suitably protected, coated, covered or boxed and crated to prevent damage or deterioration during transit, handling and storage at Site till the time of erection. While packing all the materials, the limitation from the point of view of the sizes of railway wagons available in India should be taken account of. The Bidder shall be responsible for any loss or damage during transportation, handling and storage due to improper packing. The Bidder shall ascertain the availability of Railway wagon sizes from the Indian Railways or any other agency concerned in India well before effecting despatch of equipment. Before despatch it shall be ensured that complete processing and manufacturing of the components is carried out at shop, only restricted by transport limitation, in order to ensure that site works like grinding, welding, cutting & preassembly to bare minimum. The Employer's Inspector shall have right to insist for completion of works in shops before despatch of materials for transportation.

### **3.17 CLAMPS AND CONNECTORS INCLUDING TERMINAL CONNECTORS**

- 3.17.1 The material of clamps and connectors shall be Aluminium alloy casting conforming to designation A6 of IS:617 for connecting to equipment terminals and

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conductors of aluminium. In case the terminals are of copper, the same clamps/connectors shall be used with 2mm thick bimetallic liner.

- 3.17.2 The material of clamps and connectors shall be Galvanised mild steel for connecting to shield wire.
- 3.17.3 Bolts, nuts and plain washers shall be hot dip galvanised mild steel for sizes M12 and above. For sizes below M12, they shall be electro-galvanised mild steel. The spring washers shall be electro-galvanised mild steel.
- 3.17.4 All castings shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be rounded off to meet specified corona and radio interference requirements.
- 3.17.5 They shall have same current rating as that of the connected equipment. All current carrying parts shall be at least 10 mm thick. The connectors shall be manufactured to have minimum contact resistance.
- 3.17.6 Flexible connectors, braids or laminated strips shall be made up of copper/aluminium.
- 3.17.7 Current rating and size of terminal/conductor for which connector is suitable shall be put on a suitable sticker on each component which should last atleast till erection time.

### **3.18 SPACERS**

- 3.18.1 Spacers shall conform to IS: 10162. They shall be of non-magnetic material except nuts and bolts, which shall be of hot dip galvanised mild steel.
- 3.18.2 Spacers shall generally meet the requirements of clamps and connectors as specified above. Its design shall take care of fixing and removing during installation and maintenance.
- 3.18.3 In addition to the type tests as per IS: 10162, clamp slip test should have been conducted. In this test the sample shall be installed on test span of twin/quad bundle string at a tension of 44.2kN (4500 kg). One of the clamps when subjected to a longitudinal pull of 2.5kN (250 kg) parallel to the axis of conductor shall not slip, i.e. permanent displacement between conductor and clamp after test shall not exceed 1.0 mm. This test should have been performed on all other clamps of the sample.

### **3.19 BUSHINGS, HOLLOW COLUMN INSULATORS, SUPPORT INSULATORS, AND DISC INSULATORS**

- 3.19.1 Bushings shall be manufactured and tested in accordance with IS: 2099 & IEC: 60137 while hollow column insulators shall be manufactured and tested in accordance with IEC62155/IS 5284. The support insulators shall be manufactured

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and tested as per IS: 2544/IEC 60168/IEC 60273. The insulators shall also conform to IEC 60815 as applicable having alternate long and short sheds.

Support insulators/ bushings/ hollow column insulators shall be designed to have ample insulation, mechanical strength and rigidity for the conditions under which they will be used.

- 3.19.2 Porcelain used shall be homogenous, free from laminations, cavities and other flaws or imperfections that might affect the mechanical or dielectric quality and shall be thoroughly vitrified, tough and impervious to moisture.
- 3.19.3 Glazing of the porcelain shall be uniform brown in colour, free from blisters, burns and other similar defects.
- 3.19.4 The design of the insulator shall be such that stresses due to expansion and contraction in any part of the insulator shall be lead to deterioration. All ferrous parts shall be hot dip galvanised.
- 3.19.5 Post type insulators shall consist of a porcelain part permanently secured in metal base to be mounted on supporting structures. They shall be capable of being mounted upright. They shall be designed to withstand all shocks to which they may be subjected to during operation of the associated equipment.
- 3.19.6 Bushing porcelain shall be robust and capable of withstanding the internal pressures likely to occur in service. The design and location of clamps, the shape and the strength of the porcelain flange securing the bushing to the tank shall be such that there is no risk of fracture. All portions of the assembled porcelain enclosures and supports other than gaskets, which may in any way be exposed to the atmosphere shall be composed of completely non hygroscopic material such as metal or glazed porcelain.
- 3.19.7 All iron parts shall be hot dip galvanised and all joints shall be air tight. Surface of joints shall be trued, porcelain parts by grinding and metal parts by machining. Insulator/ bushing design shall be such as to ensure a uniform compressive pressure on the joints.
- 3.19.8 In accordance with the requirement stipulated elsewhere, bushing, hollow column insulators and support insulators shall conform to type tests and shall be subjected to routine tests and acceptance test/sample test in accordance with relevant standards.

### **3.20 CONTROL CABINETS, JUNCTION BOXES, TERMINAL BOXES & MARSHALLING BOXES FOR OUTDOOR EQUIPMENT.**

- 3.20.1 All types of control cabinets, junction boxes, marshalling boxes, lighting panels, terminal boxes, operating mechanism boxes, Kiosks etc. shall generally conform to IS:5039, IS:8623 and IEC:60439 as applicable.

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- 3.20.2 They shall be of Stainless steel or Aluminium. The thickness of Stainless steel shall be minimum 1 mm. The thickness of aluminium shall be minimum 3 mm and shall provide rigidity. Top of the boxes shall be sloped towards the rear of the box.

### 3.20.3 BAY MARSHALLING BOX

Bay Marshalling Box located at a convenient location to receive and distribute cables shall be provided as required. It shall meet all the requirements as specified for cabinets/boxes.

It shall have three separate distinct compartments for following purposes:

- To receive two incoming 415V, three phase, AC supplies controlled by 100A four pole MCBs with auto changeover provision, and to distribute five (5) three phase ac supplies controlled by 32A four pole MCBs. It shall also be provided with 63A, 3 phase 4 pin industrial grade receptacle with rotary switch.
- To receive three phase incoming from first compartment and to distribute ten (10) single phase ac supplies controlled by 16A two pole MCBs.
- 150 nos. terminal blocks in vertical formation for interlocking facility.

### 3.20.4 AUXILIARY SWITCH

The auxiliary switch shall conform of following type tests:

- a) Electrical endurance test - A minimum of 1000 operations for 2A. D.C. with a time constant greater than or equal to 20 milliseconds with a subsequent examination of mV drop/ visual defects/ temperature rise test.
- b) Mechanical endurance test - A minimum of 5000 operations with a subsequent checking of contact pressure test/ visual examination
- c) Heat run test on contacts
- d) IR/HV test, etc.

### 3.21 CABLE GLANDS AND LUGS/FERRULES

- 3.21.1 Cable shall be terminated using double compression type cable glands. Testing requirements of Cable glands shall conform to BS:6121 and gland shall be of robust construction capable of clamping cable and cable armour (for armoured cables) firmly without injury to insulation. Cable glands shall be made of heavy duty brass machine finished and nickel chrome plated. Thickness of plating shall not be less than 10 micron. All washers and hardware shall also be made of brass with nickel chrome plating Rubber components shall be of neoprene or better synthetic material and of tested quality. Cable glands shall be suitable for the sizes of cable supplied/erected.

- 3.21.2 Cable lugs/ferrules for power cables shall be tinned copper solderless crimping type suitable for aluminium compacted conductor cables. Cable lugs and ferrules for

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control cables shall be tinned copper type. The cable lugs for control cables shall be provided with insulating sleeve and shall suit the type of terminals provided on the equipment. Cable lugs and ferrule shall conform to DIN standards.

### **3.22 CONDUITS, PIPES AND ACCESSORIES**

- 3.22.1 The bidder shall supply and install all rigid conduits, mild steel pipes, flexible conduits, Hume pipes, etc. including all necessary sundry materials, such as tees, elbows, check nuts, bushing reduces, enlargers, wooden plugs, coupling caps, nipples, gland sealing fittings, pull boxes, etc.
- 3.22.2 The size of the conduit/pipe shall be selected to limit the fill to a maximum of 40%. All conduits/pipes shall have their ends closed by caps until cables are pulled. After cables are pulled, the ends of conduits/pipes shall be sealed in an approved manner to prevent damage to threaded portions and entrance of moisture and foreign materials.
- 3.22.3 PVC conduits shall be of high impact, heavy gauge (at least class 2) conduit conforming to BS-4607.
- 3.22.4 The outer surface of the steel conduits shall be coated with hot-dip zinc and chromate conversion coatings. The inner surface shall have silicone epoxy ester coating for easy cable pulling. Mild steel pipes shall be hot-dip galvanized. All rigid conduits/pipes shall be of a reputed make.
- 3.22.5 The hume pipes and accessories shall be of reinforced concrete conforming to class NP2 of IS-458. All tests on hume pipes shall be conducted as per IS-458.
- 3.22.6 Flexible conduits shall be of heat-resistant lead coated steel, water-leak, fire and rust proof.

### **3.23 MOTORS**

The voltage level for motors shall be as follows:

- |                                  |                                          |
|----------------------------------|------------------------------------------|
| a) Upto 0.2 KW                   | : Single phase 240V AC / 3 phase 415V AC |
| b) Above 0.2 KW and upto 200 KW  | : 3 phase, 415V AC                       |
| c) Above 200 KW and upto 1500 KW | : 3 phase, 3.3 kV AC                     |
| d) Above 1500 KW                 | : 11 kV                                  |

The bidder may adopt 415V/3.3 KV for the drives rated in the range of 160-210 KW.

The voltage rating of the drives indicated above is for basic guideline.

- 3.23.1 All motors shall conform to IEC-60034-5 / IS Standard and with principal dimensions in accordance with IEC 60072-1 (1991), IEC 60072-2 (1990) and IEC 60072-3 (1994).

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3.23.2 All equipment shall be suitable for rated frequency of 50 Hz with a variation of +3% & -5%, and 10% combined variation of voltage and frequency unless specifically brought out in the specification

3.23.3 Paint shade shall be as per RAL 5012 (Blue) for indoor and outdoor equipment.

3.23.4 Degree of Protection

Degree of protection for various enclosures as per IEC60034-05 shall be as follows:

Indoor motors - IP 54

Outdoor motors - IP 55

Cable box-indoor area - IP 54

Cable box-Outdoor area - IP 55

3.23.5 Type:

#### **AC Motors:**

- a) Squirrel cage induction motor suitable for direct-on-line starting.
- b) Continuous duty LT motors upto 200 KW Output rating (at 50 deg.C ambient temperature), shall be Premium Efficiency class-IE3, conforming to IS 12615, or IEC:60034-30.
- c) Crane duty motors shall be squirrel cage Induction motor as per the requirement.
- d) Motor operating through variable frequency drives shall be suitable for inverter duty. Also these motors shall comply the requirements stipulated in IEC: 60034-18-41 and IEC: 60034-18-42 as applicable.

**DC Motors** Shunt wound

### **3.24 AUXILIARY SWITCH**

The auxiliary switch shall conform of following type tests:

- a) Electrical endurance test - A minimum of 1000 operations for 2A. D.C. with a time constant greater than or equal to 20 milliseconds with a subsequent examination of mV drop/ visual defects/ temperature rise test.
- b) Mechanical endurance test - A minimum of 5000 operations with a subsequent checking of contact pressure test/ visual examination
- c) Heat run test on contacts
- d) IR/HV test, etc.

### **3.25 LAMPS AND SOCKETS**

3.25.1 Lamps:

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All incandescent lamps shall use a socket base as per IS-1258, except in the case of signal lamps.

### 3.25.2 Sockets

All sockets (convenience outlets) shall be suitable to accept both 5 Amp & 15 Amp pin round Standard Indian plugs. They shall be switched sockets with shutters.

### 3.25.3 Hand Lamp:

A 240 Volts, single Phase, 50 Hz AC plug point shall be provided in the interior of each cubicle with ON-OFF Switch for connection of hand lamps.

## 3.26 SWITCHES & FUSES:

Each control panel shall be provided with necessary arrangements for receiving, distributing, isolating and fusing of DC and AC supplies for various control, signaling, lighting and space heater circuits. The incoming and sub-circuits shall be separately provided with switch-fuse units. Selection of the main and sub-circuit fuse ratings shall be such as to ensure selective clearance of sub-circuit faults. Potential circuits for relaying and metering shall be protected by HRC fuses.

All fuses shall be of HRC cartridge type conforming to IS 9228 mounted on plug-in type fuse bases. Miniature circuit breakers with thermal Protection and alarm contacts will also be accepted. All accessible live connection to fuse bases shall be adequately shrouded. Fuses shall have operation indicators for indicating blown fuse condition. Fuse carrier base shall have imprints of the fuse rating and voltage.

All control switches shall be of rotary type. Toggle/piano switches shall not be accepted.

## 3.27 TYPE, ROUTINE & ACCEPTANCE TESTS:

### 3.27.1 TYPE TEST REQUIREMENTS FOR EQUIPMENTS OTHER THAN CIRCUIT BREAKER:

a) All equipment to be supplied shall be of type tested design. During detail engineering, the contractor shall submit for Owner's approval the reports of all the type tests as listed in this specification. The validity period (counted from 02 May 2024) of reports shall be as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical Equipment in power Transmission- May2020 (with latest amendments). These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a Client.

b) However if contractor is not able to submit report of the type test(s) conducted as per CEA Guidelines for the validity period of Type test(s) conducted on Major Electrical equipment in power Transmission-May2020 (with latest amendments) from

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02 May 2024, or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor shall conduct all such tests under this contract at no additional cost to the owner either at third party lab or in presence of client/ owners representative and submit the reports for approval.

c) All acceptance and routine tests as per the specification and relevant standards shall be carried out. Charges for these shall be deemed to be included in the equipment price.

### **3.27.2 TYPE TEST REQUIREMENTS FOR CIRCUIT BREAKER:**

a) The Contractor shall carry out the type tests as listed in this specification on the equipment to be supplied under this contract. The Bidder shall indicate the charges for each of these type tests separately in the relevant schedule of BPS and the same shall be considered for the evaluation of bids. The type test charges shall be paid as per the charges quoted for each of these type tests separately in the relevant schedule of BPS (Bid Proposal Sheet) & no qty variation is allowed. only for the test(s) conducted successfully under the contract and upon certification by the Employer's engineer.

b) The type tests shall be carried out in the presence of the Employer's representative, for which minimum 60 days' notice shall be given by the Contractor. The Contractor shall obtain the Employer's approval for the type test procedure before conducting the type test. The type test procedure shall clearly specify the test set up, instrument to be used, procedure, acceptance norms, recording of various parameters, interval of recording, precautions to be taken etc. for the type test(s) to be carried out.

c) In case the Contractor has conducted such specified type test(s) according to the relevant standard and / or specification as per CEA Guidelines for the validity period (counted from 02 May 2024) of Type test(s) conducted on Major Electrical equipment in power Transmission-May2020 & with latest amendments, submit the type test reports to the Employer for waiver of conductance of such type test(s). These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a Client. The Employer reserves the right to waive conducting of any or all the specified type tests(s) under this contract. In case the type tests are waived, the type test charges shall not be payable to the Contractor.

However, if any type test report is found not meeting the specification requirements, bidder shall conduct all such type tests successfully according to relevant standards without any cost and delivery implication to BHEL.

#### **3.27.2.1 Type tests to be conducted on AIS Circuit Breaker**

- a) Di-electric tests
- b) Radio interference voltage test
- c) Temperature rise test

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- d) short time withstand current, peak withstand test, short circuit test duties, short line fault test
- e) Mechanical endurance
- f) Out of phase making & breaking test
- g) Line charging current breaking test
- h) Corona test for 765kV only
- i) IP: 55 test on each type of box
- j) Seismic withstand test with structure for 765kV only
- k) Test for reactor switching duty for 765kV CB only (applicable for Bus reactor CB)

### **3.28 CORONA AND RIV TESTS AND SEISMIC WITHSTAND TEST:**

- a) The corona and RIV tests shall conform to the requirements as per Annexure A.
- b) The seismic withstand test shall conform to requirements as per Annexure B.

### **3.29 Enclosures:**

1. ANNEXURE-A: CORONA AND RADIO INTERFERENCE VOLTAGE (RIV) TEST
2. ANNEXURE-B: SEISMIC WITHSTAND TEST
3. ANNEXURE-I: MQP (NTPC format)
4. ANNEXURE-II: QUALITY ASSURANCE FOR SWITCHYARD

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## ANNEXURE – A

**CORONA AND RADIO INTERFERENCE VOLTAGE (RIV) TEST (For 765 & 400kV AIS only)****1.0 General:**

Unless otherwise stipulated, all equipment together with its associated connectors where applicable shall be tested for external corona both by observing the voltage level for the extinction of visible corona under falling power frequency voltage and measurement of radio interference voltage (RIV).

**2.0 Test Methods for RIV:**

2.1 RIV tests shall be made according to measuring circuit as per International Special-committee on Radio Interference (CISPR) Publication 16 -1 (1993) Part – I. The measuring circuit shall preferably be tuned to frequency with 10 % of 0.5 MHz but other frequencies in the range of 0.5 MHz to 2 MHz may be used, the measuring frequency being recorded. The result shall be in microvolts. Alternatively, RIV tests shall be in accordance with NEMA standard Publication No. 107 – 1964 except otherwise noted herein. In measurement of RIV only standard fittings of identical type supplied with the equipment and a simulation of the connections as used in the actual installation will be permitted in the vicinity within 3.5 meters of terminals.

2.2 Ambient noise shall be measured before and after each series of tests to ensure that there is no variation in ambient noise level. If variation is present, the lowest ambient noise level will form basis for the measurements. RIV levels shall be measured at increasing and decreasing voltages of 85%, 100%, 115% and 130% for the specified RIV test voltage for all equipment unless otherwise specified. The specified RIV test voltage for 765 & 400KV is listed in the detailed specification together with maximum permissible RIV level in microvolts.

**3.0 Test Methods for visible Corona:**

The purpose of this test is to determine the corona extinction voltage of the apparatus, connectors etc. The test shall be carried out in the same manner as RIV test described above with the exception that RIV measurements are not required during test and a search technique shall be used near the onset and extinction voltage, when the test voltage is raised and lowered to determine their precise values. The test voltage shall be raised to 130 % of RIV test voltage and maintained there for five minutes. In case corona inception does not take place at 130 %, the voltage level shall be raised till inception of corona or rated voltage whichever is lower. The voltage will then be decreased slowly until all visible corona disappears. The test procedure shall be repeated at least 4 times with corona inception and extinction voltage recorded each time. The corona extinction voltage for purposes of determining compliance with the specification shall be the lowest of the four values at which the visible corona (negative or positive polarity) disappears.

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## ANNEXURE – B

**SEISMIC WITHSTAND TEST (for 765 & 400 kV AIS only)**

- a) The seismic withstand test on the complete equipment (except BPI) shall be carried out along with supporting structure.
- b) The seismic level specified shall be applied at the base of the structure. The accelerometers shall be provided at the terminal pad of the equipment and at any other point as agreed by the owner. The seismic test shall be carried out in all possible combinations of the equipment. The seismic test procedure shall be furnished for approval of the purchaser.

ANNEXURE-I

|                 |                                    |                                   |                   |                |
|-----------------|------------------------------------|-----------------------------------|-------------------|----------------|
| MFGR.'s<br>LOGO | MANUFACTURER'S NAME AND<br>ADDRESS | <b>MANUFACTURING QUALITY PLAN</b> |                   | PROJECT :      |
|                 |                                    | ITEM :                            | QP NO.:           | PACKAGE :      |
|                 |                                    | SUB-SYSTEM:                       | REV.NO.:          | CONTRACT NO. : |
|                 |                                    |                                   | DATE:             | MAIN-SUPPLIER: |
|                 |                                    |                                   | PAGE: .... OF.... |                |

| SL.<br>NO                     | COMPONENT &<br>OPERATIONS | CHARACTERISTICS | CLASS                                                                                                                                                                                                                                                                                              | TYPE OF<br>CHECK | QUANTUM<br>OF CHECK |       | REFERENCE<br>DOCUMENT | ACCEPTANCE<br>NORMS                                                                                             | FORMAT OF<br>RECORD |    | AGENCY      |                   |                  | REMARKS |
|-------------------------------|---------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-------|-----------------------|-----------------------------------------------------------------------------------------------------------------|---------------------|----|-------------|-------------------|------------------|---------|
|                               |                           |                 |                                                                                                                                                                                                                                                                                                    |                  | M                   | C / N |                       |                                                                                                                 |                     |    | M           | C                 | N                |         |
| 1.                            | 2.                        | 3.              | 4.                                                                                                                                                                                                                                                                                                 | 5.               | 6.                  |       | 7.                    | 8.                                                                                                              | 9.                  | D* | ** 10.      |                   |                  | 11.     |
|                               |                           |                 |                                                                                                                                                                                                                                                                                                    |                  |                     |       |                       |                                                                                                                 |                     |    |             |                   |                  |         |
|                               |                           |                 | <b>LEGEND:</b> * RECORDS, INDENTIFIED WITH “TICK” ( ✓ ) SHALL BE ESSENTIALLY INCLUDED BY SUPPLIER IN QA DOCUMENTATION.<br>** M: MANUFACTURER/SUB-SUPPLIER C: MAIN SUPPLIER, N: NTPC<br>P: PERFORM W: WITNESS AND V: VERIFICATION. AS APPROPRIATE,<br>CHP: NTPC SHALL IDENTIFY IN COLUM “N” AS ‘ W” |                  |                     |       |                       | <br><br>FOR<br>NTPC<br>USE | DOC. NO.:           |    |             | REV..... CAT..... |                  |         |
| MANUFACTURER/<br>SUB-SUPPLIER |                           | MAIN-SUPPLIER   |                                                                                                                                                                                                                                                                                                    |                  |                     |       |                       |                                                                                                                 |                     |    |             |                   |                  |         |
| SIGNATURE                     |                           |                 |                                                                                                                                                                                                                                                                                                    |                  |                     |       |                       |                                                                                                                 | REVIEWED BY         |    | APPROVED BY |                   | APPROVAL<br>SEAL |         |

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## ANNEXURE-II

| ENDORSEMENT SHEET FOR QP<br>REFERENCE / STANDARD / FIELD QUALITY PLAN (RQP / SQP/RFQP/SFQP)                                                                                                |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| TO BE FILLED IN BY SUPPLIER AT TIME OF SUBMISSION                                                                                                                                          |                 |                                                                                                                                                                                                                                                                                                                                                             | To be filled in by NTPC |
| PROJECT NAME                                                                                                                                                                               |                 | <b>REVIEW &amp; ENDORSEMENT BY NTPC PROJECT</b><br><b>SPECIFIC QP NUMBER ALLOTTED</b><br><b>QP NO.:</b><br><br><b>REV. NO.:      DATE:</b><br><b>** The RQP/SQP/RFQP/SFQP once endorsed for a particular contract shall remain valid even though the original QP may have expired or revised, unless / otherwise mutually agreed with the supplier. ①</b>                                                                                      |                         |
| CONTRACT NO.:                                                                                                                                                                              |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| MAIN SUPPLIER                                                                                                                                                                              |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| MANUFACTURER WORKS & ADDRESS                                                                                                                                                               | M/S             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |
| ITEM /EQUIPMENT / SYSTEM/<br>SUB-SYSTEM DETAILS i.e. MODEL TYPE /<br>SIZE /RATING etc.                                                                                                     |                 | <b>REV. NO.:      DATE:</b><br><b>** The RQP/SQP/RFQP/SFQP once endorsed for a particular contract shall remain valid even though the original QP may have expired or revised, unless / otherwise mutually agreed with the supplier. ①</b>                                                                                                                                                                                                     |                         |
| APPROVED QP NO.: RQP/SQP/RFQP/SFQP                                                                                                                                                         | 0000-999-QV - - | REV. NO.:                                                                                                                                                                                                                                                                                                                                                                                                                                      | DATED**:                |
| Confirmation by Main Supplier (TICK WHICHEVER APPLICABLE)                                                                                                                                  |                 | (TICK APPLICABLE)                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
| I. That the item/ component is identical to that considered for QP approval. OR.                                                                                                           |                 | The QP is endorsed for this project without any change                                                                                                                                                                                                                                                                                                                                                                                         |                         |
| II. That there are minor changes in the item/ component with respect to that considered for QP approval, however the same do not affect the contents of QP. OR                             |                 | The QP is endorsed for this project with changes as indicated.                                                                                                                                                                                                                                                                                                                                                                                 |                         |
| III. That there are minor changes in the item/ component with respect to that considered for QP approval, however the same affect the QP slightly, as indicated below / in attached sheet. |                 | <b><u>DISTRIBUTION OF ENDORSEMENT OF</u></b><br><b>A) RQP/SQP:</b><br><b>1. MAIN SUPPLIER (WITH A COPY OF QP)</b><br><b>2. MANUFACTURER</b><br><b>3. RIO</b><br><b>4. CQA-SPL</b><br><b>5. CQA-O/C</b><br><br><b>B) RFQP/SFQP:</b><br><b>1. MAIN SUPPLIER (with a copy of QP)</b><br><b>2. MANUFACTURER</b><br><b>3. NTPC FQA (with a copy of QP)</b><br><b>4. NTPC Erection (with a copy of QP)</b><br><b>5. CQA-SPL</b><br><b>6. CQA-O/C</b> |                         |
| SIGN.: (Main Supplier)                                                                                                                                                                     | DATE            | SIGN.: (Manufacturer)                                                                                                                                                                                                                                                                                                                                                                                                                          | DATE:                   |
|                                                                                                                                                                                            |                 | NTPC (Reviewed /Approved by/ Date & Seal)                                                                                                                                                                                                                                                                                                                                                                                                      |                         |

CLAUSE NO.

QUALITY ASSURANCE

**SWITCHYARD**

| Attributes /<br>Characteristics<br><br>Items/Components<br>Sub Systems              | Make, model,<br>Type & Rating,<br>Test Certificate | Routine &<br>Acceptance<br>Test as per IS /<br>IEC | Functional<br>requirements as per<br>NTPC<br>Specification |
|-------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|------------------------------------------------------------|
| 765 kV GIS (IEC:62271-203)                                                          | Y                                                  | Y                                                  | Y                                                          |
| 400 kV GIS (IEC:62271-203)                                                          | Y                                                  | Y                                                  | Y                                                          |
| 220 kV GIS (IEC:62271-203)                                                          | Y                                                  | Y                                                  | Y                                                          |
| 132 kV GIS (IEC:62271-203)                                                          | Y                                                  | Y                                                  | Y                                                          |
| Circuit Breaker (IEC:62271-100)                                                     | Y                                                  | Y                                                  | Y                                                          |
| Isolator (IEC:62271-102)                                                            | Y                                                  | Y                                                  | Y                                                          |
| Current Transformer<br>(IEC:60044/BS:3938/IS2705/ IEC:<br>61869)                    | Y                                                  | Y                                                  | Y                                                          |
| Capacitor Voltage Transformer<br>(IEC:186A /<br>358/IS3156/IEC60044/ IEC:<br>61869) | Y                                                  | Y                                                  | Y                                                          |
| Potential transformer (IEC 60044 /<br>IS3156)                                       |                                                    |                                                    |                                                            |
| Surge Arrestor (AIS) (IEC:99-<br>4/IS:3070)                                         | Y                                                  | Y                                                  | Y                                                          |
| Wave Trap<br>(IEC:353 / IS:8792 / 8793)                                             | Y                                                  | Y                                                  | Y                                                          |
| Sub Station Automation system<br>(IEC 61850)                                        | Y                                                  | Y                                                  | Y                                                          |
| Protection Relays                                                                   | Y                                                  | Y                                                  | Y                                                          |
| Energy meter                                                                        | Y                                                  | Y                                                  | Y                                                          |
| Bus Post Insulator<br>(IEC:168 / 815 / IS:2544)                                     | Y                                                  | Y                                                  | Y                                                          |
| Disc, Pin & String Insulator<br>(IEC:383 / IS:731)                                  | Y                                                  | Y                                                  | Y                                                          |
| Aluminum Tube<br>(IS:5082 / 2673 / 2678)                                            | Y                                                  | Y                                                  | Y                                                          |
| Conductor (IS:398)                                                                  | Y                                                  | Y                                                  | Y                                                          |
| Hardware fittings for Insulator<br>(IS:2486 / BS:3288)                              | Y                                                  | Y                                                  | Y                                                          |
| Hollow insulator<br>(IEC:233/ IS:5621)                                              | Y                                                  | Y                                                  | Y                                                          |
| Spacers, Clamps & Connector<br>(IS:10162 / 5561/ 617)                               | Y                                                  | Y                                                  | Y                                                          |
| Galvanized Steel Structures<br>(IS:2062/2629/4759/6745)                             | Y                                                  | Y                                                  | Y                                                          |
| Vibration Damper (IS:9708)                                                          | Y                                                  | Y                                                  | Y                                                          |

**EPC PACKAGE FOR DARLIPALI STPP,  
STAGE-II (1X800 MW)**

**TECHNICAL SPECIFICATIONS  
SECTION – VI, PART-B  
BID DOC. NO. CS-9551-001-2**

**SUB SECTION E45  
SWITCHYARD**

**PAGE  
1 OF 2**

CLAUSE NO.

QUALITY ASSURANCE



| Attributes /<br>Characteristics<br><br>Items/Components<br>Sub Systems                                                                                                                                                                                                                                        | Make, Type<br>Rating, and<br>Model, Test<br>Certificates | Routine &<br>Acceptance<br>Test as per<br>relevant IS/IEC | Functional<br>requirements<br>as per NTPC<br>Specification |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                               |                                                          |                                                           |                                                            |
| Sag Compensating Spring<br>DIN:2089/2096 IS:3195 / 7906                                                                                                                                                                                                                                                       | Y                                                        | Y                                                         | Y                                                          |
| Long rod Insulator                                                                                                                                                                                                                                                                                            | Y                                                        | Y                                                         | Y                                                          |
| SF6 Gas filling & evacuating plant                                                                                                                                                                                                                                                                            | Y                                                        | Y                                                         | Y                                                          |
| SF6 Gas Leak Detector                                                                                                                                                                                                                                                                                         | Y                                                        | Y                                                         | Y                                                          |
| Leakage Current Analyzer                                                                                                                                                                                                                                                                                      | Y                                                        | Y                                                         | Y                                                          |
| Nitrogen Gas Filling Device                                                                                                                                                                                                                                                                                   | Y                                                        | Y                                                         | Y                                                          |
| Event Logger                                                                                                                                                                                                                                                                                                  | Y                                                        | Y                                                         | Y                                                          |
| Operation Analyzer                                                                                                                                                                                                                                                                                            | Y                                                        | Y                                                         | Y                                                          |
| Disturbance Recorder                                                                                                                                                                                                                                                                                          | Y                                                        | Y                                                         | Y                                                          |
| Synchronizing Trolley                                                                                                                                                                                                                                                                                         | Y                                                        | Y                                                         | Y                                                          |
| Relay Test Kit                                                                                                                                                                                                                                                                                                | Y                                                        | Y                                                         | Y                                                          |
| <p>Notes: 1) This is an indicative list of test/checks. The manufacture is to furnish a detailed Quality Plan indicating the practice and procedure along with relevant supporting documents during QP finalisation for all items.</p> <p>2) All major Bought Out Items will be subject to NTPC approval.</p> |                                                          |                                                           |                                                            |

EPC PACKAGE FOR DARLIPALI STPP,  
STAGE-II (1X800 MW)

TECHNICAL SPECIFICATIONS  
SECTION – VI, PART-B  
BID DOC. NO. CS-9551-001-2

SUB SECTION E45  
SWITCHYARD

PAGE  
2 OF 2

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 765kV Isolator with Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

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#### **SECTION 4: ANNEXURES**

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| ANNEXURE-C: | Technical Checklist                                  |
| ANNEXURE-D: | Format for Guaranteed Technical Particulars          |

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 765kV Isolator with Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

### ANNEXURE-A: Compliance Certificate of Technical Specification

The bidder shall confirm compliance to the following by signing and stamping this compliance certificate and furnishing same with the offer.

1. The scope of supply, technical details, construction features, design parameters etc. shall be as per technical specification & there are no exclusion/ deviation with regard to same.
2. There are no deviation(s) with respect to specification other than those furnished in the schedule of deviations.
3. Only those technical submittals which are specifically asked for in Notice Inviting Tender (NIT) to be submitted at tender stage shall be considered as part of offer. Any other submission, even if made, shall not be considered as part of technical offer.
4. Any comments/ clarifications on technical/ inspection requirements furnished as part of bidder's covering letter shall not be considered by BHEL, and bidder's offer shall be construed to be in conformance with the specification.
5. Any changes made by the bidder in the price schedule with respect to the description/ quantities from those given in 'BOQ' of the specification shall not be considered (i.e., technical description & quantities as per the specification shall prevail).

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 765kV Isolator with Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

### ANNEXURE-B: Deviation/ Change Request of Technical Specification

Bidder shall list out all technical potential deviation/ change request (s) along with clause with respect to technical specifications.

| Sl. No. | Page No. | Clause No. | Deviation | Reason/ Justification(s) |
|---------|----------|------------|-----------|--------------------------|
|---------|----------|------------|-----------|--------------------------|

Any deviation not specifically brought out in this section shall not be admissible for any commercial implication at later stage. Except to the technical deviations listed in this schedule, bidder's offer shall be considered in full compliance to the tender specifications irrespective of any such deviation indicated / taken elsewhere in the submitted offer.

Date:

Bidder's Stamp & Signature

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 765kV Isolator with Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

### ANNEXURE-C: Technical Checklist

| Sl. No.  | Particulars                                                                                                                                                                                                                                                  | Confirmation | by Bidder |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|
| <b>1</b> | <b>Technical Qualifying Requirement</b>                                                                                                                                                                                                                      |              |           |
| 1.1      | The bidder to furnish relevant documents for meeting the qualifying requirement. Performance certificates shall be submitted in English. Translated pages should be attested by the ultimate customer, if attested only by the bidder it shall be notarized. | Confirmed    | Yes/ No   |
| 1.2      | The bidder's scope includes supply and services such as Supervision of installation, Testing and commissioning.                                                                                                                                              | Confirmed    | Yes/ No   |
| <b>2</b> | <b>Un-priced BOQ</b>                                                                                                                                                                                                                                         |              |           |
| 2.1      | Confirm that all items have been quoted separately. (If any item has not been quoted, the same shall be specifically brought out with technical reasons thereof. Record the same in schedule of technical deviations.                                        | Confirmed    | Yes/ No   |
| <b>3</b> | <b>Technical</b>                                                                                                                                                                                                                                             |              |           |
| 3.1      | Catalogues, indicative OGA of the offered equipment is attached.                                                                                                                                                                                             | Confirmed    | Yes/ No   |
| 3.2      | Minimum Number of auxiliary contacts on each isolator - besides requirement of this specification/ Equipment Schematics. The bidder shall wire up 12 NO +12 NC+ 4 MBB to Terminal Block (reversible) for purchaser 's use.                                   | Confirmed    | Yes/ No   |
| 3.3      | Minimum Number of auxiliary contacts on each earthing switch - besides requirement of this specification/ Equipment Schematics. The bidder shall wire up 12 NO +12 NC + 4 MBB to Terminal Block (reversible) for purchaser 's use.                           | Confirmed    | Yes/ No   |
| 3.4      | Bidder to provide flexible Copper Braid for Earthing System for Isolator and earth switch MOM Box (2 number per box), size equivalent to 75X12 mm GS flat.                                                                                                   | Confirmed    | Yes/ No   |
| <b>4</b> | <b>Technical Deviations</b>                                                                                                                                                                                                                                  |              |           |
| 4.1      | Confirm that the Complete systems have been offered as per the requirements of Technical Specification and Technical Deviation sheet has been submitted. Deviations mentioned elsewhere in the bid shall not be considered.                                  | Confirmed    | Yes/ No   |
| <b>5</b> | <b>Guaranteed Technical Particulars</b>                                                                                                                                                                                                                      |              |           |
| 5.1      | All equipment being supplied shall conform to Guaranteed Technical Particulars as per technical specification and                                                                                                                                            | Confirmed    | Yes/ No   |

## Bharat Heavy Electricals Limited

**Project:** 765/132kV AIS Switchyard Extn. at NTPC DARLIPALI STPP Stage-II (1x800 MW)

**Customer:** NTPC Limited

**Technical Specification:** 765kV Isolator with Earth Switches

**Document No.** TB-436-316-TS-004B Rev 00

|          |                                                                                                                                                                                                                                                                                                                                                                                                    |           |         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|
|          | applicable IS/ IEC.                                                                                                                                                                                                                                                                                                                                                                                |           |         |
| <b>6</b> | <b>Type Tests Requirements</b>                                                                                                                                                                                                                                                                                                                                                                     |           |         |
| 6.1      | All equipment being supplied shall conform to type tests as per technical specification and shall be subject to routine tests in accordance with requirements stipulated under respective sections.                                                                                                                                                                                                | Confirmed | Yes/ No |
| 6.2      | The validity of type test reports shall be as per the latest CEA guidelines (amended time to time) as date of bid opening (i.e. 10.07.2024). In case, where type test certificates are older than period as per latest CEA guidelines (amended time to time), bidder/ manufacturer shall carry out the type tests prior to dispatch of equipment without any commercial implication on BHEL/ NTPC. | Confirmed | Yes/ No |

Date:  
Signature

Bidder's Stamp &

| CLAUSE<br>NO. | Bidder's Name .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|               | <p style="text-align: center;"><b>DB:12B</b></p> <p style="text-align: center;"><b>EHV DISCONNECTORS/ISOLATORS</b></p> <p>(Bidder shall furnish the data for each type of DS separately)</p> <p>1. General .....</p> <p>a) Name &amp; country of the Manufacturer .....</p> <p>b) Manufacturer's type designation .....</p> <p>c) Standard Applicable .....</p> <p>d) Rated Voltage (KV) .....</p> <p>e) Rated Current (Amps) .....</p> <p>    i) Under normal .....</p> <p>    ii) Under site conditions at 50 deg C ambient .....</p> <p>f) Number of poles .....</p> <p>g) Whether all 3 poles are ganged mechanically .....</p> <p>h) Pole to pole spacing .....</p> <p>2. Guaranteed Ratings .....</p> <p>a) Rated short time current of Isolator .....</p> <p>    i) for 1 sec. (kA rms) .....</p> <p>    ii) for 3 sec. (kA rms) .....</p> <p>    iii) Dynamic current .....</p> <p>b) Opening Time of .....</p> <p>    i) Isolator (sec) .....</p> <p>    ii) Earth switch (sec) .....</p> |                                                                                     |
| Section-4     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page-6                                                                              |

| CLAUSE<br>NO. | Bidder's Name .....                                                                                                        |  |
|---------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 3.            | c) Closing time of .....                                                                                                   |                                                                                     |
|               | i) Isolator (sec) .....                                                                                                    |                                                                                     |
|               | ii) Earth switch (sec) .....                                                                                               |                                                                                     |
|               | Dielectric with-stand capacity of Completely assembled Isolator and earth switch .....                                     |                                                                                     |
|               | a) One minute power frequency withstand test voltage (kV) rms .....                                                        |                                                                                     |
|               | i) against ground (kV peak) .....                                                                                          |                                                                                     |
|               | - dry (kV rms) .....                                                                                                       |                                                                                     |
|               | - Wet (kV rms) .....                                                                                                       |                                                                                     |
|               | b) 1.2/50 micro second impulse withstand test voltage .....                                                                |                                                                                     |
|               | i) against ground (kV peak) .....                                                                                          |                                                                                     |
|               | ii) across open contacts (kV peak) .....                                                                                   |                                                                                     |
|               | c) 250/2500 micro second switching surge withstand test voltage (dry & wet) .....                                          |                                                                                     |
|               | i) against ground (kV peak) .....                                                                                          |                                                                                     |
|               | ii) across open contacts (kV peak) .....                                                                                   |                                                                                     |
|               | d) Corona extinction voltage (kV rms) .....                                                                                |                                                                                     |
|               | e) Radio interference level at $1.1 \times U_m / \sqrt{3}$ (in micro volts) for frequency between 0.5 MHz to 2.0 MHz ..... |                                                                                     |
|               | f) Total creepage distance to ground (mm) .....                                                                            |                                                                                     |

| CLAUSE<br>NO. | Bidder's Name .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 4.            | Operating Mechanism .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                     |
|               | a) Type and rating (KW) of motor .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |
|               | b) Rated voltage of motor .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                     |
| 4.1           | Interlockings .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                     |
|               | a) Whether mechanical/<br>constructional interlock<br>between Isolator and Earth<br>switch provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Yes/No<br>.....                                                                     |
|               | b) Whether interlock coil is<br>continuously rated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | .....                                                                               |
| 5             | Constructional Features<br><br>a) Main contacts .....<br>i) Type of contacts .....<br><br>ii) Contact area (cm <sup>2</sup> ) .....<br><br>iii) Material of contacts .....<br><br>iv) Contact pressure (Kg/cm <sup>2</sup> ) .....<br><br>v) Maximum current density<br>under normal current<br>carrying capacity<br>(Amp/cm <sup>2</sup> ) .....<br><br>vi) Thickness of silver<br>plating .....<br><br>b) Auxiliary contacts on<br>Isolator .....<br><br>i) Total number .....<br>i) NO .....<br>iii) NC .....<br><br>iv) Adjustable .....<br><br>v) Make before break ..... |                                                                                     |

| CLAUSE<br>NO. | Bidder's Name .....                                               |  |
|---------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|               | vi) Rated voltage (volts) .....                                   |                                                                                     |
|               | vii) Rated continuous current (Amps) .....                        |                                                                                     |
|               | i) Rated DC breaking current with 20 ms time constant (A) .....   |                                                                                     |
|               | c) Auxiliary contacts on earth switch .....                       |                                                                                     |
|               | i) Total number .....                                             |                                                                                     |
|               | i) NO .....                                                       |                                                                                     |
|               | iii) NC .....                                                     |                                                                                     |
|               | iv) Adjustable .....                                              |                                                                                     |
|               | v) Rated Voltage (volts) .....                                    |                                                                                     |
|               | vi) Rated continuous current (Amps) .....                         |                                                                                     |
|               | vii) Rated DC breaking current with 20 ms time constant (A) ..... |                                                                                     |
|               | e) Whether details of constructional interlock enclosed           | Yes/No                                                                              |

| <b>CLAUSE<br/>NO.</b> | <b>Bidder's Name .....</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>A.</b>             | <p style="text-align: center;"><b>DB:12G</b></p> <p style="text-align: center;"><b>EHV SWITCHYARD<br/>AUXILIARY ITEMS</b></p> <p><b>CONTROL CABINETS</b></p> <p>(Bidder shall furnish this data for each equipment marshalling box/operating mechanism box separately i.e.Bay Marshalling Box, cabinets of breakers, disconnecting switches, instrument transformers, lighting panels, etc.)</p> <ol style="list-style-type: none"> <li>1.     Manufacturer's Name and country of origin .....</li> <li>2.     Design ambient air temperature (°C) .....</li> <li>3.     Standards Applicable .....</li> <li>4.     Thickness of sheet steel (mm) and whether cold rolled or hot rolled .....</li> <li>5.     Degree of protection provided .....</li> <li>6.     Shede of finish paint as per RAL .....</li> <li>        a)     Outside .....</li> <li>        b)     Inside .....</li> </ol> |                                                                                     |
|                       | <p><b>B.     TERMINAL CLAMPS AND CONNECTORS, AND SPACERS</b></p> <p>(Bidder shall furnish this data for Terminal clamps and connector separately for each type and equipment)</p> <ol style="list-style-type: none"> <li>1.     Manufacturer's Name and address .....</li> <li>2.     Applicable Standards .....</li> </ol> <p><b>C.     INSULATOR HARDWARE</b></p> <p>(Bidder shall furnish this data separately for each type and rating)</p> <ol style="list-style-type: none"> <li>1.     Manufacturer .....</li> </ol>                                                                                                                                                                                                                                                                                                                                                                    |                                                                                     |